

SE250B4 Smart Module

Hardware Design Guide

1VV0301817 Rev.9

2026-06-08

Released

Confidential

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1 Applicability Table

Table 1: Applicability Table

Products
SE250B4-NA
SE250B4-EU
SE250B4-WE

2 Introduction

2.1 Scope

This document describes electrical specifications, mechanical information, interfaces application and manufacturing information about Telit SE250B4 Smart module. With the help of this document and other application notes or user guides, users can understand Telit SE250B4 Smart module well and develop various products quickly.

2.2 Audience

This document is intended for system integrators that are using the Telit SE250B4 module in their products.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com
- TS-LOCATION@telit.com

Alternatively, use: <https://www.telit.com/contact-us/>

For Product information and technical documents, visit: <https://www.telit.com>

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and Conditions

Refer to <https://www.telit.com/terms-conditions/>.

2.6 Disclaimer

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3 General Product Description

3.1 Overview

Telit SE250B4 Smart module is a multi-mode and multi-band wireless smart module, which is based on the Qualcomm QCM2290 platform. It integrates baseband, memory, RF front end, and the necessary circuitry to support advanced multimedia features, global location-based services, wireless connectivity, and various air interface standards such as GSM, WCDMA, LTE, WiFi and Bluetooth.

With its high level of integration, the Telit SE250B4 Smart module aims to reduce PCB surface area, shorten time-to-market, and lower BoM cost, thereby promoting the adoption of wireless products across various industries worldwide. The operating bands vary among the different variants of the Telit SE250B4 Smart module.

3.2 Product Variants and Frequency Bands

Telit SE250B4 Smart module comes in three variants, each tailored to different regional requirements:

- SE250B4-NA
- SE250B4-EU
- SE250B4-WE

For detailed information on the differences between these two variants, please refer to the specific product documentation.



Figure 1: SE250B4 NA/EU/WE SKU

Table 2: Product Variants and Frequency Bands

Configuration		SE250B4-NA	SE250B4-EU	SE250B4-WE
CPU		2.0GHz	2.0GHz	2.0GHz
Memory	RAM	2GB/3GB	2GB/3GB	2GB/3GB
	Flash	16GB/32GB	16GB/32GB	16GB/32GB
Standards & bands				
GSM	900MHz		P	
	1800MHz		P	
WCDMA	B1		P	
	B2			
	B4			
	B5		P	
	B8		P	
LTE FDD	B1		P	
	B2	P		
	B3		P	
	B4	P		
	B5	P	P	
	B7	P	P	
	B8		P	
	B12	P		
	B13	P		
	B14	P		
	B17	P		
	B20		P	
	B25	P		
	B26	P		
	B28A		P	
	B28B		P	
	B66	P		
	B71	P		
LTE TDD	B38		P	
	B40		P	
	B41	P	P	
WLAN	2.4G/5GHz; 802.11a/b/g/n/ac	P	P	P
BT	BT 5.0 + BR/EDR + BLE	P	P	P
GNSS	GPS	P	P	
	GLONASS	P	P	
	Galileo	P	P	
	BEIDOU	P	P	

3.3 Target Market

Telit SE250B4 Smart module offers tamper-resistant, confidential, secure, and authentic end-to-end connectivity, making it ideal for telematics applications. It is one of the key components among the M2M applications, such as:

- CPE (Customer Premises-based Equipment)
- POS machine
- Smart metering
- Router
- Data card
- Digital signage
- Alarm panel
- Security
- Industrial PDA.

3.4 Main Features

Table 3: Main Features

Function	Features
Application processor	Quad ARM Cortex-A53 cores up to 2.0GHz, 64-bit Arm Cortex
Memory RAM	2GB, 3GB 1804MHz clock (2x16 bits), LPDDR4X SDRAM
Memory Flash	16GB, 32GB SE250B4-NA / SE250B4-EU / SE250B4-WE External memory via SDC2 Support SD3.0 flash devices, up to 128GB
Graphics	Adreno 702; up to 845MHz with 64-bit addressing
Operating system	Android 13 go version and above / Yocto Linux (Kernel 5.X) *
Software upgrade	Upgrade via USB interface; support forced download
Power supply	Voltage range: 3.5V ~ 4.4V, support single-cell lithium battery power supply
Charge management	Integrated up to 1.85A for fast charging
Display support	4-lane MIPI_DSI, 1.5Gbps each; HD+(1680x720), 60fps
Camera interfaces	MIPI_CSI – up to 3 cameras (4/2/1), 25MP total max
Video applications performance	Encode: 1080p 30 fps: H265 Decode: 1080p 30 fps: HEVC, H.265, VP9 Encode + Decode: 720p 30 fps + 1080p 30 fps
Audio supply	Three analog inputs that support single-ended configurations OR two data input Three outputs: earpiece, stereo headphones, and mono speaker driver
Audio codec support	MP3; AAC; He-AAC v1, v2; WMA 9/Pro; Dolby AC-3, eAC-3, DTS-HD M6 and DTS-HD M8
Voice codec support	EVRC, -B, -WB; G.711; G.729A,-AB; GSM-FR,-EFR,-HR; ARM-NB,-WB; eAMR; BeAMR
USB	USB 3.1/2.0 compliant
UART	Supports up to 3*UART, up to 4MHz
I2C	Supports up to 6*I2C, used to cameras, sensors, TP, and so forth
SPI	Supports up to 5*SPI, up to 50MHz (Master only)

Function	Features
ADC	Supports up to 2*ADC, typical input range: 0.3~4.5V
UIM card	Dual cards dual standby; 1.8V/2.95V dual voltage adaptation
LTE features	Support 3GPP R12 CAT4 FDD and TDD Support 1.4 to 20MHz RF bandwidth Cat 4 FDD Datarate Max 150Mbps (DL) / Max 50Mbps (UL) Cat 4 TDD Datarate Max 130Mbps (DL) / Max 30Mbps (UL)
UMTS features	Support 3GPP R9 DC-HSDPA/HSPA+/HSDPA/HSUPA/WCDMA Support 16-QAM, QPSK DC-HSDPA: Max 42Mbps (DL) HSUPA: Max 5.76Mbps (UL) WCDMA: Max 384Kbps (DL) / Max 384Kbps (UL)
GSM features	R99: CSD: 9.6kbps, 14.4kbps GPRS: Support GPRS multi-slot level 33 (default 33) coding schemes: CS-1, CS-2, CS-3, CS-4 Max 85.6Kbps (UL) / Max 107Kbps (DL) EDGE: Support EDGE multi-slot level 33 (default 33) Support GMSK and 8-PSK Modulation coding DL coding schemes: CS 1-4 and MCS 1-9 UP coding schemes: CS 1-4 and MCS 1-9 Max 236.8Kbps (UL) / Max 296Kbps (DL)
WLAN features	Support SoftAP Function, 802.11 a/b/g/n/ac Encryption: WPA WPA2 Qos: WPA WMM, WMM PS Wi-Fi bands: 2.4GHz/5GHz
BT Features	Bluetooth 5.0 and earlier Specification: V2.1+EDR, 3.0+HS, V4.1 BLE Tx power levels: Class 1 & 2

Note: *Voice calls are currently not supported in Linux OS.

3.5 Tx Output Power

Table 4: TX Output Power

Band	Power class
EGSM 900MHz	Class 4
DCS 1800	Class 1
WCDMA/HSPA+	Class 3
LTE All Bands	Class 3

3.6 Mechanical Specifications

3.6.1 Dimensions

The overall dimensions of the Telit SE250B4 Smart module family are:

- Length: 43 mm, +/- 0.15 mm tolerance
- Width: 41 mm, +/- 0.15 mm tolerance
- Thickness: 3 mm, +/- 0.25 mm tolerance

3.7 Temperature Range

Table 5: Temperature Range

Temperature range	type	Temperature range	Note
Operating Temperature		-20°C /+75°C	The module is fully functional in the complete temperature range, and it fully meets the 3GPP specifications. EU RED certified at -20°C/+75°C, GCF/PTCRB certified at -10°C/+55°C
Extended Temperature		-30°C ~ +85°C	Telit guarantees full functionality within this range as well. However, possibly there may be some performance deviations in this extended range related to 3GPP requirements, which means that some RF parameters may deviate from the 3GPP specification on the receiver or the maximum output power may be slightly degraded. Even so, all functionalities, such as Connection to calls, SMS, USB communication, UART activation and so on, will be maintained, and the effect of such degradations will not lead to malfunctions.
Storage and non-operating		-40°C ~ +90°C	

3.8 Block Diagram

The architecture block diagram is shown below.

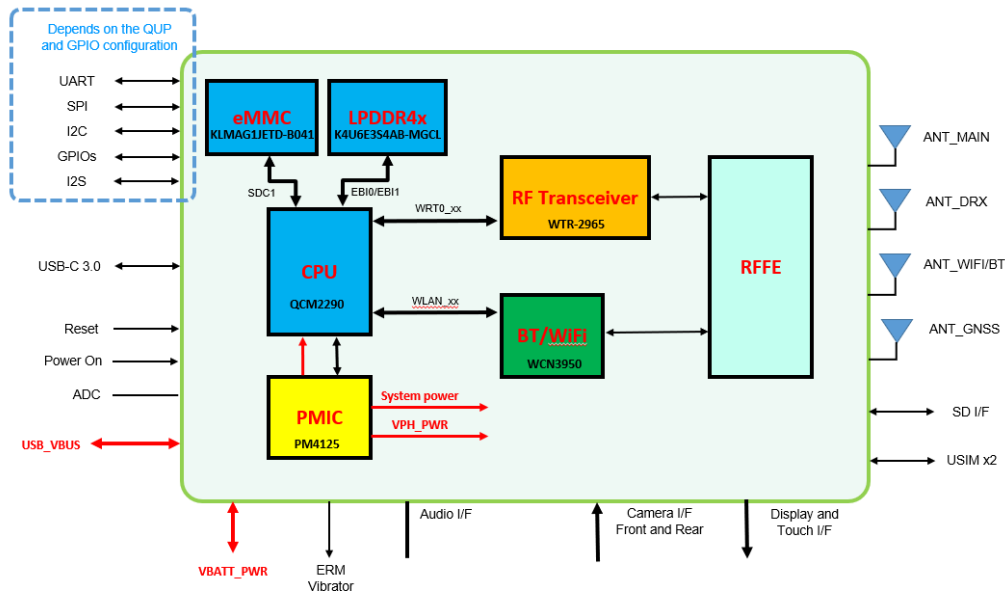


Figure 2: SE250B4 block diagram

The number of GPIOs and enabled peripherals is configurable via software and depends on the customer's application. With default settings, the device supports the following:

- 1 x USB
- 2 x SIM
- 1 x UARTs
- 1 x DEBUG UART
- 1 x SD CARD
- 26 GPIOs, with 23 of them configurable for alternate functions, including:
 - 8 for NFC
 - 3 for sensors
 - 3 for HDMI
 - 2 for PM8008 RST_N + interrupt
 - 7 for LCM (3 EN + 2 ID + 2 BIAS)

4 Pins Allocation

4.1 LGA Pads Layout

Top View

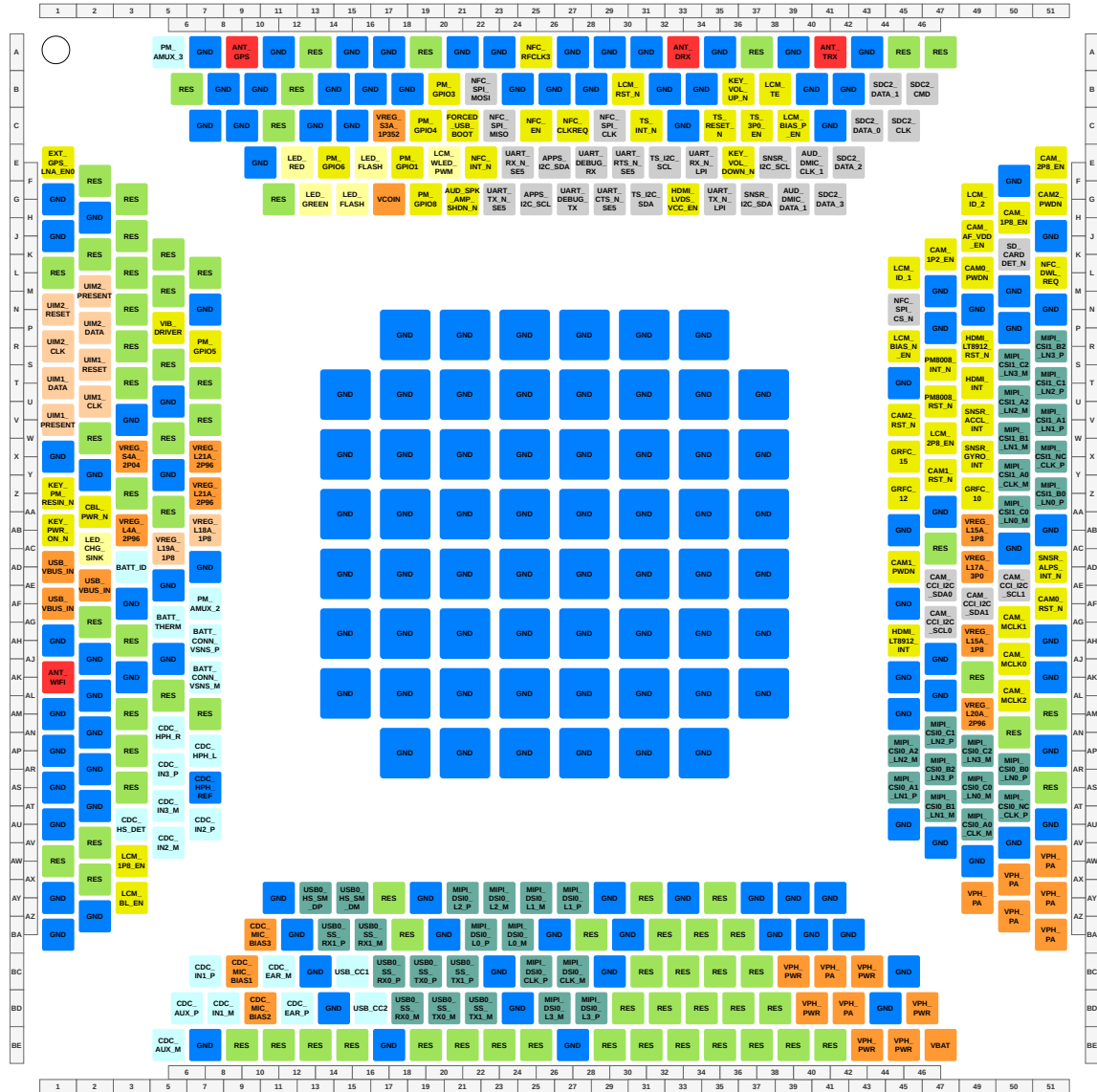


Figure 3: LGA Pads Layout

- SUPPLY AND CONTROL
- SIM CARD
- ANALOG FUNCTIONALITY
- GROUND
- DIGITAL FUNCTIONALITY
- DIGITAL FUNCTIONALITY (high current)
- DIGITAL COMMUNICATION
- HIGH SPEED DIGITAL COMMUNICATION
- RF SIGNALS
- NO CONNECT / RESERVED FOR FUTURE USE

4.2 Pin-out

4.2.1 USB Interface

Table 6: PIN-out description

PIN	Signal	I/O	Function	DC Characteristics	Power Domain / Comment	GPIO
C21	FORCED_USB_BOOT	DI			GPIO power domain.	GPIO_95
AY15	USB0_HS_SM_DM	AI/O	USB differential (-)		90Ω differential impedance with 10% tol; compliant to USB 2.0 standard	
AY13	USB0_HS_SM_DP	AI/O	USB differential (+)			
AD1, AE2, AF1	USB_VBUS_IN	PI	Charging power input and power output for OTG devices USB/adaptor insertion detection	Vmax = 6.0V Vmin = 4.0V Vnom = 5.0V	The Max output current is 500mA	
BD18	USB0_SS_RX0_M	AI/O	USB 3.1 channel 1 superspeed receive (-)		90Ω differential impedance, compliant to USB 3.1 standard	
BC17	USB0_SS_RX0_P	AI/O	USB 3.1 channel 1 superspeed receive (+)			
BD20	USB0_SS_TX0_M	AI/O	USB 3.1 channel 1 superspeed transmit (+)			
BC19	USB0_SS_TX0_P	AI/O	USB 3.1 channel 1 superspeed transmit (-)			
BA16	USB0_SS_RX1_M	AI/O	USB 3.1 channel 2 superspeed receive (-)			
BA14	USB0_SS_RX1_P	AI/O	USB 3.1 channel 2 superspeed receive (+)			
BD22	USB0_SS_TX1_M	AI/O	USB 3.1 channel 2 superspeed transmit (+)			

PIN	Signal	I/O	Function	DC Characteristics	Power Domain / Comment	GPIO
BC21	USB0_SS_TX1_P	AI/O	USB 3.1 channel 2 superspeed transmit (-)			
BC15	USB_CC1	AI	USB Type-C detect 1			
BD16	USB_CC2	AI	USB Type-C detect 2			

4.2.2 USIM Interface

Table 7: USIM Interface

PIN	Signal	I/O	Function	DC Characteristics	Power Domain / Comment	GPIO
U2	UIM1_CLK	DO	(U)SIM1 card clock		USIM1 power domain.	GPIO_77
V1	UIM1_PRESENT	DI	(U)SIM1 card hot-plug detect	VILmax = 0.63V VIHmin = 1.17V	GPIO power domain. Active low. External pull up to VREG_L15A_1P8 required.If unused, keep it open (for use as GPIO please contact support).	GPIO_79
T1	UIM1_DATA	DI/O	(U)SIM1 card data	VILmax = 0.2 × USIM1_VDD VIHmin = 0.7 × USIM1_VDD VOLmax = 0.4V VOHmin = 0.8 × USIM1_VDD	USIM1 power domain.	GPIO_76
S2	UIM1_RESET	DO	(U)SIM1 card reset	VOLmax = 0.4V VOHmin = 0.8 × USIM1_VDD	USIM1 power domain.	GPIO_78
AB7	VREG_L18A_1P8	PO	USIM1_VDD	IOmax = 67mA 1.8V (U)SIM: Vmax = 1.85V Vmin = 1.75V 2.95 V (U)SIM: Vmax = 3.1V Vmin = 2.8V	Either 1.8V or 2.95V (U)SIM card is supported.	
R1	UIM2_CLK	DO	(U)SIM2 card clock		USIM2 power domain.	GPIO_73

PIN	Signal	I/O	Function	DC Characteristics	Power Domain / Comment	GPIO
M2	UIM2_PRESENT	DI	(U)SIM2 card hot-plug detect	VILmax = 0.63V VIHmin = 1.17V	GPIO power domain. Active low. External pull up to VREG_L15A_1P8 required. If unused, keep it open (for use as GPIO please contact support).	GPIO_75
P2	UIM2_DATA	DI/O	(U)SIM2 card data	VILmax = $0.2 \times$ USIM2_VDD VIHmin = $0.7 \times$ USIM2_VDD VOLmax = 0.4V VOHmin = $0.8 \times$ USIM2_VDD	USIM2 power domain.	GPIO_72
N1	UIM2_RESET	DO	(U)SIM2 card reset	VOLmax = 0.4V VOHmin = $0.8 \times$ USIM2_VDD	USIM2 power domain.	GPIO_74
AC5	VREG_L19A_1P8	PO	USIM2_VDD	IOmax = 67mA 1.8 V (U)SIM: Vmax = 1.85V Vmin = 1.75V 2.95 V (U)SIM: Vmax = 3.1V Vmin = 2.8V	Either 1.8V or 2.95V (U)SIM card is supported.	

4.2.3 UART Interface

Table 8: UART Interface

PIN	Signal	I/O	Function	DC Characteristics	Power Domain / Comment	GPIO
E34	UART_RX_N_LPI	DI	UART2 receive	VILmax = 0.63V VIHmin = 1.17V	GPIO power domain. If unused, keep it open. LPI UART pins are reserved for internal pupose and should left open.	LPI_GPIO_26
G35	UART_TX_N_LPI	DO	UART2 transmit	VOLmax = 0.45V VOHmin = 1.35V		LPI_GPIO_25
E24	UART_RX_N_SE5	DI	UART1 receive	VILmax = 0.63V VIHmin = 1.17V		GPIO_17
G23	UART_TX_N_SE5	DO	UART1 transmit	VOLmax = 0.45V VOHmin = 1.35V		GPIO_16
E28	UART_DEBUG_RX	DI	DEBUG UART RECEIVE	VILmax = 0.63V VIHmin = 1.17V		GPIO_13
G27	UART_DEBUG_TX	DO	DEBUG UART TRANSMIT	VOLmax = 0.45V VOHmin = 1.35V		GPIO_12

PIN	Signal	I/O	Function	DC Characteristics	Power Domain / Comment	GPIO
G29	UART_CTS_N_SE5	DI	DCE clear to send signal from DTE	VILmax = 0.63V VIHmin = 1.17V		GPIO_14
E30	UART_RTS_N_SE5	DO	DCE request to send signal to DTE	VOLmax = 0.45V VOHmin = 1.35V		GPIO_15

4.2.4 SDIO/SD card Interface

Table 9: SDIO/SD card Interface

PIN	Signal	I/O	Function	DC characteristics	Power Domain	GPIO
C45	SDC2_CLK	DO			SDC2P95 / SDC1P8	
B46	SDC2_CMD	DO				
C43	SDC2_DATA_0	DI/O				
B44	SDC2_DATA_1	DI/O				
E42	SDC2_DATA_2	DI/O				
G41	SDC2_DATA_3	DI/O				
K50	SD_CARD_DET_N	DI				GPIO_88
AB3	VREG_L4A_2P96	PO	SD-Card_LDO line			
X7, Z7	VREG_L21A_2P96	PO	SD-CARD_VDD	2.95V/841mA		

4.2.5 Display Interface

Table 10: Display Interface

PIN	Signal	I/O	Function	Power Domain	GPIO
BC27	MIPI_DSI0_CLK_M	AO	LCD MIPI clock (-)	DSI	-
BC25	MIPI_DSI0_CLK_P	AO	LCD MIPI clock (+)	DSI	-
BA24	MIPI_DSI0_L0_M	AO	LCD MIPI lane 0 data (-)	DSI	-
BA22	MIPI_DSI0_L0_P	AO	LCD MIPI lane 0 data (+)	DSI	-
AY25	MIPI_DSI0_L1_M	AO	LCD MIPI lane 1 data (-)	DSI	-
AY27	MIPI_DSI0_L1_P	AO	LCD MIPI lane 1 data (+)	DSI	-
AY23	MIPI_DSI0_L2_M	AO	LCD MIPI lane 2 data (-)	DSI	-
AY21	MIPI_DSI0_L2_P	AO	LCD MIPI lane 2 data (+)	DSI	-
BD26	MIPI_DSI0_L3_M	AO	LCD MIPI lane 3 data (-)	DSI	-
BD28	MIPI_DSI0_L3_P	AO	LCD MIPI lane 3 data (+)	DSI	-
B30	LCM_RST_N	DO	LCD reset	GPIO	-
B38	LCM_TE	DO	LCD tearing effect	GPIO	GPIO_81
R45	LCM_BIAS_N_EN	DO	LCM bias EN (-)	GPIO	GPIO_106
C39	LCM_BIAS_P_EN	DO	LCM bias EN (+)	GPIO	GPIO_107
L45	LCM_ID1	DI	LCM_ID1 display	GPIO	GPIO_111
G49	LCM_ID2	DI	LCM_ID2 display	GPIO	GPIO_112
AW3	LCM_1P8_EN	DO	LCM supply enable	GPIO	GPIO_114
W47	LCM_2P8_EN	DO	LCM supply Enable	GPIO	GPIO_60
AY3	LCM_BL_EN	DO	LCM BL enable	GPIO	GPIO_113
AH45	HDMI_LT8912_INT	DI	INTerrupt for MIPI DSI bridge	GPIO	GPIO_28

PIN	Signal	I/O	Function	Power Domain	GPIO
T49	HDMI_INT	DI	Interrupt for HDMI	GPIO	GPIO_34
R49	HDMI_LT8912_RST_N	DO	HDMI_Reset	GPIO	GPIO_108
G33	HDMI_LVDS_VCC_EN	DO	LVDS supply enable	GPIO	GPIO_105

4.2.6 TP Interface

Table 11: TP Interface

PIN	Signal	I/O	Function	Power Domain	Comment	GPIO
E32	TS_I2C_SCL	DO	TP I2C clock	GPIO	Need to be pulled up to 1.8V externally. Can be used for other I2C devices.	GPIO_7
G31	TS_I2C_SDA	DO	TP I2C data			GPIO_6
C31	TS_INT_N	DI	TP interrupt	GPIO	1.8V power domain.	GPIO_80
C35	TS_RESET_N	DO	TP reset	GPIO	1.8V power domain. Active low.	GPIO_71
C37	TS_3P0_EN	DI	TP Enable	GPIO		GPIO_104

4.2.7 Camera Interface

Table 12: Camera Interface

PIN	Signal	I/O	Function	Power Domain	GPIO
AJ50	CAM_MCLK0	DO	Master clock of camera 0	GPIO	GPIO_20
L49	CAM0_PWDN	DO	Power down of camera 0	GPIO	GPIO_97
AF51	CAM0_RST_N	DO	Reset of camera 0	GPIO	GPIO_18
AG50	CAM_MCLK1	DO	Master clock of camera 1	GPIO	GPIO_21
AD45	CAM1_PWDN	DO	Power down of camera 1	GPIO	GPIO_102
Y47	CAM1_RST_N	DO	Power down of camera 1	GPIO	GPIO_19
AL50	CAM_MCLK2	DO	Master clock of camera 2	GPIO	GPIO_27
G51	CAM2_PWDN	DO	Power down of camera 2	GPIO	GPIO_37
V45	CAM2_RST_N	DO	Power down of camera 2	GPIO	GPIO_24
AG47	CAM_CCI_I2C_SCL0	DO	I2C clock of camera 0	GPIO	GPIO_23
AE47	CAM_CCI_I2C_SDA0	DI/O	I2C data of camera 0	GPIO	GPIO_22
AE50	CAM_CCI_I2C_SCL1	DO	I2C clock of camera 1	GPIO	GPIO_30
AF49	CAM_CCI_I2C_SDA1	DI/O	I2C data of camera 1	GPIO	GPIO_29
J49	CAM_AF_VDD_EN	DO		GPIO	GPIO_38
K47	CAM_1P2_EN	DO		GPIO	GPIO_58
H50	CAM_1P8_EN	DO		GPIO	GPIO_39
E51	CAM_2P8_EN	DO		GPIO	GPIO_42
AU49	MIPI_CSI0_A0_CLK_M	AI	MIPI CSI0 clock (+)	CSI	-
AT50	MIPI_CSI0_NC_CLK_P	AI	MIPI CSI0 clock (-)	CSI	-
AS49	MIPI_CSI0_C0_LN0_M	AI	MIPI CSI0 lane 0 data (-)	CSI	-
AR50	MIPI_CSI0_B0_LN0_P	AI	MIPI CSI0 lane 0 data (+)	CSI	-
AT47	MIPI_CSI0_B1_LN1_M	AI	MIPI CSI0 lane 1 data (-)	CSI	-
AS45	MIPI_CSI0_A1_LN1_P	AI	MIPI CSI0 lane 1 data (+)	CSI	-
AP45	MIPI_CSI0_A2_LN2_M	AI	MIPI CSI0 lane 2 data (-)	CSI	-

PIN	Signal	I/O	Function	Power Domain	GPIO
AN47	MIPI_CSI0_C1_LN2_P	AI	MIPI CSI0 lane 2 data (+)	CSI	-
AP49	MIPI_CSI0_C2_LN3_M	AI	MIPI CSI0 lane 3 data (-)	CSI	-
AR47	MIPI_CSI0_B2_LN3_P	AI	MIPI CSI0 lane 3 data (+)	CSI	-
Y50	MIPI_CSI1_A0_CLK_M	AI	MIPI CSI1 clock (-)	CSI	-
X51	MIPI_CSI1_NC_CLK_P	AI	MIPI CSI1 clock (+)	CSI	-
AA50	MIPI_CSI1_C0_LN0_M	AI	MIPI CSI1 lane 0 data (-)	CSI	-
Z51	MIPI_CSI1_B0_LN0_P	AI	MIPI CSI1 lane 0 data (+)	CSI	-
W50	MIPI_CSI1_B1_LN1_M	AI	MIPI CSI1 lane 1 data (-)	CSI	-
V51	MIPI_CSI1_A1_LN1_P	AI	MIPI CSI1 lane 1 data (+)	CSI	-
U50	MIPI_CSI1_A2_LN2_M	AI	MIPI CSI1 lane 2 data (-)	CSI	-
T51	MIPI_CSI1_C1_LN2_P	AI	MIPI CSI1 lane 2 data (+)	CSI	-
S50	MIPI_CSI1_C2_LN3_M	AI	MIPI CSI1 lane 3 data (-)	CSI	-
R51	MIPI_CSI1_B2_LN3_P	AI	MIPI CSI1 lane 3 data (+)	CSI	-

4.2.8 Keypad Interface

Table 13: Keypad Interface

PIN	Signal	I/O	Function	Power Domain	GPIO
AB1	KEY_PWR_ON_N	DI	Turn on/off the module	MSM	
Z1	KEY_PM_RESIN_N	DI	Reset the module	MSM	
E36	KEY_VOL_DOWN_N	DI	Volume down	GPIO	GPIO_103
B36	KEY_VOL_UP_N	DI	Volume up	GPIO	GPIO_96

4.2.9 Sensor Interface

Table 14: Sensor interface

PIN	Signal	I/O	Function	Power Domain	GPIO
V49	SNSR_ACCL_INT	DI	Inerrupt for ACC sensor	GPIO	GPIO_32
X49	SNSR_GYRO_INT	DI	Interrupt for gyro sensor	GPIO	GPIO_33
AD51	SNSR_ALPS_INT_N	DI	Interrupt for Lux/proximity sensor	GPIO	GPIO_35
E38	SNSR_I2C_SCL	DO	I2C clock for external sensors	GPIO	GPIO_110
G37	SNSR_I2C_SDA	DI/O	I2C data for external sensors	GPIO	GPIO_109
E26	APPS_I2C_SDA	DI/O	For PM8008/MIPI DSI bridge IC/NFC/ LCD BIAS IC/Backlight driver IC	GPIO	GPIO_4
G25	APPS_I2C_SCL	DO		GPIO	GPIO_5

4.2.10 RF Interface

Table 15: RF interface

PIN	Signal	I/O	Function	Comment
A41	ANT_TRX	AI/O	Main antenna interface	
A33	ANT_DRX	AI	Diversity antenna interface	
A9	ANT_GPS	AI	GNSS antenna interface	
AK1	ANT_WIFI	AI/O	Wi-Fi/Bluetooth antenna interface	

4.2.11 Audio Interface

Table 16: Audio Interface

PIN	Signal	I/O	Function	Power Domain	Comment
BC9	CDC_MIC_BIAS1	AO	Bias voltage 1 output for microphone		Vomin = 1.6V Vnom=1.8V Vomax = 2.85 V I=3mA
BD10	CDC_MIC_BIAS2	AO	Bias voltage 2 output for microphone		
BA10	CDC_MIC_BIAS3	AO	Bias voltage 3 output for microphone		
BD8	CDC_IN1_M	AI	Microphone input for channel 1 (-)		1 kHz input. Input signal level required to get 0 dBFS digital output; gain = 0 dB +/-0.5 dBV
BC7	CDC_IN1_P	AI	Microphone input for channel 1 (+)		
AV5	CDC_IN2_M	AI	Microphone input for channel 2 (-)		
AU7	CDC_IN2_P	AI	Microphone input for channel 2 (+)		
AT5	CDC_IN3_M	AI	Microphone input for channel 3 (-)		
AR5	CDC_IN3_P	AI	Microphone input for channel 3 (+)		
BC11	CDC_EAR_M	AO	Earpiece output (-)		The typical output voltage is 2 Vrms.
BD12	CDC_EAR_P	AO	Earpiece output (+)		
AS7	CDC_HPH_REF	GND	Headphone reference Ground		
AP7	CDC_HPH_L	AO	Headphone left channel output		
AN5	CDC_HPH_R	AO	Headphone right channel output		
AU3	CDC_HS_DET	AI	Headset hot-plug detect		
BD6	CDC_AUX_P	AO	Audio line differential output (+)		
BE5	CDC_AUX_M	AO	Audio line differential output (-)		
E40	AUD_DMIC_CLK_1	DO	Clock for digital mic	GPIO	GPIO_98 (QCM2290)

PIN	Signal	I/O	Function	Power Domain	Comment
G39	AUD_DMIC_DATA_1	DI	data for digital mic	GPIO	GPIO_99 (QCM2290)
G21	AUD_SPK_AMP_SHDN_N	DO	SHDN for audio amp	IOP_LV	

4.2.12 Battery Charger

Table 17: Battery Charge

PIN	Signal	I/O	Function	Notes
AD3	BATT_ID	AI	Battery type detect	
AG5	BATT_THERM	AI	Battery temperature detect	
AH7	BATT_CONN_VSNS_P	AI	Battery voltage detect (+)	
AK7	BATT_CONN_VSNS_M	AI	Battery voltage detect (-)	

4.2.13 Mix Functions

Table 18: Mix Function

PIN	Signal	I/O	Function	Comment	PowerDomain	GPIO
AA2	CBL_PWR_N	DO		Initiate power-on when grounded	MSM	
P5	VIB_DRIVER	PO	Vibration motor driver output control	V _{omin} = 1.5V V _{Omax} = 3.3V I _{Omax} = 100mA		
U47	PM8008_RST_N	DI/O	Reset for PM8008		GPIO	GPIO_26
S47	PM8008_INT_N	DI/O	Interrupt for PM8008		GPIO	GPIO_25
G17	VCOIN	PI/O	Power supply for RTC	V _{omax} = 3.2V V _{Imin} = 2.5V V _{Imax} = 3.2V		
Z49	GRFC_10	DI/O	Generic RF controller		GPIO	GPIO_47
Z45	GRFC_12	DI/O	Generic RF controller		GPIO	GPIO_49
X45	GRFC_15	DI/O	Generic RF controller		GPIO	GPIO_52
AF7	PM_AMUX_2	ADC	General-purpose ADC interface	V _{min} = 0 V _{max} = 1.875V		
A5	PM_AMUX_3	ADC	General-purpose ADC interface	V _{min} = 0 V _{max?} = 1.875V		
E1	EXT_GPS_LNA_EN0	DO	GNSS LNA enable control		GPIO	GPIO_63

4.2.14 GPIO

Table 19: GPIO

PIN	Signal	I/O	Function	PowerDomain	Comment
B20	PM_GPIO3	DI/O	General-purpose input/output	TBD	PM_GPIOx are internally connected to power management IC. These PM_GPIO's are not recommend as normal GPIO. These pins should be left open.
C19	PM_GPIO4			TBD	
G19	PM_GPIO8			MSM	
E18	PM_GPIO1			TBD	
R7	PM_GPIO5			MSM	
E14	PM_GPIO6			MSM	

4.2.15 NFC/SPI

Table 20: NFC/SPI

PIN	Signal	I/O	Function	PowerDomain	GPIO
C23	NFC_SPI_MISO	DI	NFC/SPI Master Input Slave Output	GPIO	GPIO_0
B22	NFC_SPI_MOSI	DO	NFC/SPI Master Output Slave Input	GPIO	GPIO_1
C29	NFC_SPI_CLK	DO	NFC/SPI_CLK	GPIO	GPIO_2
N45	NFC_SPI_CS_N	DO	NFC/SPI_Chip Select	GPIO	GPIO_3
C25	NFC_EN	DO	NFC_EN	GPIO	GPIO_69
E22	NFC_INT_N	DI	NFC_INTERRUPT	GPIO	GPIO_70
L51	NFC_DWL_REQ	DI	Reserved NFC chip connection	GPIO	GPIO_31
C27	NFC_CLKREQ	DI	Reserved NFC chip connection	GPIO	GPIO_86
A25	NFC_RFCLK3	DO	Reserved NFC chip connection	PM1P875	

4.2.16 LED

Table 21: LED

PIN	Signal	I/O	Function	GPIO
AC2	LED_CHG_SINK	DO	Indicate the module's charging status	
E20	LCM_WLED_PWM	DO	PWM output	PM_GPIO_02
E16	LED_FLASH	DO	Flash/torch driver output (see 5.22)	
G15	LED_FLASH	DO		
G13	LED_GREEN	AI	PM4125 battery charger. Max voltage VPH_PA, sink current 9mA	
E12	LED_RED	AI		

4.2.17 Power

Table 22: Power

PIN	Signal	I/O	Function	Comment
BC39, BC43, BD40, BE43, BE45, BD46	VPH_PWR	PO	3.8V up 1A of continuous power output supported	The SE250B4 can support up to 3.8W, we do not recommend to exceed these limits.
AW51, AX50, AY49, AY51, AZ50, BA51, BC41, BD42	VPH_PA	PI	Power supply for the Module, battery side	It can adsorb up to 3A. Please consider adding TVS for protection.
BE47	VBAT	PI	Power supply for BB	Internally short circuited to VPH_PA
AB49, AH49	VREG_L15A_1P8	PO	1.8V output	Vnom = 1.8V IOmax = 200mA
AD49	VREG_L17A_3P0	PO	3.0V output	Vnom = 3.0V IOmax = 192mA
AM49	VREG_L20A_2P96	PO	2.96V output	Vnom = 2.96V IOmax = 500mA internal eMMC used. internal used for RF function
X3	VREG_S4A_2P04	PO	2.04V	Vnom = 2.04V IOmax = 600mA Internal PMIC LDOs input Reserved for external LDOs input on camera power supply on EVB(TLB2).
C17	VREG_S3A_1P352	PO	Vout=1.352V	Vout=1.352V up 600mA, Internal PMIC LDOs input input voltage of PM8008 on camera power supply in QTI reference design.

4.2.18 GND

Table 23: GND

PIN	Signal	Function	Comment
A11, A15, A17, A21, A23, A27, A29 A31, A35, A39, A43, A7, AA5, AA47, AB45, AB51, AC50, AD7, AE5, AF3, AF45, AG2, AH1, AH51, AJ2, AJ47, AJ5, AK3, AK45, AK51, AL2, AL47, AM1, AM45, AN2, AN50, AP1, AP51, AR2, AS1, AT2, AU1, AU45, AU51, AV47, AV50, AW49, AY1, AY11, AY17, AY19, AY29, AY33, AY37, AY39, AY41, AZ2, B10, B14, B16, B18, B24, B26, B28, B32, B34, B40, B42, B8, BA1, BA12, BA20, BA26, BA30, BA38, BA40, BA42, BC13, BC23, BC29, BC37, BC45, BD14, BD24, BD38, BE39, BE41, BD44, BE11, BE17, BE21, BE27, BE7, C13, C15, C33, C41, C7, C9, E10, F50, G1, H2, J1, J51, M47, M50, N49, N51, N7, P47, P50, R3, T45, U5, V3, X1, Y2, Y5	GND		
1,2,3,4,5,6,7,8,9,10 11,12,13,14,15,16,17,18,19,20 21,22,23,24,25,26,27,28,29,30 31,32,33,34,35,36,37,38,39,40 41,42,43,44,45,46,47,48,49,50 51,52,53,54,55,56,57,58,59,60	GND (thermal dissipation)		

4.2.19 RFU

Table 24: RFU

PIN	Signal	Function	Comment
A13, A19, A37, A45, A47, AC47, AH3, AK49, AL5, AM3, AM51, AM7, AP3, AS3, AS51, AV2, AW1, AX2, AY31, AY35, B6, B12, BA18, BA28, BA32, BA34, BA36, BC31, BC33, BC35, BD30, BD32, BD34, BD36, BE13, BE15, BE19, BE23, BE25, BE29, BE31, BE33, BE35, BE37, BE9, C11, F2, G3, G11, J3, K2, K5, L1, L3, L7, M5, N3, S5, T3, T7, V7, W2, W5, Z3	RFU		

Note: Reserved Pins must not be connected.

4.3 PIN Description

Table 25: Pin-type description

Symbol	Description
PI	Power input
PO	Power output
PI/O	Power input/output
AI	Analog input
AO	Analog output
AI/O	Analog input/output
DI	Digital input
DO	Digital output
DI/O	Digital input/output

Note: Some pins support alternate functions other than generic GPIO. For the detailed list of alternate functions supported by module pins please contact your local support.

4.4 Pin Characteristics by Power Domains

Table 26: GPIO Power Domain - Electrical Characteristics

Parameter	Description	Min	Max	Units
V _{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	1.17	2.1	V
V _{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	-0.3	0.63	V
V _{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	1.26	2.1	V
V _{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	-0.3	0.54	V
V _{OH}	High-level output voltage, CMOS	1.35	1.8	V
V _{OL}	Low-level output voltage, CMOS	0.0	0.45	V
V _{SHYS}	Schmitt hysteresis voltage (HIHYS_EN= low)	100	-	mV
V _{SHYS}	Schmitt hysteresis voltage (HIHYS_EN= high)	300	-	mV
I _{OH}		TBD		mA
I _{OL}			TBD	mA

Table 27: USIM1 / USIM2 Power Domain - Electrical Characteristics

Parameter	Description	Min	Max	Units
V _{USIMx_2P95}	USIM VDD voltage level for 2.95V compatible SIM	1.17	2.1	V
V _{USIMx_1P8}	USIM VDD voltage level for 1.8V compatible SIM	-0.3	0.63	V
V _{IH}	High-level input voltage	0.7 * V _{USIMx,y}	V _{USIMx,y} + 0.3	V
V _{IL}	Low-level input voltage	-0.3	0.2 * V _{USIMx,y}	V
V _{OH}	High-level output voltage	0.8 * V _{USIMx,y}	V _{USIMx,y}	V
V _{OL}	Low-level output voltage	0.0	0.4	V
V _{SHYS}	Schmitt hysteresis voltage	100	-	mV
I _{OH}		TBD		mA
I _{OL}			TBD	mA

Table 28: SDC2P95 (for 3V SD Card) Power Domain - Electrical Characteristics

Parameter Description		Min	Max	Units
V _{IH}	High-level input voltage	0.625 * V _{SDC2P95}	V _{SDC2P95} + 0.3	V
V _{IL}	Low-level input voltage	-0.3	0.25 * V _{SDC2P95}	V
V _{OH}	High-level output voltage	0.75 * V _{SDC2P95}	V _{SDC2P95}	V
V _{OL}	Low-level output voltage	0.0	0.125 * V _{SDC2P95}	V
V _{SHYS}	Schmitt hysteresis voltage	100	-	mV
I _{OH}		TBD		mA
I _{OL}			TBD	mA

Table 29: SDC1P8 (for 1V8 SD Card) Power Domain - Electrical Characteristics

Parameter Description		Min	Max	Units
V _{IH}	High-level input voltage	1.27	2	V
V _{IL}	Low-level input voltage	-0.3	0.25 * V _{SDC1P8}	V
V _{OH}	High-level output voltage	1.4	-	V
V _{OL}	Low-level output voltage	-	0.45	V
V _{SHYS}	Schmitt hysteresis voltage	100	-	mV
I _{OH}		TBD		mA
I _{OL}			TBD	mA

Table 30: MSM / IOP_LV Power Domain - Electrical Characteristics

Parameter Description		Min	Max	Units
V _{IH}	High-level input voltage	1.17	2.1	V
V _{IL}	Low-level input voltage	-0.3	0.63	V
V _{OH}	High-level output voltage, CMOS	1.35	1.8	V
V _{OL}	Low-level output voltage, CMOS	0.0	0.45	V
V _{SHYS}	Schmitt hysteresis voltage	15	-	mV

Table 31: PM1P875 Power Domain - Electrical Characteristics

Parameter Description		Min	Max	Units
V _{IH}	High-level input voltage	1.22	2.175	V
V _{IL}	Low-level input voltage	-0.3	0.65	V
V _{OH}	High-level output voltage, CMOS	1.425	1.8	V
V _{OL}	Low-level output voltage, CMOS	0.0	0.45	V
V _{SHYS}	Schmitt hysteresis voltage	15	-	mV

CSI / DSI Power Domain - Electrical Characteristics:

CSI : Signal levels according to MIPI C-PHY v1.0 specification

DSI : Signal levels according to MIPI D-PHY v1.2 specification, supports only unidirectional data receiving.

5 Power Supply

The power supply circuitry and board layout are critical components of the overall product design and significantly impact the product's performance, hence read carefully the requirements and the guidelines to achieve an optimal design.

For high power transmission, it is recommended to use a power supply capable of supporting up to 3A.

5.1 Power Supply Requirements

Table 32: Power Supply Requirements

Power Supply	Value
Nominal Supply Voltage	3.8V
Operating Voltage Range	3.50V ÷ 4.40V
Extreme Voltage Range	3.50V ÷ 4.75V

Note: The Extreme Operating Voltage Range MUST never be exceeded.

If the power supply is not properly designed, it can cause a significant voltage drop.

The hardware shutdown voltage of the module is 2.55V. If the voltage drops below 2.55V, the module will shut down.

5.2 Power Consumption

Table 33: Power Consumption

Parameter	Conditions	Min	Typical (RMS)	Max	Unit
Leakage current	Off mode	-	70	-	µA
Flight mode	Flight mode	-	4.3	-	mA
Standby GSM/GPRS	BS-PA-MFRMS=9	-	4.8	-	mA
	BS-PA-MFRMS=5	-	4.9	-	mA
	BS-PA-MFRMS=2	-	5.6	-	mA
Standby WCDMA	2.56s, DRX=8	-	4.7	-	mA
Standby LTE FDD	2.56s, DRX=8	-	4.8	-	mA
Standby LTE TDD	1.28s, DRX=7	-	5.1	-	mA
Voice Call 2G*	EGSM900 PCL= 5@ 31.97dB Channel=62	-	225	-	mA
	EGSM1800 PCL= 0@ 28.95dB Cannel=698	-	175	-	mA
Voice Call 3G*	B1 power@ 21dB Channel=10700	-	505	-	mA
	B8 power@ 21dB, Channel=3012	-	400	-	mA
Data Call LTE	FDD B1 power@ 22dB BW=10MHz	-	575	-	mA

Parameter	Conditions	Min	Typical (RMS)	Max	Unit
	Ch=18300				
	FDD B3 power@ 22dB BW=10MHz Ch=19575	-	615	-	mA
	FDD B5 power@ 22dB BW=10MHz Ch=20525	-	475	-	mA
	FDD B8 power@ 22dB BW=10MHz Ch=21625	-	480	-	mA
	TDD B38 power@ 23dB BW=10MHz Ch=37800	-	290	-	mA
	TDD B40 power@ 23dB BW=10MHz Ch=38700	-	318	-	mA
	TDD B41 power@ 23dB BW=10MHz Ch=40740	-	270	-	mA
Peak Current	-	-	1155	-	mA
SPECIAL CASES ON FLIGHT MODE					
TEST CASE		Current mA			
Flight mode on with WiFi re-enabled		6.0			
Flight mode on with BT re-enabled		5.9			
Flight mode on with WiFi/BT re-enabled		7.4			

Note:

- **Leakage Current:** The current supplied by VBAT during shutdown.
- **Flight Mode:** In flight mode, the backlight and display are powered off, and wireless connections are disabled.
- *Voice calls are currently not supported in Linux OS.
- Power consumption values reported in the above table refer to the radio-modem part.

5.3 GNSS Performance

Table 34: SE250B4-EU GNSS Performances

Parameter	Value (typ.)	Unit
Acquisition sensitivity*	-147	dBm
Reacquisition sensitivity*	-156	dBm
Tracking sensitivity*	-158	dBm
Accuracy** CEP	0.78	m
TTF cold start	28	s
TTF warm start	23	s
TTF hot start	1	s
*Test conditions: Static scenario, Full Power mode		
**Test conditions: 24-hr Static, Full Power mode, Live (rooftop antenna)		
*** Test Conditions: Static scenario, -130 dBm, Full Power mode		

Table 35: SE250B4-NA GNSS Performances

Parameter	Value (typ.)	Unit
Acquisition sensitivity*	-147	dBm
Reacquisition sensitivity*	-156	dBm
Tracking sensitivity*	-158	dBm
Accuracy** CEP	0.8	m
TTFF cold start	28	s
TTFF warm start	23	s
TTFF hot start	1	s
*Test conditions: Static scenario, Full Power mode		
**Test conditions: 24-hr Static, Full Power mode, Live (rooftop antenna)		
*** Test Conditions: Static scenario, -130 dBm, Full Power mode		

Note: SE250B4-WE does not support GNSS

5.4 General Design Rules

The principal guidelines for the Power Supply Design embrace three different design steps:

- Electrical design
- Thermal design
- PCB layout

5.5 Electrical Design Guidelines

The electrical design of the power supply strongly depends on a drained power source. The electrical design guidelines categories are:

- +5V input (typically PC internal regulator output)
- +12V input (typically automotive)
- Battery

5.6 LDO Power Supply Guidelines

The power supply range of the module is between 3.5V – 4.4V, with a recommended value of 3.8V. The power supply performance, such as load capacity and voltage ripple, directly affect the module's performance and stability. Under worst conditions, the transient peak current of the module may surge up to 3A.

During these high peak current periods, the voltage may drop from the nominal value. If the power supply network is not properly designed, the voltage could fall below 2.55V, causing the module to shut down automatically. Refer to the figure below for a graphic representation of this electric phenomenon.

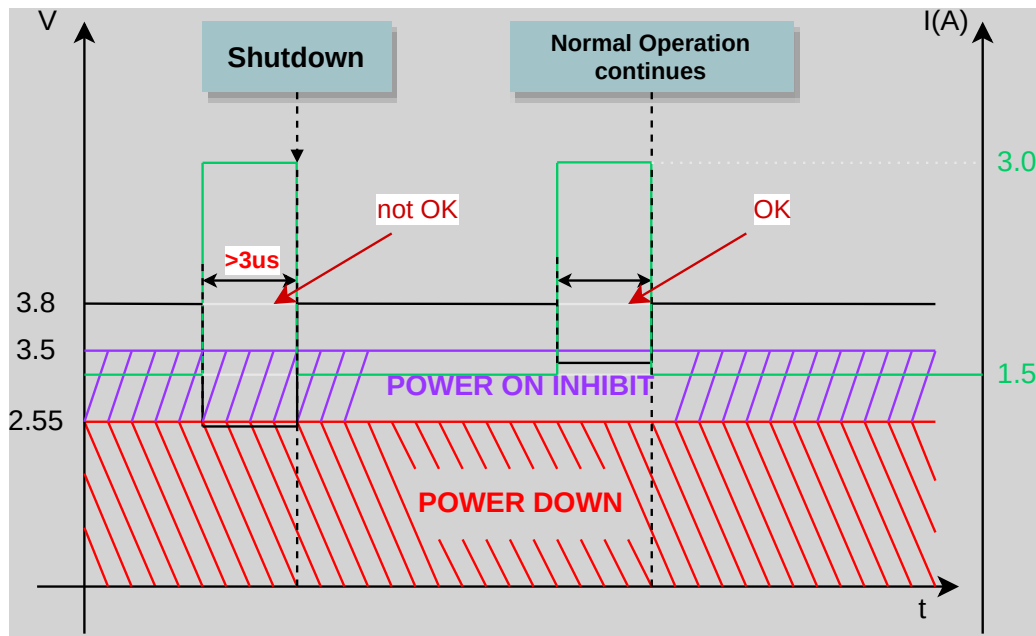


Figure 4: Power Down Voltage Sequence

Voltage drops can be reduced using a bypass capacitor of about 100 μ F with low ESR ($<0.7\Omega$) and a multi-layer ceramic chip capacitor array consisting of at least three ceramic capacitors of 100nF, 33pF, and 10pF respectively. These should be placed as close as possible to the VBAT and VPH_PA pins. Additionally, use a 4.7 μ F capacitor in parallel to further minimize voltage ripple.

The schematic for the recommended capacitor bank is shown in the figure below.

SE 250 B4 Module

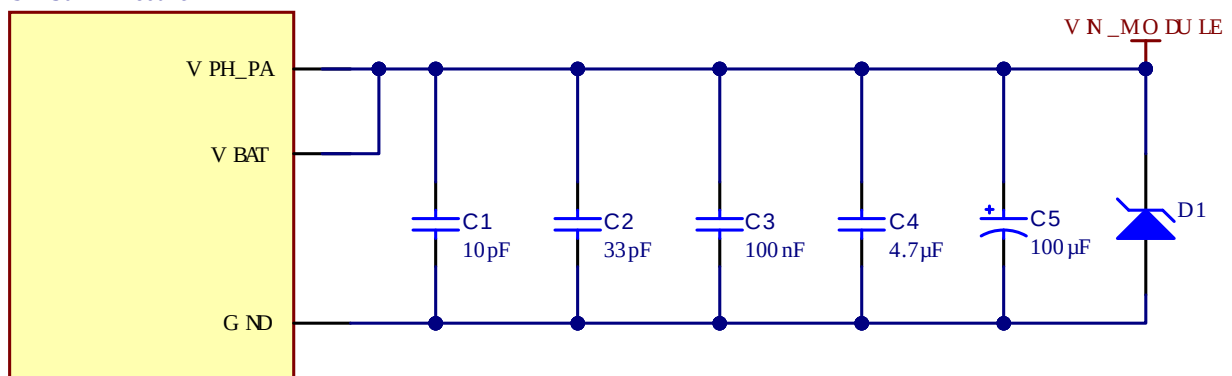


Figure 5: Reference design for power supply

5.7 Reference Design for Power Supply

The design of the power supply network is crucial, as the module's performance largely depends on the power supply source. The supply source should be able to provide at least 3A of current at 3.8V.

To obtain 3.8V, different design techniques can be used depending on the ratio between the input and the output voltages and on the use of a battery as primary power source.

For non-battery powered applications, if the DC input voltage is +5V, then a power efficiency optimization is not mandatory. In this case, consider using a high-current, low-dropout regulator, such as the MIC3932.

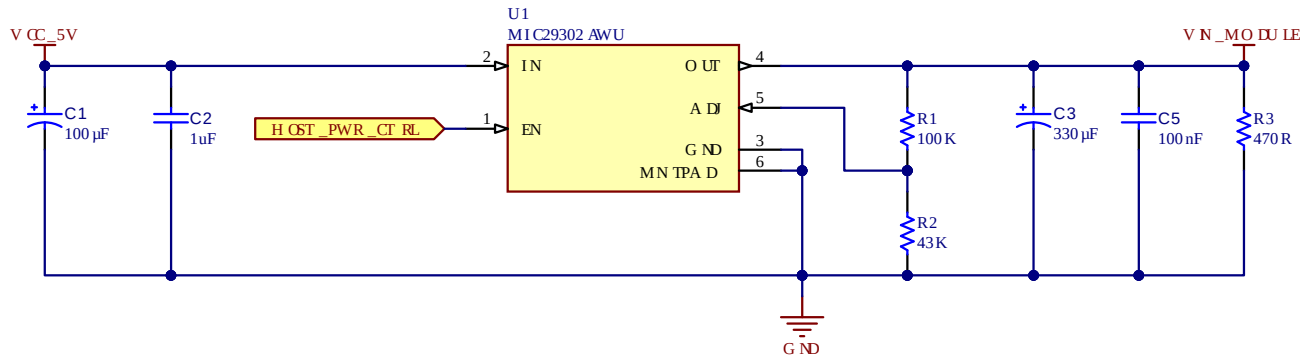


Figure 6: Reference design for non-battery solutions

Note:

- To ensure proper behavior of the regulator under minimal load, an extra minimum load is required. The Telit SE250B4 Smart module in sleep mode and power off has a trifling current adsorption.
- For details on the minimum load, please refer to the MIC29302 specifications.

For battery-powered devices, achieving high efficiency conversion is crucial. In this case, it is recommended to use a buck converter.

The EVB is designed to facilitate the testing of battery-powered applications.

For most applications it is suggested the following design of a buck converter that provides up to 3A continuous output current at 3.8V. For applications including a battery it is suggested to use a reverse current blocking diode and to adjust the output voltage (modify the voltage divider resistors on the buck converter FB pin) to accommodate for the diode forward drop under standard operating conditions.

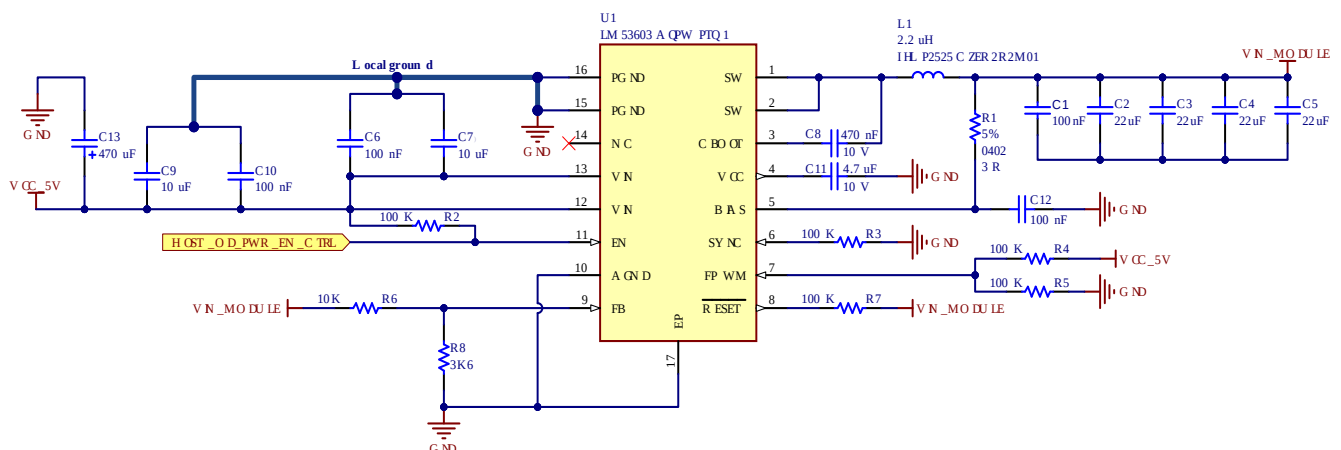


Figure 7: STEP-DOWN DC-DC power supply reference circuit

HOST_CD_PWR_EN_CTRL is supposed to be driven by the host board in open drain configuration. If this signal is driven directly by an IC please ensure the component can withstand the input voltage provided on VCC_5V. If enable control is not required the HOST_CD_PWR_EN_CTRL can be left floating and the EN signal will be driven high by the 100kohm pull-up.

5.7.1 Thermal Design Guidelines

The thermal design for the power supply heat sink must be performed taking into account the following specifications:

- Average current consumption during RF transmission at maximum power level as shown in the Telit SE250B4 Smart module current consumption.
- Average current consumption during Class10 GPRS transmission at maximum power level as shown in Telit SE250B4 Smart module current consumption.
- Average GPS current consumption during GPS tracking (LTE @ idle): mA (TBD mA).

5.8 RTC Bypass Out

When the module is powered down and there is no power supply on VBAT the module can keep the RTC (real time clock) if an external coin cell is connected to the module VCOIN pad. To ensure RTC support in the absence of VBAT, a qualified coin cell or keep-alive capacitor must be connected to the VCOIN PIN. When VBAT is present and valid, coin cell charging is enabled through software control and powered from VBAT.

- If the RTC fails in keeping the time, the module can synchronize the RTC clock time , aligning with the clock of the network cell, after power on.
- Refer to VCOIN characteristics for RTC hardware parameters.
- The voltage value range for VCOIN power supply is between 2.0-3.25V and the typical value is 3.0V. Under these conditions, the average current consumption is about 5 μ A when VBAT is disconnected, and RTC is supplied directly from VCOIN only.
- RTC error is 50ppm when the module is powered by VBAT; RTC error is 200 ppm when the module is powered by VCOIN and the modem is OFF.
- When a rechargeable coin battery is connected externally, the ESR of the coin battery shall be less than 2k Ω . It is recommended to use a Seiko's MS621FE-FL11E battery.
- If the VCOIN PIN is connected to a large capacitance externally, the recommended capacitance value is 100 μ F, which can keep the real-time clock for about 45 seconds.

The following is the circuit diagram:

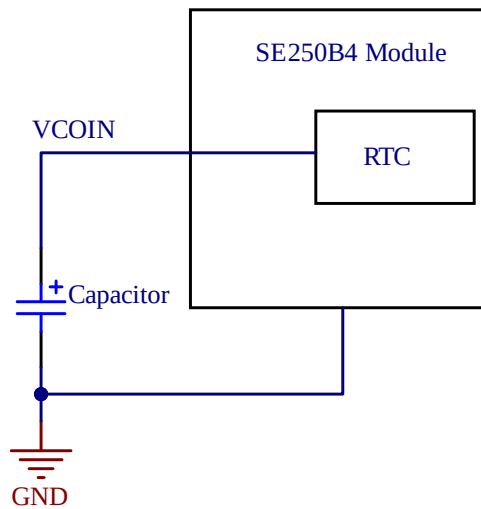


Figure 8: Keep Alive Capacitor

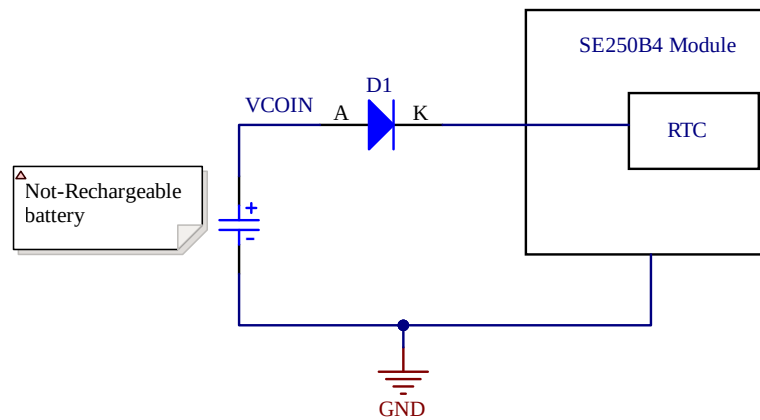


Figure 9: Not-rechargeable Battery

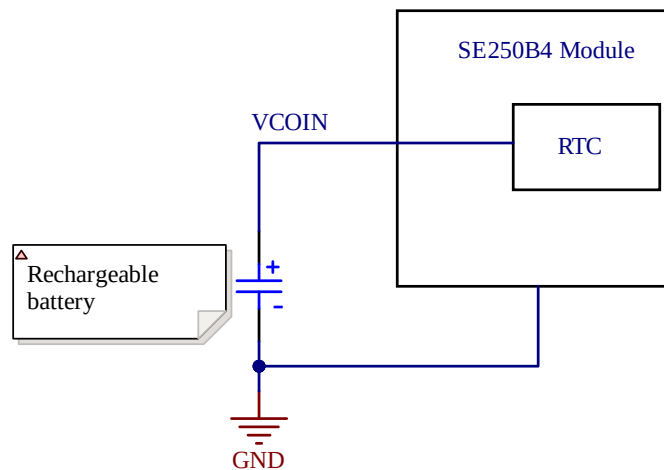


Figure 10: Rechargeable battery

Typically, VCOIN operates at a voltage of 3.0V, with a current consumption of approximately 7.5 μ A when VBAT is absent.

Table 36: RTC Bypass Out

Parameter	Description	Min	Type	Max
VCOIN-IN	VCOIN input voltage	2.0	3.0	3.25
ICOIN-IN	VCOIN current consumption	-	7.5	-
VCOIN-OUT	VCOIN output voltage	2.5	3.1	3.2
ICOIN-OUT	VCOIN output current	-	-	2

Note: If the VBAT supply connected to the module is never removed, or the RTC functionality is not required, then the VCOIN PIN can be left floating. The software needs to be configured to turn off the VCOIN charge and discharge function.

Table 37: VCOIN absolute maximum

Parameter	Description	Min	Max
VCOIN-MAX	VCOIN absolute maximum input value	2.0	3.25

5.9 Battery Charge

Telit SE250B4 Smart module supports battery charging. The embedded IC supports trickle charging, pre-charging, constant current charging and constant voltage charging modes with optimized procedures and thresholds for Li-ion batteries.

The charging process can be divided in the following steps depending on the battery voltage:

- **Trickle charging (Trickle-A):** When the battery voltage is below 2.1V, a constant charging current of 90mA is applied to the battery.
- **Pre-charging (Trickle-B):** In the range between 2.1 and 3.4V, the current is set up to 300mA as maximum value.
- **Constant current mode (CC mode):** It occurs when the battery voltage is between 3.4V and 4.2V, in this case the maximum charging current is 1.85A when an adapter is used and 450mA for USB charging.
- **Constant voltage mode (CV mode):** It starts when the battery voltage reaches the final value 4.2V the system will switch to CV mode and the charging current will decrease gradually and will stop when the charging level will reach 100%.
- Battery charging is controlled by a PMIC state-machine. The first step in the automated charging process determines if trickle charging is needed. Charging a severely depleted battery must begin with trickle charging to limit the current inflow and protect the battery from more charging current than it can handle. Once a minimum battery voltage is established using trickle charging and pre-charging, constant-current charging is enabled to charge the battery quickly – this mode is sometimes called fast charging. Once the battery approaches its target voltage, the charge is completed using constant-voltage charging.

The following is the charge control diagram:

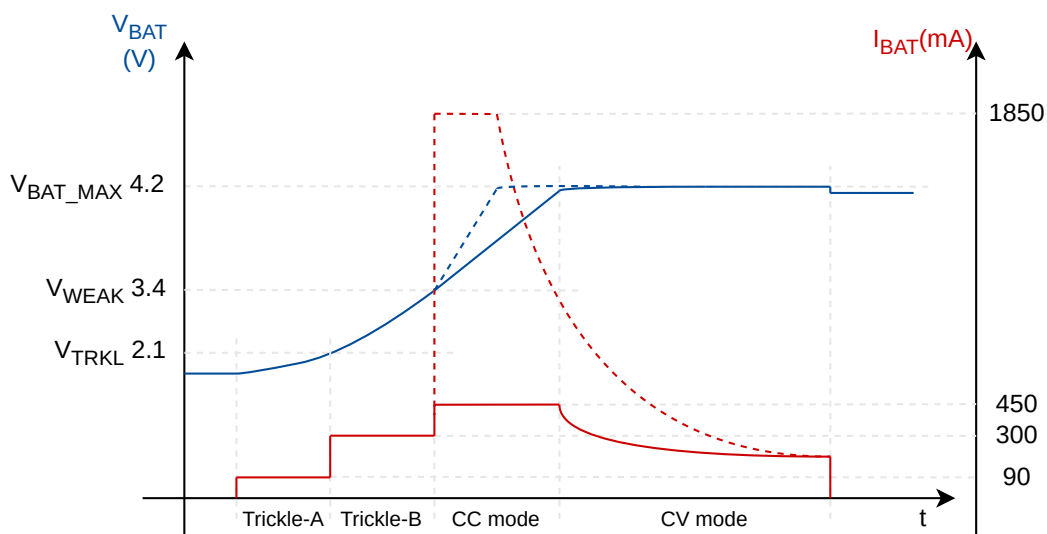


Figure 11: Voltage/Current charging measurements (dotted red and blue curves are show only for reference and not on scale on the t axis)

Table 38: Voltage/Current charging measurements

Parameter	Comments	Min	Typical	Max	Units
ITRKL-A	Trickle-A Charging current	90	100	110	mA
ITRKL-B	Trickle-B Charging current	250	300	350	mA
VTRKL	Trickle-B threshold voltage range Programmable	2.0	2.1	2.2	V
VWEAK	Weak battery threshold range Programmable	3.2	3.4	3.6	V
VBAT_MAX	Maximum battery voltage Programmable, 20 mV steps	3.6	4.2	4.6	V
IBAT_MAX	Fast charging current range Programmable, 100 mA steps	0	450/1850	2000	mA

5.9.1 Battery Charger Pads

Table 39: Battery Charger PADS

PIN	Signal	I/O	Function	Comment
AD3	BATT_ID	AI	Battery type detect	When no battery ID pin is used, a 100 k Ω external pull-down resistor to ground is strongly recommended.
AG5	BATT_THERM	AI	Battery temperature detect	If unused, connect it to GND with a 47k Ω resistor.
AH7	BATT_CONN_VSNS_P	AI	Battery voltage detect (+)	Cannot be kept open.
AK7	BATT_CONN_VSNS_M	AI	Battery voltage detect (-)	

The module supports battery temperature detection, under the condition that the battery integrates a thermistor (47k Ω 1% NTC thermistor with a B-constant of 4050 K by default) and the thermistor is connected to BATT_THERM pin. If the BAT_THERM pin is not connected, there will be malfunctions such as battery charging failure, battery level display error etc. The battery charge temperature range varies with different types of batteries.

A reference design for the battery charging circuit is shown below, for battery packs not providing a dedicated ID resistor, user should connect BATT_ID to ground via 100k Ω external pull-down resistor.

Avoid using ESD protection devices that introduce excessive capacitance on the BATT_ID and BATT_THERM lines.

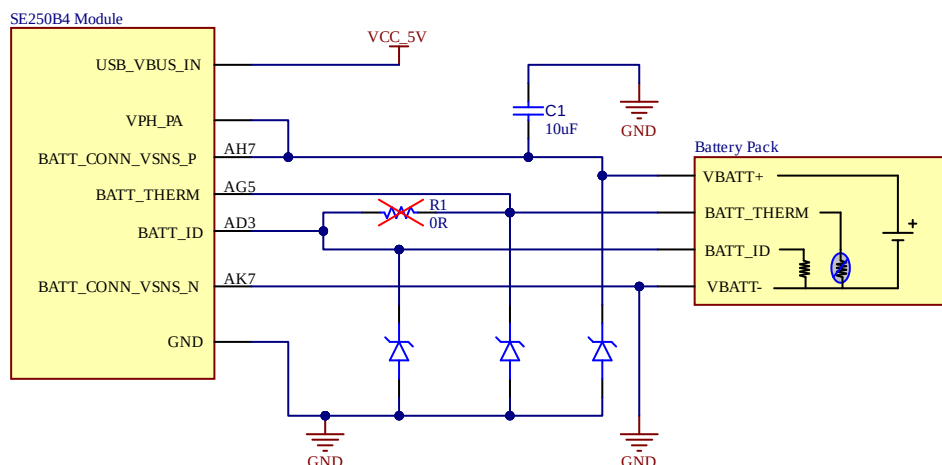


Figure 12: Battery charging circuit example

Mobile devices such as mobile phones or handheld POS systems are powered by batteries. To optimize performance, it's necessary to adjust the charging and discharging curve according to the specific battery type being used.

If the thermistor is not available in the battery, the designer needs to connect the BATT_THERM pad to GND via a 47k Ω resistor otherwise the system may mistakenly

judge that the battery temperature is abnormal and therefore cause battery charging failure.

BATT_CONN_VSNS_P and BATT_CONN_VSNS_M ,must be connected otherwise exceptions in voltage detection will be caused, with associated problems of turn on/off and battery charging/discharging.

If BATT_THERM functionality is not desired, the pin shall be grounded using 47k Ω resistor, and the software battery temperature feature should be disabled. Similarly, if an external charger is used, and no battery is required in the application design, then BATT_THERM functionality should be disabled.

BATT_THERM is used for battery-temperature monitoring (BTM) and battery-presence detection (BPD), either in conjunction with BATT_ID or for application that do not provide a ID resistor in the battery pack.

The module, depending by SW configuration can support NTC resistors as defined in [Table 40](#) . Suggested NTC values are 10k (B-Constant = 3380K $\pm$ 1%) or 47k (B-Constant = 4050K) , such resistor should be integrated inside the battery, and the cold / hot comparator threshold setting should be 70% / 35%. The allowable charging temperature range is -2 $^{\circ}$ C to 52 $^{\circ}$ C, with an accuracy of $\pm$ 2 $^{\circ}$ C.

BATT_CONN_VSNS_P is used for battery voltage sensing, the typical input range is 3.0V~4.75V.

BATT_CONN_VSNS_P PIN cannot be left floating. When using a lithium battery for module power supply, it must be connected to the battery's positive PIN. Alternatively, when powered from an LDO or DCDC, it should be connected to the module's VPH_PA.

Table 40: Battery Gauge Specification

Parameter	Min	Typical	Max	Units
Supported thermistor beta range	3300	-	4400	-
Thermistor supported range	10	-	100	k Ω
BAT_ID resistor supported range	7.5	-	450	k Ω

5.10 Charging LED

Telit SE250B4 Smart module supports a charging LED indicator.

Table 41: Charging LED

PIN name	PIN number	I/O	Description
LED_CHG_SINK	AC2	DO	charging indicator

The following is the circuit diagram:

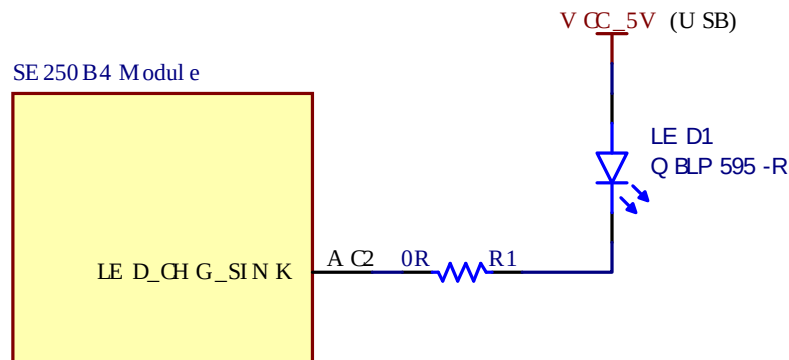


Figure 13: LED Reference Circuit

Note: While charging, the LED_CHG_SINK pin controls the LED indicator.

5.11 Power On

Powering on the Telit SE250B4 Smart module is achieved by holding down the KEY_PWR_ON_N PIN for more than 2 seconds, then release it. This PIN is internally pulled up to 1.8V, eliminating the need for an external pull-up resistor.

Table 42: Power ON voltage Levels

Parameters	Description	Min	Type	Max	Unit
VIH	High-level input voltage	1.4	-	-	V
VIL	Low-level input voltage	-	-	0.6	V

The power circuitry can be designed using a Push button or an open collector with a single transistor. Refer to the diagrams below for more details:

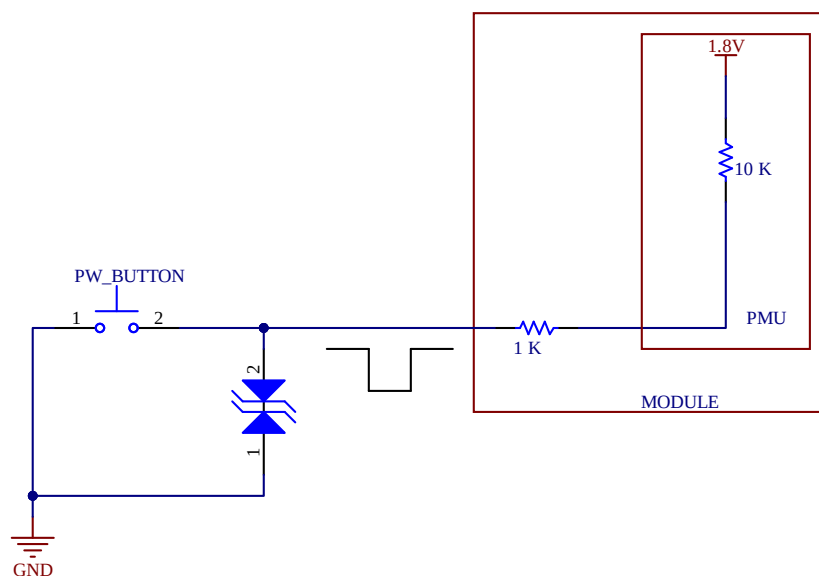


Figure 14: Power on/off using the power button

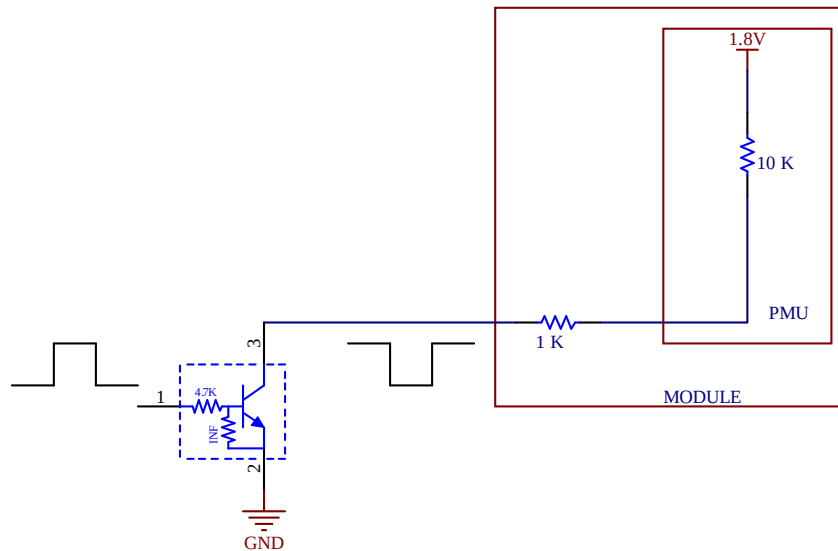


Figure 15: Power On/Off using the transistor

5.11.1 Power-On Sequence

The following is the power on sequence diagram:

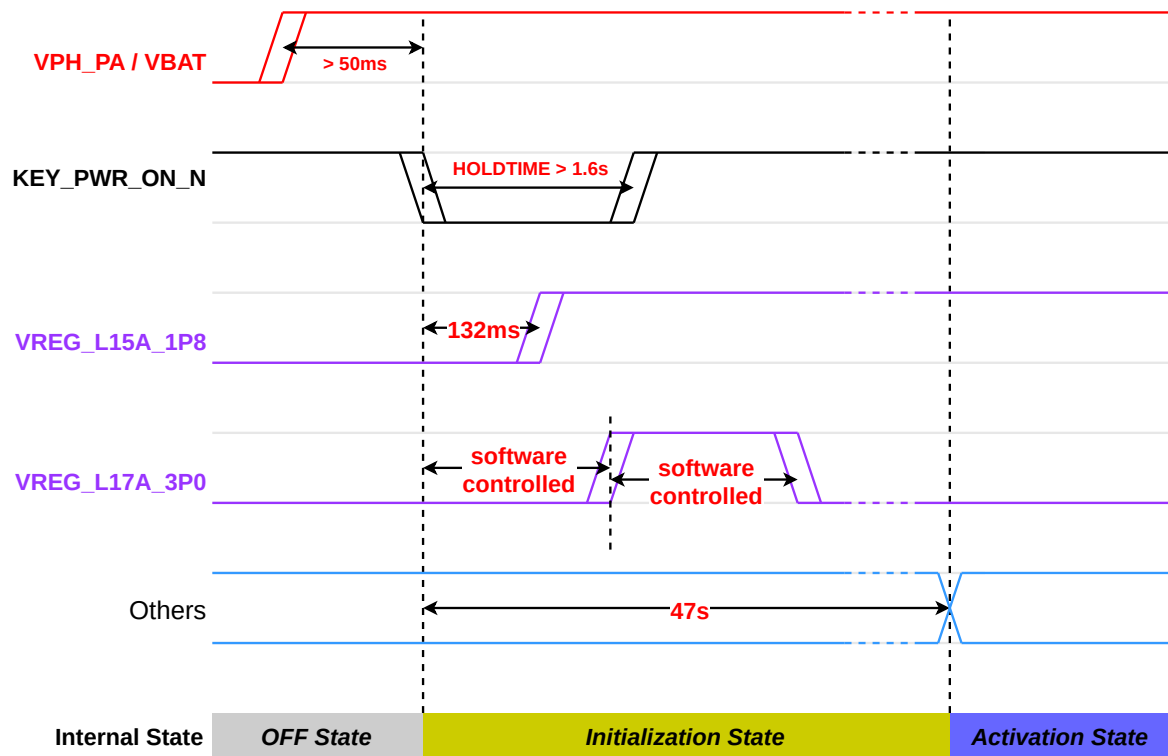


Figure 16: Turn-on Timing

Note:

- 1 When the module is turned on for the first time the power on sequence may be different from the one shown above.
- 2 Make sure the VBAT remains stable at 3V8 for at least 50ms before pulling down KEY_PWR_ON.

5.12 Power Off

The SE250B4 Smart Module supports power-off through the KEY_PWR_ON_N and KEY_VOL_UP_N pins.

To power off the module:

- 1 Pull KEY_PWR_ON_N and KEY_VOL_UP_N low simultaneously.
- 2 Hold both pins low for more than 1 second.
- 3 Upon detecting both pins at a logic low, the module displays a shutdown confirmation prompt on the display.
- 4 The module powers off after the shutdown action is confirmed through the user interface.

Additionally, the module can also be forced to Shutdown/Restart by pulling down KEY_PWR_ON_N for more than 8 seconds.

Refer to the diagram below for unconditional shutdown by KEY_PWR_ON_N pressure:

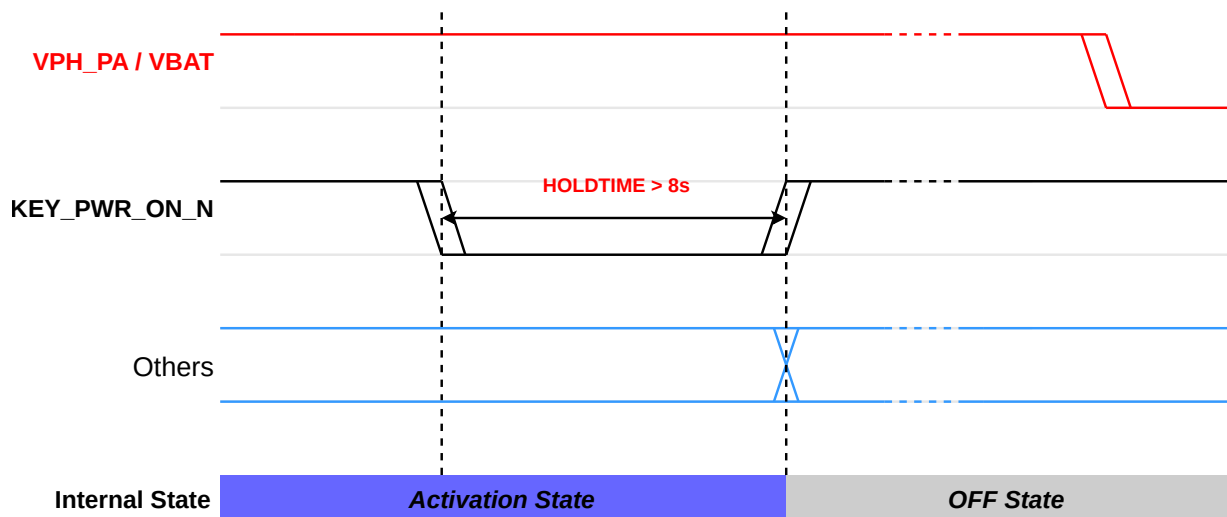


Figure 17: Forced Turn-off / Restart Timing

Note:

- 1 The VPH_PA and VBAT power supply circuit of the module can be cut off in the customer's hardware design after module shutdown procedure is complete.
- 2 It is recommended to add a low-cost MCU capable of controlling the KEY_PWR_ON_N to power on and power off the module, as well as the hardware watchdog to protect the normal operation.
- 3 Do not directly cut off the power supply of VPH_PA and VBAT while the module is in operation to prevent potential corruption of the internal flash. It is recommended to shut down the module through KEY_PWR_ON_N or AT command* before disconnecting the power supply VPH_PA / VBAT of the module.

Note:

*AT port is available only in userdebug build. AT port is not available in user build FW, typically used for production. Hence it is recommended not to use AT commands, except for debug and evaluation.

5.13 Unconditional Restart

To initiate an unconditional restart of the Telit SE250B4 Smart module, users can reset the device by pulling down the KEY_PM_RESIN_N PIN for more than 200ms. After the module detects that the KEY_PM_RESIN_N is at a low level, the device will perform an unconditional restart.

5.14 Communication Ports

5.14.1 USB Interface

The module provides USB interfaces compliant with both USB 3.1 and USB 2.0 specifications, supporting the following data rates:

- Superspeed (5Gbps)
- High-Speed (480 Mbps)
- Full Speed (12Mbps)

The USB interface also supports USB OTG and can be used for AT commands* and data transmission, Software debugging and Firmware upgrading.

The module only support USB Type-C connector. The USB interface comprises one USB 2.0 compliant high-speed differential channel (USB0_HS_SM_DP, USB0_HS_SM_DM) and one USB 3.1 compliant superspeed differential channel (USB0_SS_RX0_P /M, USB0_SS_TX0_P /M and USB0_SS_RX1_P/M, USB0_SS_TX1_P/M).

The following table shows the pin definition of USB interface:

Table 43: Communication PORTS

PIN	Signal	I/O	Function	Comment
C21	FORCED_USB_BOOT	AI		
AY15	USB0_HS_SM_DM	AI/O	USB differential (-)	90Ω differential impedance with 10% tol; compliant to USB 2.0 standard
AY13	USB0_HS_SM_DP	AI/O	USB differential (+)	
AD1, AE2, AF1	USB_VBUS_IN USB_VBUS_IN	PI PI	Charging power input and power output for OTG devices USB/adaptor insertion detection	The Max output current is 500mA
BD18	USB0_SS_RX0_M	AI/O	USB 3.1 channel 1 superspeed receive (-)	90Ω differential impedance, compliant to USB 3.1stand ard
BC17	USB0_SS_RX0_P	AI/O	USB 3.1 channel 1 superspeed receive (+)	
BD20	USB0_SS_TX0_M	AI/O	USB 3.1 channel 1 superspeed transmit (+)	
BC19	USB0_SS_TX0_P	AI/O	USB 3.1 channel 1 superspeed transmit (-)	

PIN	Signal	I/O	Function	Comment
BA16	USB0_SS_RX1_M	AI/O	USB 3.1 channel 2 superspeed receive (-)	
BA14	USB0_SS_RX1_P	AI/O	USB 3.1 channel 2 superspeed receive (+)	
BD22	USB0_SS_TX1_M	AI/O	USB 3.1 channel 2 superspeed transmit (+)	
BC21	USB0_SS_TX1_P	AI/O	USB 3.1 channel 2 superspeed transmit (-)	
BC15	USB_CC1	AI	USB Type-C detect 1	should be brought to test point while using USB2.0
BD16	USB_CC2	AI	USB Type-C detect 2	should be brought to test point while using USB2.0

Note:

*AT port is available only in the userdebug build. AT port is not available in user build FW, typically used for production. Hence it is recommended not to use AT commands, except for debug and evaluation.

A basic connection Scheme is shown in the figure below:

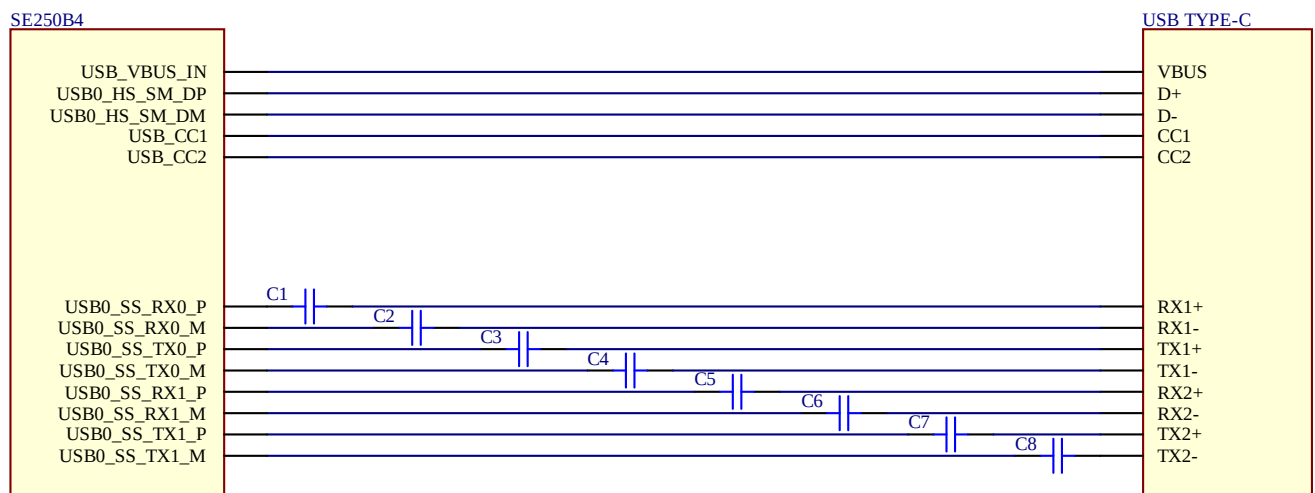


Figure 18: USB connector

Note:

USB2.0 usage

For both upstream (SE250 USB device) and downstream (SE250 USB Host), the USB_CC2 signal can be left open (floating).

With pull-up 56kΩ on USB_CC1 to USB_5V, SE250 is USB 2.0 device. When connected USB cable to a PC, it shows up the modem and debug ports (if available)

With pull-down 5.1kΩ on USB_CC1 to ground, SE250 is a USB 2.0 host.

As per the USB 3.x specification, AC coupling capacitors with a value ranging from 75nF (minimum) to 265nF (maximum) are mandatory for all transmitters.

It is recommended to set C3, C4, C7 and C8 = 100nF, and the capacitors on the RX lines to a value in the range of 330nF to 470nF so that the series of these capacitors and the ones on the connected device (assuming a proper design with 100nF) still complies with USB specification.

To ensure USB performance, comply with the following rules when designing the USB interface:

- Route USB signal traces as differential pairs, avoid routing them above cuts or slots on their reference GND plane, the differential impedance of each differential pair should be controlled to $90\Omega \pm 10\%$
- Keep the ESD protection components as close as possible to the USB connector. Be careful to the influence of junction capacitance of ESD protection component on USB lines. Typically, the capacitance value should be less than 2pF for USB 2.0 and less than 0.5pF for USB 3.1
- Do not route signals traces under crystals, oscillators, magnetic devices, or RF signal traces. Route the USB differential traces in the inner layer with ground on all four sides of the USB track (TOP, BOTTOM, and shield left and right with VIA grounding) → this means that a stack up with at least 4 layers should be used.
- Make sure the intra-pair length difference doesn't exceed 2mm, for USB 2.0 and 0.7mm for USB 3.1 (TX and RX)
- The spacing between USB signals and all other signals should be at least 4 times the trace width, while the spacing between Rx and Tx should be at least 3 times the trace width.
- For USB 3.1, it is recommended to perform some signals integrity simulation after design, the quality of transmitted and received signals, can be affected by cable or numbers of vias that could provide a parasitic not negligible effect, in this case the simulation will help to better tune the design in order to keep good performances.
- If there are test points, place them on the trace to keep stubs as short as possible.
- If USB connector is used as the charger input, USB_VBUS node must be routed to the module using extremely wide traces or sub planes.

Table 44: USB lines length inside the module

Pin Number	Signal	Length (mm)	Length Difference DP-DM
AY15	USB0_HS_SM_DM	22.939	0.024
AY13	USB0_HS_SM_DP	22.915	
BD18	USB0_SS_RX0_M	25.38	0.009
BC17	USB0_SS_RX0_P	25.371	
BA16	USB0_SS_RX1_M	25.278	0.013
BA14	USB0_SS_RX1_P	25.291	
BD20	USB0_SS_TX0_M	17.415	0.066
BC19	USB0_SS_TX0_P	17.481	
BD22	USB0_SS_TX1_M	16.148	0.389

BC21	USB0_SS_TX1_P	15.759	
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5.14.2 Forced USB BOOT

FORCED_USB_BOOT is used to activate the emergency download interface. If FORCED_USB_BOOT is pulled up to VREG_L15A_1P8 at boot time, the module enters emergency download mode. This mode is used when the product fails to start properly. To facilitate the subsequent software upgrades and debugging, it is recommended to place a test point on these lines.

The following is the circuit diagram:

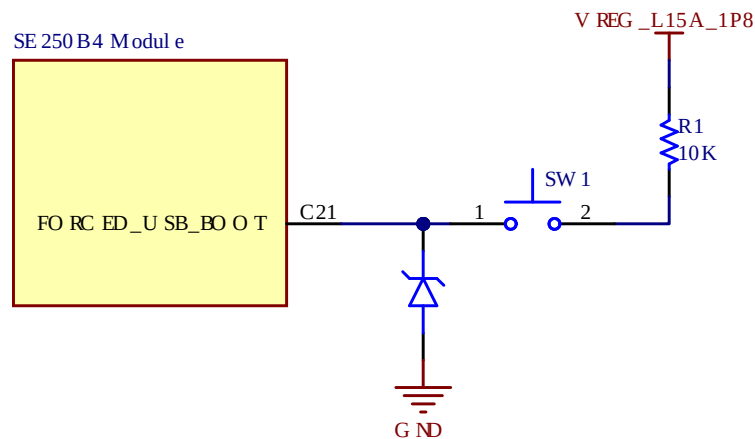


Figure 19: Forced Download Circuit

Table 45: Forced USB

PIN name	PIN	I/O	Description	Note
FORCED_USB_BOOT	C21	I	USB forced download signal, short-circuit to VREG_L15A_1P8 at power-on to enter forced download mode.	Reserved test point
VREG_L15A_1P8	AB49, AH49	PO	1.8V LDO output	

5.14.3 UART Interfaces

The module provides three UART interfaces:

- **UART0 (debug UART):** A two wire UART interface, used for debugging by default.
- **UART1:** A four-wire UART interface with RTS/CTS HW flow control.
- **UART2:** A two-wire UART LPI interface which is reserved for internal purposes and cannot be used for application processors. Leave these pins open.

Table 46: UART PADS

PIN	Signal	I/O	Function	DC Characteristics	Comment
E28	UART_DEBUG_RX	DI	DEBUG UART RECEIVE	VILmax = 0.63V VIHmin = 1.17V	GPIO power domain. If unused, keep it open.
G27	UART_DEBUG_TX	DO	DEBUG UART TRANSMIT	VOLmax = 0.45V VOHmin = 1.35V	
E24	UART_RX_N_SE5	DI	UART1 receive	VILmax = 0.63V VIHmin = 1.17V	

PIN	Signal	I/O	Function	DC Characteristics	Comment
G23	UART_TX_N_SE5	DO	Uart1 transmit	VOLmax = 0.45V VOHmin = 1.35V	
G29	UART_CTS_N_SE5	DI	DCE clear to send signal from DTE	VILmax = 0.63V VIHmin = 1.17V	
E30	UART_RTS_N_SE5	DO	DCE request to send signal to DTE	VOLmax = 0.45V VOHmin = 1.35V	
E34	UART_RX_N_LPI	DI	UART2 receive	VILmax = 0.63V VIHmin = 1.17V	These LPI UART are internally connected ADSP and can not be used for application processor. Keep it open.
G35	UART_TX_N_LPI	DO	UART2 transmit	VOLmax = 0.45V VOHmin = 1.35V	

Telit SE250B4 Smart module provides multiple sets of GPIOs, available as BLSP (BAM-enabled low-speed peripheral) interfaces that can be configured to support various interface combinations. The operational voltage is 1.8V.

- **UART:** Support 3*UART; up to 4 Mbps.
- **SPI:** Supports 4*SPI; master-only mode; up to 50MHz.
- **I2C:** Support 4*I2C; master-only mode; up to 3.4MHz, external 2.2kΩ pull-up resistors are needed.

For customization, please refer to the multiplexed interfaces listed below, which can be configured to support alternate functions as required.

Table 47: Multiplexed interfaces for alternate functions.

PIN	Serial engine	Telit Primary Function	Alternative Function 1	Alternative Function 2	Alternative Function 3	GPIO
C23	SE0	NFC_SPI_MISO	SPI_MISO	UART_CTS_N	I2C_SDA	GPIO_0
B22		NFC_SPI_MOSI	SPI MOSI	UART_RTS_N	I2C_SCL	GPIO_1
C29		NFC_SPI_CLK	SPI_SCLK	UART_TX_N		GPIO_2
N45		NFC_SPI_CS_N	SPI_CS_0_N	UART_RX_N		GPIO_3
B30		LCM_RST_N	SPI_CS_1_N			GPIO_82
C27		NFC_CLKREQ	SPI_CS_2_N			GPIO_86
E26	SE1	APPS_I2C_SDA	SPI_MISO	UART_CTS_N	I2C_SDA	GPIO_4
G25		APPS_I2C_SCL	SPI MOSI	UART_RTS_N	I2C_SCL	GPIO_5
C25		NFC_EN	SPI_SCLK	UART_TX_N		GPIO_69
E22		NFC_INTERRUPT	SPI_CS_0_N	UART_RX_N		GPIO_70
G31	SE2	TS_I2C_SDA	SPI_MISO	UART_CTS_N	I2C_SDA	GPIO_6
E32		TS_I2C_SCL	SPI MOSI	UART_RTS_N	I2C_SCL	GPIO_7
C35		TS_RESET_N	SPI_SCLK	UART_TX_N		GPIO_71
C31		TS_INT_N	SPI_CS_0_N	UART_RX_N		GPIO_80
B36	SE4	KEY_VOL_UP_N	SPI_MISO	UART_CTS_N	I2C_SDA	GPIO_96
L49		CAM0_PWDN	SPI MOSI	UART_RTS_N	I2C_SCL	GPIO_97
G27		UART_DEBUG_TX	SPI_SCLK	UART_TX_N		GPIO_12
E28		UART_DEBUG_RX	SPI_CS_0_N	UART_RX_N		GPIO_13
G29		UART_CTS_N_SE5	SPI_MISO	UART_CTS_N	I2C_SDA	GPIO_14

PIN	Serial engine	Telit Primary Function	Alternative Function 1	Alternative Function 2	Alternative Function 3	GPIO
E30	SE5	UART_RTS_N_SE5	SPI MOSI	UART_RTS_N	I2C_SCL	GPIO_15
G23		UART_TX_N_SE5	SPI_SCLK	UART_TX_N		GPIO_16
E24		UART_RX_N_SE5	SPI_CS_0_N	UART_RX_N		GPIO_17

Note:

- When a serial engine is assigned to a specific function, it cannot be simultaneously configured or reused for an alternate function.
- Serial Engine 1 (SE1) is primarily allocated for PM8008, MIPI-DSI bridge IC, NFC, LCD bias IC, and backlight driver IC functions.
- Serial Engine 2 (SE2) is assigned to the touchscreen interface by default.
- If the default (primary) functions are not required, the respective serial engine may be reconfigured to an alternate supported function, subject to multiplexing constraints.

The serial port level of the Telit SE250B4 Smart module UART port is 1.8V. To interface with 3.3V serial ports, a level conversion chip must be used. It is recommended to use TI's TXS0104EPWR.

The following is the circuit diagram:

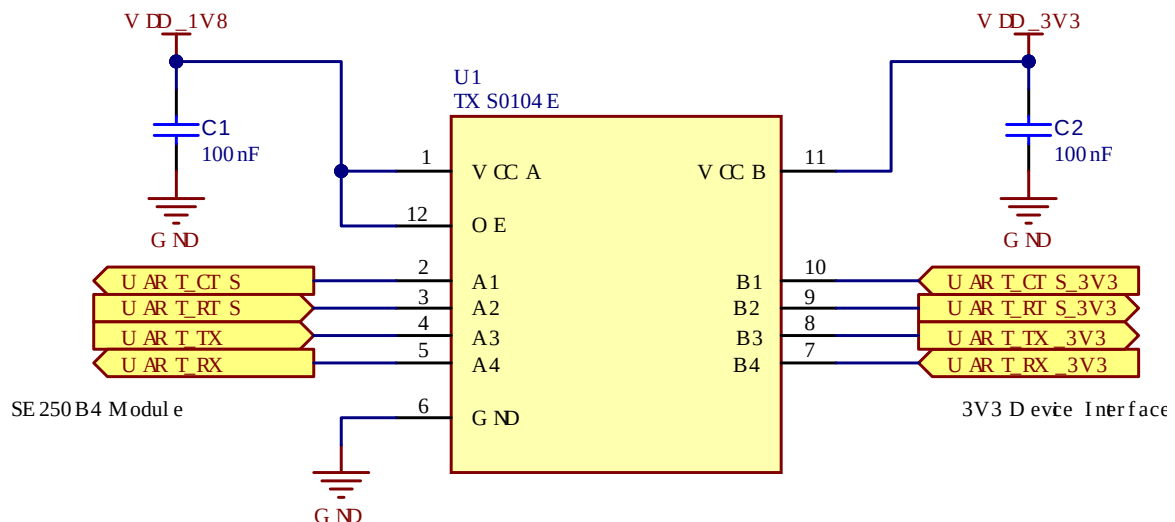


Figure 20: Reference Circuit with Level Translator Chip for UART0

The figure below represent an example of connection between the module and a PC. Using a RS-232 level translator chip between the module and PC COM PORT is mandatory to achieve a proper level for signals.

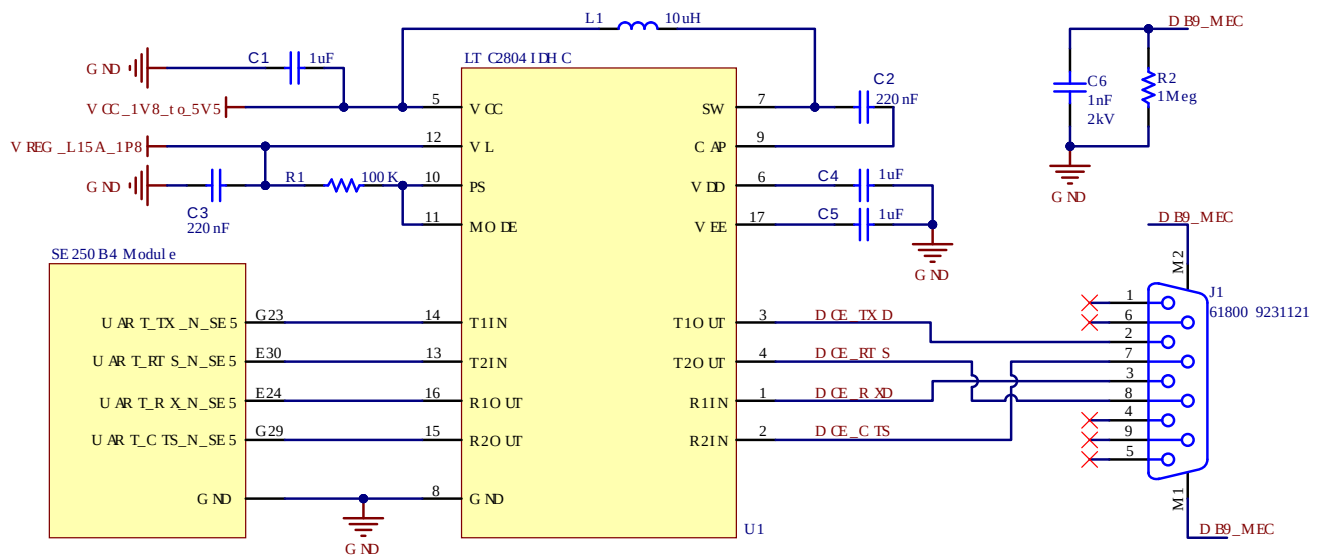


Figure 21: RS-232 Level Match Circuit for UART0

Note: UART_DEBUG are similar to UART. Please refer to the same reference design.

5.15 UIM Interface

Telit SE250B4 Smart module provides two (U)SIM interfaces compliant with ETSI and IMT-2000 requirements. Dual SIM Dual Standby functionality is supported by default. Either card support 1.8V or 2.95V and are powered by an internal power supply rail of the module.

Table 48: USIM PADS

PIN	Signal	I/O	Function	Comment
U2	UIM1_CLK	DO	(U)SIM1 card clock	50Ω characteristic impedance.
V1	UIM1_PRESENT	DI	(U)SIM1 card hot-plug detect	Active low. Externally pull it up to 1.8V. If unused, keep it open.
T1	UIM1_DATA	DI/O	(U)SIM1 card data	Cannot be multiplexed into a generic GPIO. 50Ω characteristic impedance.
S2	UIM1_RESET	DO	(U)SIM1 card reset	
AB7	VREG_L18A_1P8	PO	USIM1_VDD	Either 1.8V or 2.95V (U)SIM card is supported.
R1	UIM2_CLK	DO	(U)SIM2 card clock	50Ω characteristic impedance.
M2	UIM2_PRESENT	DI	(U)SIM2 card hot-plug detect	Active low. Externally pull it up to 1.8V. If unused, keep it open.
P2	UIM2_DATA	DI/O	(U)SIM2 card data	Cannot be multiplexed into a generic GPIO. 50Ω characteristic impedance.
N1	UIM2_RESET	DO	(U)SIM2 card reset	
AC5	VREG_L19A_1P8	PO	USIM2_VDD	Either 1.8V or 2.95V. (U)SIM card is supported.

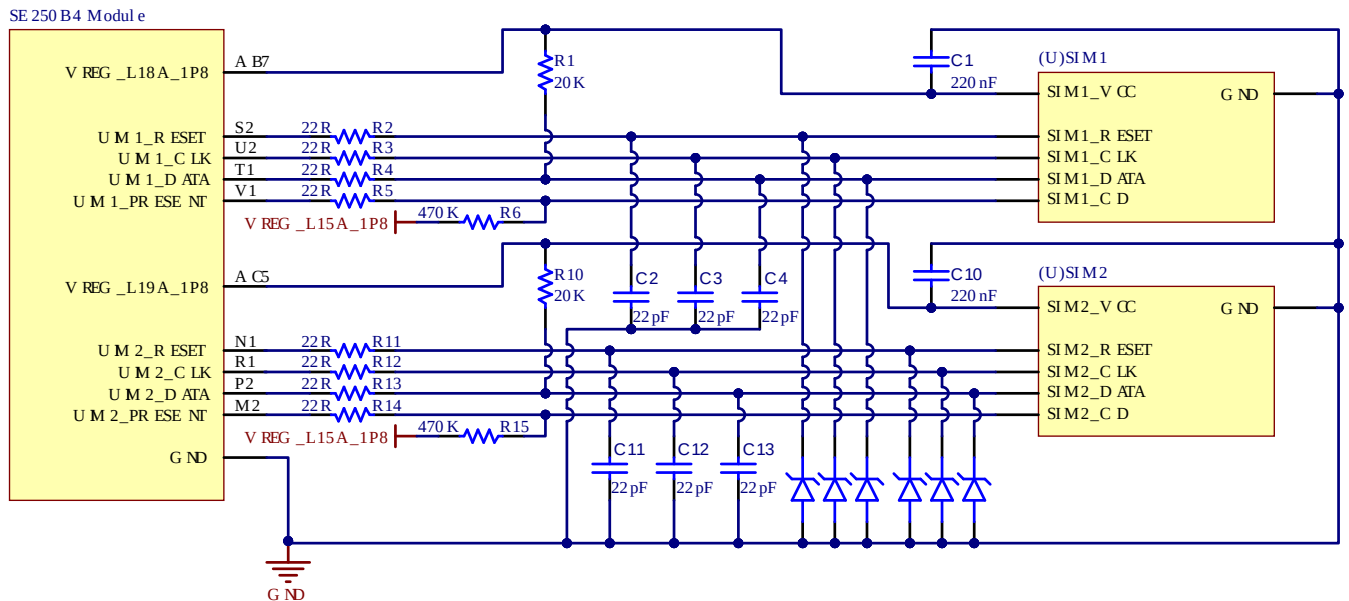


Figure 22: Reference Circuit for (U)SIM Interface with a 6-pin (U)SIM Card Connector

Note: UIM_DATA lines need to be pulled up to VREG_L18A_1P8 / VREG_L19A_1P8 voltage via a 20K resistor.

UIM1/2_Present need to be pulled up to VREG_L15A_1P8 through a 470K resistor.

On VREG_L18A_1P8 and VREG_L19A_1P8, add a ceramic 220nF bypass capacitor to reduce interference.

Please ensure that TVS diodes do not add more than 10pF capacitance to the signal paths.

To optimize performance and ensure reliability in your design, consider the following criteria:

- Place the (U)SIM card connector as close as possible to the module on your application board to keep the trace as short as possible (max 200mm)
- Keep the (U)SIM card signals away from RF and power supply traces to reduce interference.
- Place a bypass capacitor close the VREG_L18A_1P8 Pad on SE250B4 and if necessary, another bypass capacitor close to the SIM_VCC pad. Consider a capacitance value of 220nF (it will act as a high pass for frequency 1MHz (~724KHz))
- Pay attention when routing UIMx_DATA, UIMx_CLK, and USIMx_signals. UIMx_DATA and UIMx_CLK can suffer from crosstalk if not properly routed. An interference on the UIMx_RESET line might cause the SIM to malfunction, so it is advisable to route it away from aggressor signals with sharp transition.
- Add a TVS array for ensuring ESD protection on all lines and pay attention that parasitic capacitance doesn't exceed 10pF. Moreover, 22Ω series resistor can be used to suppress EMI such spurious transmission.

- At least, 4-layer through-hole PCB should be chosen for good impedance control and signal shielding.

5.16 SD CARD Interface

SD Card interface of the module supports SD 3.0 protocol. The pin definition of SD card is shown below.

Table 49: SD CARD PADS

PIN	Signal	I/O	Function	Comment
C45	SDC2_CLK	DO	SD card clock	50Ω characteristic impedance.
B46	SDC2_CMD	DO	SD card command	
C43	SDC2_DATA_0	DIO	SDIO data bit 0	
B44	SDC2_DATA_1	DIO	SDIO data bit 1	
E42	SDC2_DATA_2	DIO	SDIO data bit 2	
G41	SDC2_DATA_3	DIO	SDIO data bit 3	Active low.
K50	SD_CARD_DET_N	DI	SD card hot-plug detect	
AB3	VREG_L4A_2P96	PO	1.8/2.95 V output power for SD card pull-up circuits	
X7, Z7	VREG_L21A_2P96	PO	SD card power supply	

Our EVK includes a MICRO SD connector.

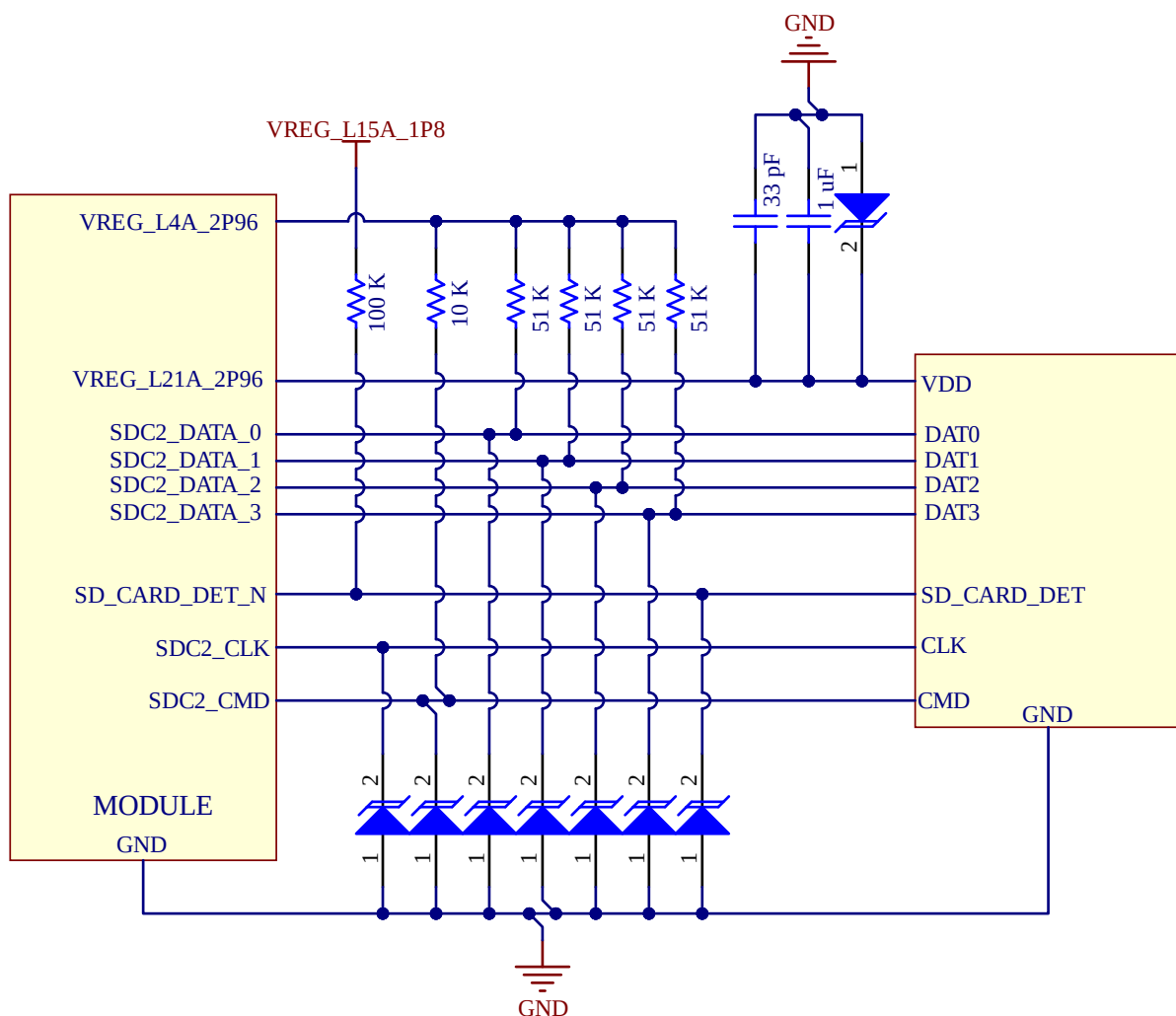


Figure 23: Reference Circuit for the SD Card Interface

VREG_L21A_2P96 is the power supply for SD card and provides up 841mA of output current. To accommodate this, it's recommended to use a trace width of at least 0.8/1mm on the PCB. Additionally, consider adding capacitors of 33pF and 1μF in parallel to enhance voltage stability.

SDC2_DATA_[0..3], SDC2_CLK, SDC2_CMD are high-speed signals and requires a characteristic impedance trace of $50\Omega \pm 10\%$. Ensure these traces do not intersect with other high-speed lines and are adequately shielded. SDC2_CLK should ideally be shielded separately.

It is recommended to use an inner layer on the PCB and maintain the data signal of the same length.

Table 50: SD signal length inside the module

PIN	Signal	Length (mm)
C45	SDC2_CLK	3.6
B46	SDC2_CMD	15.864
C43	SDC2_DATA_0	15.518
B44	SDC2_DATA_1	15.626
E42	SDC2_DATA_2	16.02
G41	SDC2_DATA_3	15.523

5.17 General Purpose I/O

Telit SE250B4 Smart module has different GPIOs that can be configured depending on customer application.

In the default configuration, the module provides limited GPIOs due to many signals being dedicated to other functions as outlined in the PINOUT section. However, customers have the flexibility to configure different GPIOs according to their hardware and software needs.

5.18 I2C Interface

The Telit SE250B4 provides four I2C interfaces. All I2C interfaces are open drain signals and therefore must be pulled up externally. The reference power domain is 1.8V. The sensor I2C interface only supports sensors for ADSP architecture. On the other hand, CAM0 I2C and CAM1 I2C signals are controlled by the Kernel code and support connection to radio-output related devices.

Table 51: I2C Interface

PIN	Signal	I/O	Function	Comment
E32	TS_I2C_SCL	DO	TP I2C clock	Need to be pulled up to 1.8V externally. Can be used for other I2C devices.
G31	TS_I2C_SDA	DO	TP I2C data	
AG47	CAM_CCI_I2C_SCL0	DO	I2C clock of camera 0	Can only be used for camera I2C devices.
AE47	CAM_CCI_I2C_SDA0	DIO	I2C data of camera 0	
AE50	CAM_CCI_I2C_SCL1	DO	I2C clock of camera 1	
AF49	CAM_CCI_I2C_SDA1	DIO	I2C data of camera 1	
E38	SNSR_I2C_SCL	DO	I2C clock for external sensors	Can only be used to for sensors.

PIN	Signal	I/O	Function	Comment
G37	SNSR_I2C_SDA	DIO	I2C data for external sensors	

5.19 ADC Converter

The Telit SE250B4 supports two Analog-to-Digital Converter (ADC) interface. The ADC interface supports resolution up to 15 bits. The pin definition is shown below.

Table 52: ADC converter

PIN	Signal	I/O	Function
AF7	PM_AMUX_2	ADC	General-purpose ADC interface
A5	PM_AMUX_3	ADC	General-purpose ADC interface

Table 53: ADC Performances

Parameter	Comments	Min	Type	Max	Unit
Input voltage range	Programmable	0.1	-	1.7	V
		0.3	-	4.5	
Resolution		-	15	-	bits
Analog input bandwidth		-	100	-	kHz
Sample rate	XO/8	-	2.4	-	MHz
INL	15-bit output	-8	-	8	LSB
DNL	15-bit output	-4	-	4	LSB
Offset error	Relative to full-scale	-1	-	1	%
Gain error	Relative to full-scale	-1	-	1	%

Note: Voltages exceeding the input voltage range might result in permanent damage to the module. Please ensure proper signal scaling and voltage level at the ADC inputs.

5.20 Vibrator

Telit SE250B4 Smart module supports silent incoming-call alarms with a vibration motor driver. The motor is connected between VIB_DRIVER and GND. The programmable motor voltage ranges from 1.2 to 3.1V in 100 mV steps.

Table 54: Vibrator

PIN	Signal	I/O	Function
P5	VIB_DRIVER	PO	Vibration motor driver output control. $I_{O_max} = 100\text{mA}$

The following is the circuit diagram:

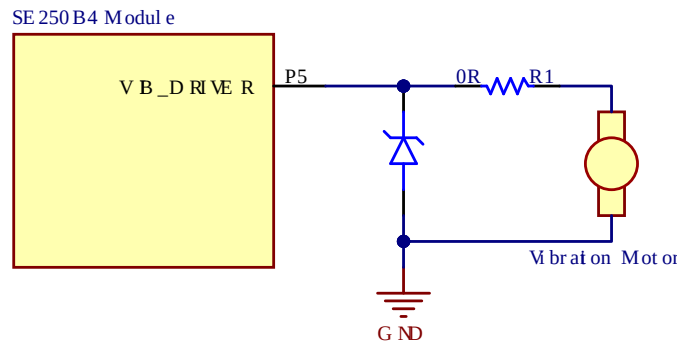


Figure 24: Reference Circuit for Motor Connection

When the motor stops working and the VIB_DRIVER is disconnected, the residual energy on the motor will be discharged by the freewheeling diode, thus avoiding component damage.

5.21 LCM Interface

The module provides one LCM interface, which complies to MIPI DSI standard. The interface supports high-speed differential data transmission and support HD+ display (1680 x 720 @60 fps). The pin definition of the LCM interface is shown below.

Table 55: LCM Interface

PIN	Signal	I/O	Function
BC27	MIPI_DSI0_CLK_M	AO	LCD MIPI clock (-)
BC25	MIPI_DSI0_CLK_P	AO	LCD MIPI clock (+)
BA24	MIPI_DSI0_L0_M	AO	LCD MIPI lane 0 data (-)
BA22	MIPI_DSI0_L0_P	AO	LCD MIPI lane 0 data (+)
AY25	MIPI_DSI0_L1_M	AO	LCD MIPI lane 1 data (-)
AY27	MIPI_DSI0_L1_P	AO	LCD MIPI lane 1 data (+)
AY23	MIPI_DSI0_L2_M	AO	LCD MIPI lane 2 data (-)
AY21	MIPI_DSI0_L2_P	AO	LCD MIPI lane 2 data (+)
BD26	MIPI_DSI0_L3_M	AO	LCD MIPI lane 3 data (-)
BD28	MIPI_DSI0_L3_P	AO	LCD MIPI lane 3 data (+)
B30	LCM_RST_N	DO	LCD reset
B38	LCM_TE	DO	LCD tearing effect
R45	LCM_BIAS_N_EN	DO	LCM bias EN (-)
C39	LCM_BIAS_P_EN	DO	LCM bias EN (+)
L45	LCM_ID1	DI	LCM_ID1 display
G49	LCM_ID2	DI	LCM_ID2 display
AY3	LCM_BL_EN	DO	LCM BL enable
E20	LCM_WLED_PWM	DO	PWM output

The reference schematic for LCM also includes Backlight driver reference design and IC BOOST reference design.

The design is in vector form so the customer can easily zoom in to better see the details.

Both the Backlight driver and IC boost contain IC address for reference and are supplied directly from the SE250B4.

This may be considered as reference for general LCM driver and reflects the design of our EVB in which we have a socket for the LCM.

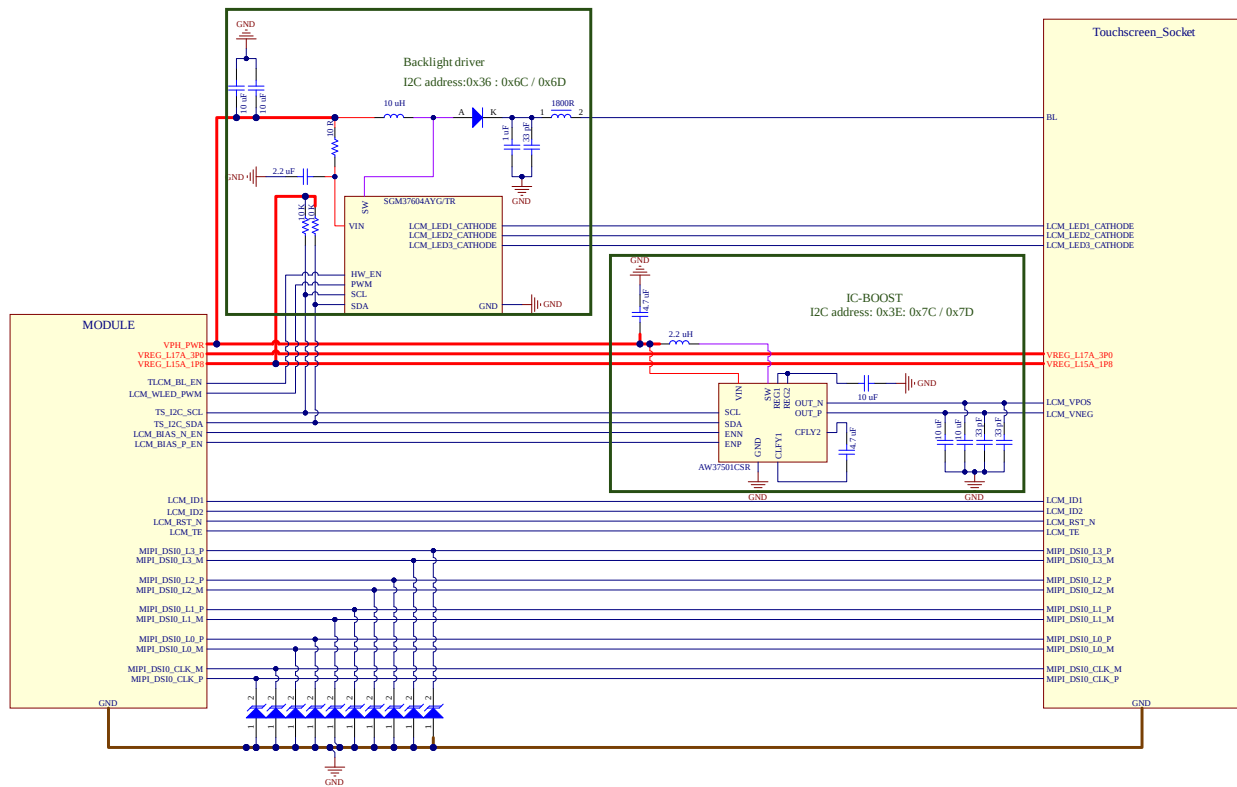


Figure 25: Reference Circuit Design for LCM Interface

5.22 LED Flash Interface

The Telit SE250B4 supports one flash LED driver with maximum output power up to 1A. Connect LED_FLASH to the anode of the LED, its cathode should be connected directly to GND.

The Flash current is programmable in step of 12.5mA (max 1A) or 5mA (max 640mA)

Table 56: LED Flash interface

PIN	Signal	I/O	Function
E16	LED_FLASH	DO	Flash/torch driver output
G15	LED_FLASH	DO	Flash/torch driver output

5.23 Touch Panel Interface

The Telit SE250B4 provides one I2C interface for the connector to a Touch Panel (TP) and also provides the corresponding power supply and interrupt pins. The definition of TP interface is shown in the table and reference schematic below.

Table 57: Touch panel Interface

PIN	Signal	I/O	Function	Comment
E32	TS_I2C_SCL	DO	TP I2C clock	Need to be pulled up to 1.8V externally. Can be used for other I2C devices.
G31	TS_I2C_SDA	DO	TP I2C data	
C31	TS_INT_N	DI	TP interrupt	1.8V power domain.

PIN	Signal	I/O	Function	Comment
C35	TS_RESET_N	DO	TP reset	1.8V power domain. Active low.
C37	TS_3P0_EN	DI	TP Enable	

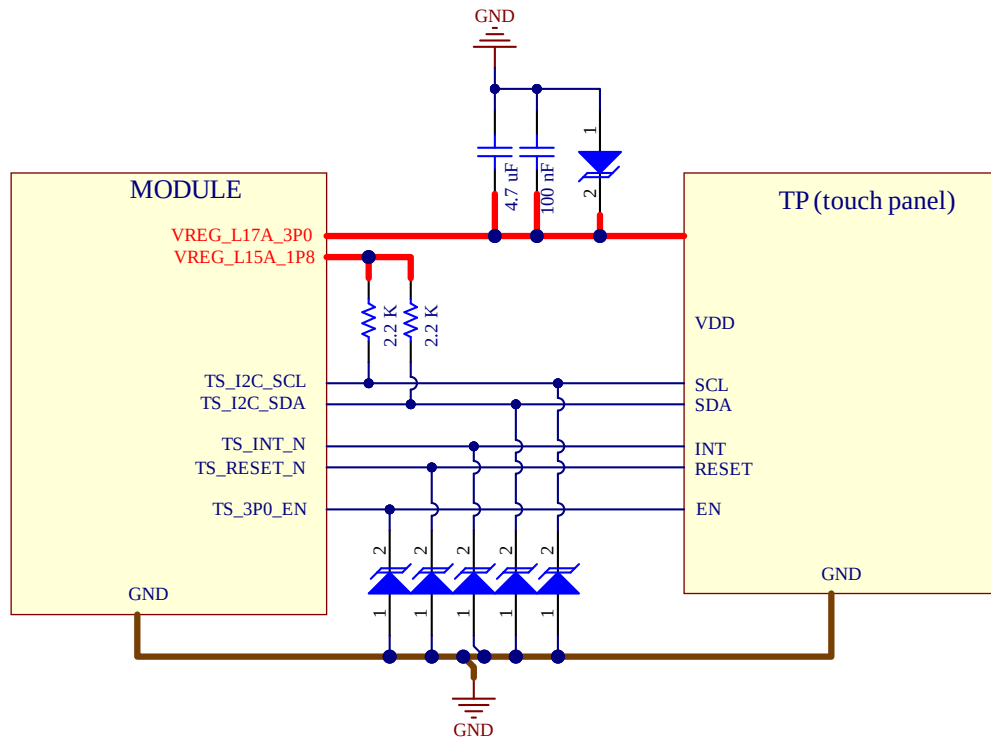


Figure 26: Reference Circuit Design for the Touch Panel Interface

5.24 Camera Interface

Based on MIPI CSI standard, the module supports two cameras (4-lane+4-lane) or three cameras (4-lane+2-lane+1-lane), and the maximum resolution is 25MPx. The video and photo quality are determined by various factors such as camera sensors and camera lens specifications. The pad list of cameras is shown below, our reference design will take into consideration two cameras respectively of 13 and 5MPx and a front camera of 2MPx. The cameras can be used also simultaneously.

Table 58: Camera Interface

PIN	Signal	I/O	Function
AJ50	CAM_MCLK0	DO	Master clock of camera 0
L49	CAM0_PWDN	DO	Power down of camera 0
AF51	CAM0_RST_N	DO	Reset of camera 0
AG50	CAM_MCLK1	DO	Master clock of camera 1
AD45	CAM1_PWDN	DO	Power down of camera 1
Y47	CAM1_RST_N	DO	Reset of camera 1
AL50	CAM_MCLK2	DO	Master clock of camera 2
G51	CAM2_PWDN	DO	Power down of camera 2
V45	CAM2_RST_N	DO	Reset of camera 2
AG47	CAM_CCI_I2C_SCL0	DO	I2C clock of camera 0
AE47	CAM_CCI_I2C_SDA0	DIO	I2C data of camera 0
AE50	CAM_CCI_I2C_SCL1	DO	I2C clock of camera 1

PIN	Signal	I/O	Function
AF49	CAM_CCI_I2C_SDA1	DIO	I2C data of camera 1
J49	CAM_AF_VDD_EN	DO	
K47	CAM_1P2_EN	DO	
H50	CAM_1P8_EN	DO	
E51	CAM_2P8_EN	DO	
AU49	MIPI_CSI0_A0_CLK_M	AI	MIPI CSI0 clock (+)
AT50	MIPI_CSI0_NC_CLK_P	AI	MIPI CSI0 clock (-)
AS49	MIPI_CSI0_C0_LN0_M	AI	MIPI CSI0 lane 0 data (-)
AR50	MIPI_CSI0_B0_LN0_P	AI	MIPI CSI0 lane 0 data (+)
AT47	MIPI_CSI0_B1_LN1_M	AI	MIPI CSI0 lane 1 data (-)
AS45	MIPI_CSI0_A1_LN1_P	AI	MIPI CSI0 lane 1 data (+)
AP45	MIPI_CSI0_A2_LN2_M	AI	MIPI CSI0 lane 2 data (-)
AN47	MIPI_CSI0_C1_LN2_P	AI	MIPI CSI0 lane 2 data (+)
AP49	MIPI_CSI0_C2_LN3_M	AI	MIPI CSI0 lane 3 data (-)
AR47	MIPI_CSI0_B2_LN3_P	AI	MIPI CSI0 lane 3 data (+)
Y50	MIPI_CSI1_A0_CLK_M	AI	MIPI CSI1 clock (-)
X51	MIPI_CSI1_NC_CLK_P	AI	MIPI CSI1 clock (+)
AA50	MIPI_CSI1_C0_LN0_M	AI	MIPI CSI1 lane 0 data (-)
Z51	MIPI_CSI1_B0_LN0_P	AI	MIPI CSI1 lane 0 data (+)
W50	MIPI_CSI1_B1_LN1_M	AI	MIPI CSI1 lane 1 data (-)
V51	MIPI_CSI1_A1_LN1_P	AI	MIPI CSI1 lane 1 data (+)
U50	MIPI_CSI1_A2_LN2_M	AI	MIPI CSI1 lane 2 data (-)
T51	MIPI_CSI1_C1_LN2_P	AI	MIPI CSI1 lane 2 data (+)
S50	MIPI_CSI1_C2_LN3_M	AI	MIPI CSI1 lane 3 data (-)
R51	MIPI_CSI1_B2_LN3_P	AI	MIPI CSI1 lane 3 data (+)

Note: In a 3-camera use case, MIPI_CSI1_B2_LN3_P and MIPI_CSI1_C2_LN3_M are assigned to the MIPI clock interface of Camera 3, while MIPI_CSI1_C1_LN2_P and MIPI_CSI1_A2_LN2_M are assigned to the MIPI data interface of Camera 3.

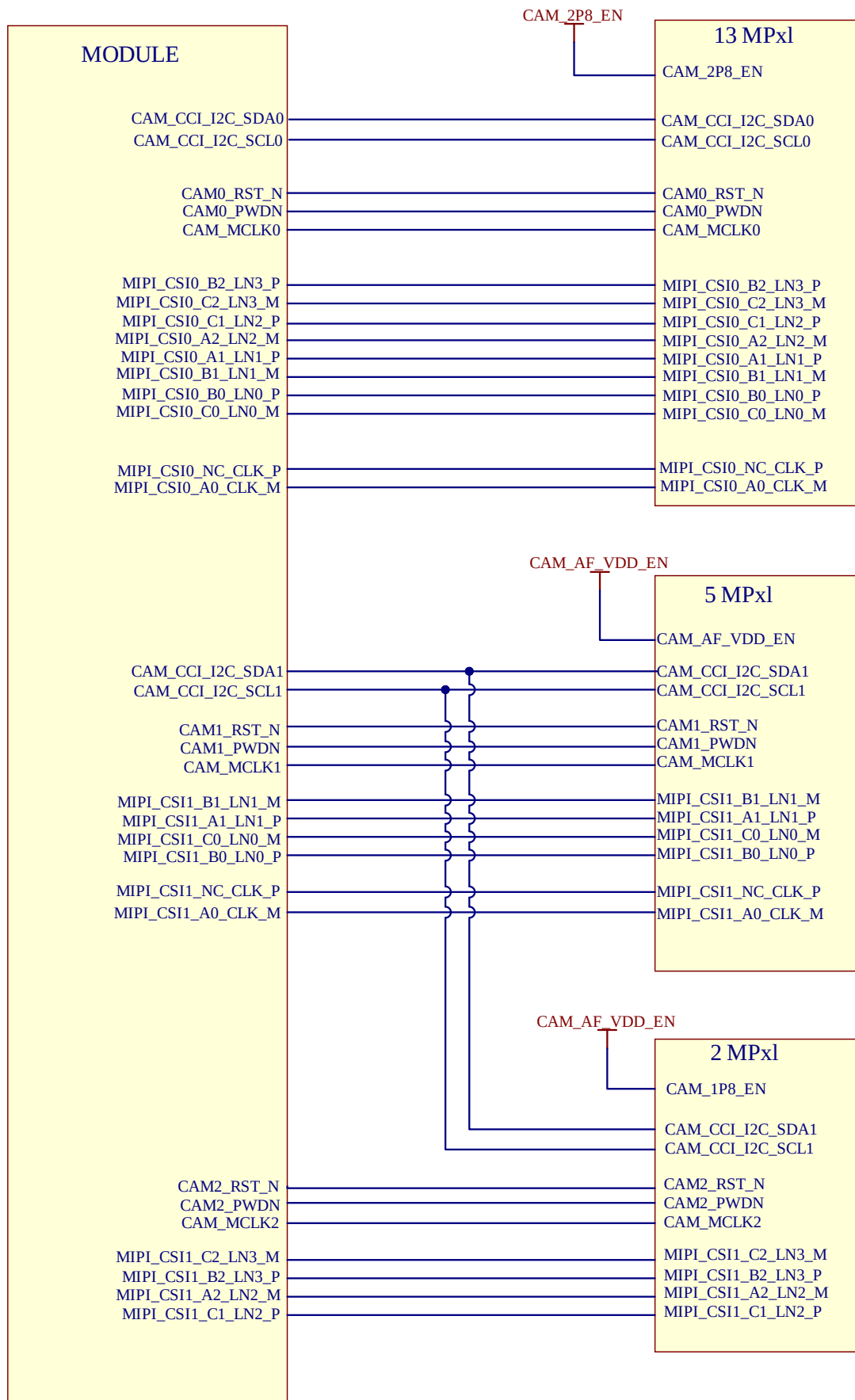


Figure 27: Reference Circuit Design for 3-Camera Applications

5.24.1 MIPI Design Consideration

Special attention must be given to the routing of high speed LCM and camera signals.

Both DSI and CSI MIPI lines are high-speed differential signal lines, with DSI supporting a maximum speed of up to 1.5 Gbps and CSI supporting a data rate of up to 2.5 Gbps.

The differential impedance should be controlled to $100\Omega \pm 10\%$.

Additionally, it is recommended to route these traces in the PCB inner layers, and avoid cross-talking with other “power”, “high speed”, or RF lines.

To maintain signal integrity, the total length of MIPI traces should not exceed 240mm, and the differential length mismatch should not exceed 0.7mm.

It is recommended to select a low-capacitance TVS for ESD protection on application board, ensuring that the total parasitic capacitance does not exceed 1pF.

The following table reports the MIPI trace length inside the module:

Table 59: MIPI lines length inside the module

PIN	Signal	Length (mm)	Lengt Difference P-N
BC27	MIPI_DSI0_CLK_M		
BC25	MIPI_DSI0_CLK_P		
BA24	MIPI_DSI0_L0_M		
BA22	MIPI_DSI0_L0_P		
AY25	MIPI_DSI0_L1_M		
AY27	MIPI_DSI0_L1_P		
AY23	MIPI_DSI0_L2_M		
AY21	MIPI_DSI0_L2_P		
BD26	MIPI_DSI0_L3_M		
BD28	MIPI_DSI0_L3_P		
AU49	MIPI_CSI0_A0_CLK_M	22.271	0.302
AT50	MIPI_CSI0_NC_CLK_P	21.969	
AS49	MIPI_CSI0_C0_LN0_M	22.369	0.081
AR50	MIPI_CSI0_B0_LN0_P	22.288	
AT47	MIPI_CSI0_B1_LN1_M	21.902	0.54
AS45	MIPI_CSI0_A1_LN1_P	22.442	
AP45	MIPI_CSI0_A2_LN2_M	22.084	0.021
AN47	MIPI_CSI0_C1_LN2_P	22.063	
AP49	MIPI_CSI0_C2_LN3_M	22.313	0.027
AR47	MIPI_CSI0_B2_LN3_P	22.286	
Y50	MIPI_CSI1_A0_CLK_M	13.775	0.022
X51	MIPI_CSI1_NC_CLK_P	13.753	
AA50	MIPI_CSI1_C0_LN0_M	13.456	0.08
Z51	MIPI_CSI1_B0_LN0_P	13.536	
W50	MIPI_CSI1_B1_LN1_M	13.579	0.023
V51	MIPI_CSI1_A1_LN1_P	13.556	
U50	MIPI_CSI1_A2_LN2_M	13.427	0.121
T51	MIPI_CSI1_C1_LN2_P	13.548	
S50	MIPI_CSI1_C2_LN3_M	13.553	0.004
R51	MIPI_CSI1_B2_LN3_P	13.549	

5.25 Sensor Interface

The module supports communication with sensors via I2C interfaces, supporting various sensors such as ALS/PS, BDS, accelerometer, gyroscope, and so on. While the EVK includes some sensors, the module also offers two general-purpose pads suitable for external sensor applications.

Table 60: Sensor Interface PADS

PIN	Signal	I/O	Function	Comment
V49	SNSR_ACCL_INT	DI	Interrupt for ACC sensor	GPIO_32
X49	SNSR_GYRO_INT	DI	Interrupt for gyro sensor	GPIO_33
AD51	SNSR_ALPS_INT_N	DI	Interrupt for Lux/proximity sensor	GPIO_35
E38	SNSR_I2C_SCL	DO	I2C clock for external sensors	GPIO_110
G37	SNSR_I2C_SDA	DI/O	I2C data for external sensors	GPIO_109
E26	APPS_I2C_SDA	DI/O	For PM8008/MIPI DSI bridge IC/NFC/LCD	GPIO_4
G25	APPS_I2C_SCL	DO	BIAS IC/Backlight driver IC	GPIO_5

5.26 Audio Interface

The Telit SE250B4 provides three analog input channels and three analog output channels. The following table shows the pin definition.

Table 61: Audio Interface

PIN	Signal	I/O	Function	Comment
BC9	CDC_MIC_BIAS1	AO	Bias voltage 1 output for microphone	Vomin = 1.6V Vnom=1.8V Vomax = 2.85V I=3mA
BD10	CDC_MIC_BIAS2	AO	Bias voltage 2 output for microphone	
BA10	CDC_MIC_BIAS3	AO	Bias voltage 3 output for microphone	
BD8	CDC_IN1_M	AI	Microphone input for channel 1 (-)	
BC7	CDC_IN1_P	AI	Microphone input for channel 1 (+)	
AV5	CDC_IN2_M	AI	Microphone input for channel 2 (-)	
AU7	CDC_IN2_P	AI	Microphone input for channel 2 (+)	
AT5	CDC_IN3_M	AI	Microphone input for channel 3 (-)	
AR5	CDC_IN3_P	AI	Microphone input for channel 3 (+)	
BC11	CDC_EAR_M	AO	Earpiece output (-)	The typical output voltage is 2 Vrms.
BD12	CDC_EAR_P	AO	Earpiece output (+)	
AS7	CDC_HPH_REF	GND	Headphone reference Ground	
AP7	CDC_HPH_L	AO	Headphone left channel output	
AN5	CDC_HPH_R	AO	Headphone right channel output	
AU3	CDC_HS_DET	AI	Headset hot-plug detect	
BD6	CDC_AUX_P	AO	Audio line differential output (+)	
BE5	CDC_AUX_M	AO	Audio line differential output (-)	
E40	AUD_DMIC_CLK_1	DO	Clock for digital mic	GPIO_98 (QCM2290)
G39	AUD_DMIC_DATA_1	DI	data for digital mic	GPIO_99 (QCM2290)
G21	AUD_SPK_AMP_SHDN_N	DO	SHDN for audio amp	

- The module offers three audio input channels.

- The output voltage range of CDC_MIC_BIAS1 and CDC_MIC_BIAS2 is programmable between 1.6V and 2.85V and the maximum output current is 3mA. CDC_MIC_BIAS3 supports 1.8V pull-up output only and is not programmable.
- The earpiece interface uses differential output.
- The lineout interface uses differential output, lineout is used as audio PA input.
- The headphone interface features stereo left and right channel outputs and supports headphone insertion detection.

5.26.1 EMC MIC

The following schematic defines the suggested implementation for an Electret microphone. R6 and R7 are optional EMI filters and recommended if the MIC is connected by long cables.

The schematic accommodates also for the differential configuration, to achieve that layout set R5 as DNI, R3 to 0 ohm and replace R1 and R4 with 1K1 resistors.

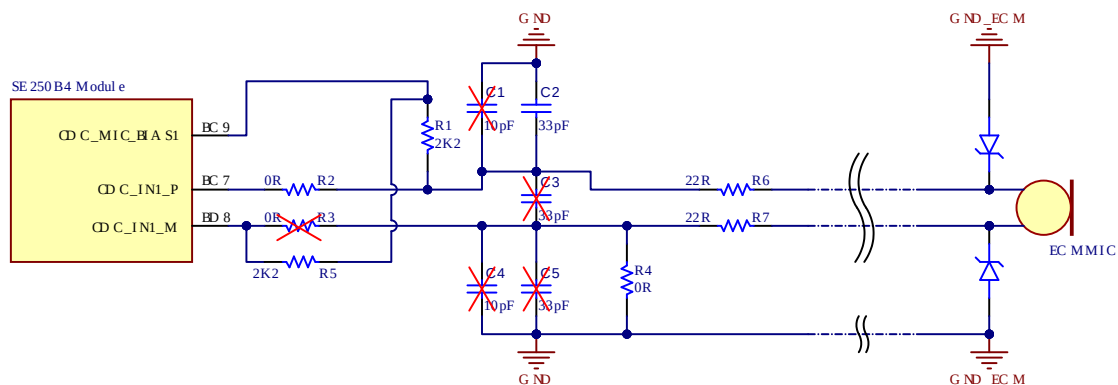


Figure 28: Reference Circuit Design for ECM Microphone Interface

- The characteristics of audio signals are significantly influenced by the microphone gain.

The tables list reference measurements corresponding to different selected gain values:

Table 62: Microphone Amplifier Gain = 0 dB (minimum gain)

Parameter	Test conditions	Min	Typical	Max	Units
Input-referred noise	Single-ended, A-weighted, capless	-	18.5	25.1	μVrms
Signal-to-noise ratio	Single-ended, A-weighted, capless	92	94	-	dB
THD+N ratio, Analog input = -1 dBV	f = 1.02kHz; single-ended input; 200Hz to 20kHz bandwidth; capless	-	-83	-70	dB

Table 63: Microphone Amplifier Gain = 6 dB

Parameter	Test conditions	Min	Typical	Max	Units
Input referred noise	Single-ended, A-weighted, capless	-	10	13	μVrms
Signal-to-noise ratio	Single-ended, A-weighted, capless	91	94	-	dB
THD+N ratio Analog input = -1 dBV	f = 1.02kHz; single-ended input; 200Hz to 20kHz bandwidth; capless	-	-82.5	-70	dB

Table 64: Microphone amplifier gain = 24 dB (maximum gain)

Parameter	Test conditions	Min	Typical	Max	Units
Input referred noise	Single-ended, A-weighted, capless	-	2.6	4.2	μVrms
Signal-to-noise ratio	Single-ended, A-weighted, capless	84	87.5	-	dB
THD+N ratio Analog input = -1 dBV	f = 1.02kHz; single-ended input; 200Hz to 20kHz bandwidth; capless	-	-82	-60	dB

The table below shows the general characteristics of microphone.

Table 65: General Microphone Specifications

Parameter	Test conditions	Min	Typical	Max	Units
Full-scale input voltage	Single-ended 1 kHz input. Input signal level required to get 0dBFS digital output	-0.5	0	0.5	dBV
Input impedance	Capless input Input disabled	1	-	-	MΩ
		3	-	-	MΩ
Input capacitance	Capless input	-	-	15	pF

Table 66: Microphone Bias Output Performance Specifications

Parameter	Test conditions	Min	Typical	Max	Units
Output voltage	No load	1.60	-	2.85	V
Output voltage accuracy	No load	-3.00	0.00	3.00	%
Output current	2 microphone loads of 1.0 to 1.5mA each	2.0	3.0	-	mA
Output switch to ground	On resistance	-	-	20	Ω
	Sink current	2.0	-	-	mA
Output noise	0.1 μF bypass	0.0	2.0	4.0	μVrms
PSRR - Power supply rejection ratio 100 mVpp applied to PMIC Vbatt input	at 20Hz	90	-	-	dB
	at 200Hz to 1kHz	90	-	-	dB
	at 5kHz	90	-	-	dB
	at 10kHz	90	-	-	dB
	at 20kHz	85	-	-	dB
Output capacitor value	External bypass mode	0.1	0.1	0.5	μF
	No external bypass mode	-	-	270	pF

5.26.2 MEMS MIC

The design depends by the chosen type of MEMS microphone.

For Analog MEMS microphones please consider the following reference designs

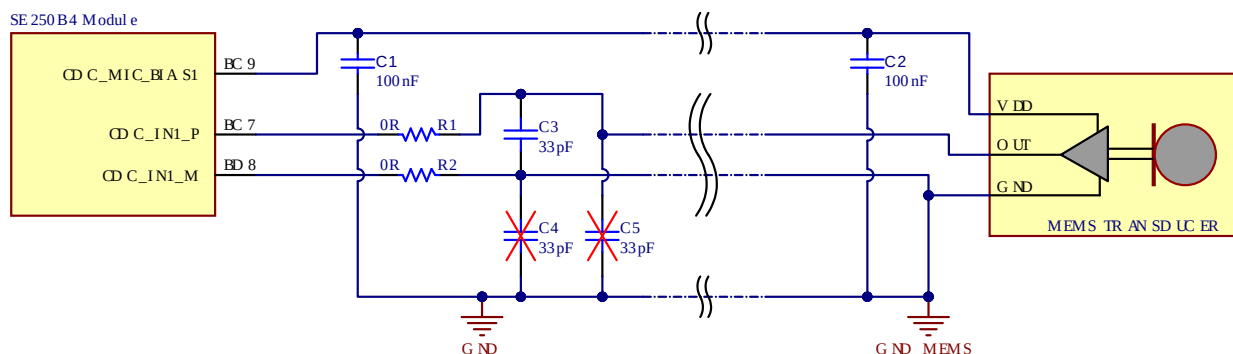


Figure 29: Reference Circuit Design for analog single ended MEMS Microphone Interface

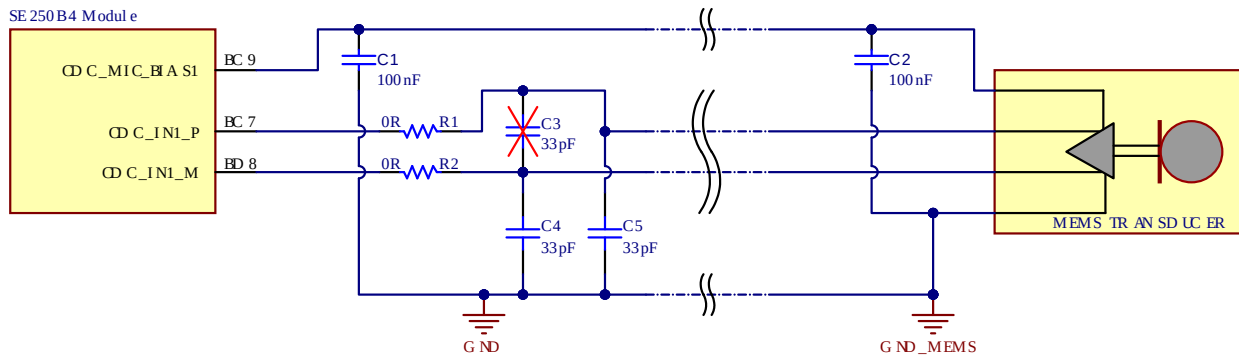


Figure 30: Reference Circuit Design for analog differential MEMS Microphone Interface

For Digital microphone, consider the following reference design.

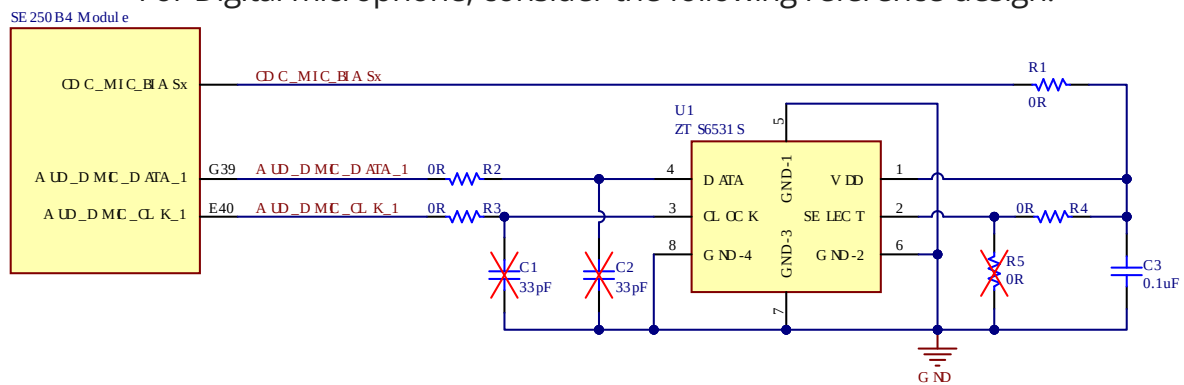


Figure 31: Reference Circuit Design for digital MEMS Microphone Interface

Typically, the Microphone SELECT pin should be connected to either its VDD or GND to define the CLK data front and communication scheme. Depending on their SELECT configuration, up to 2 microphones can communicate on the same data bus. For proper implementation, please refer to the Microphone datasheet. The following table (reference purpose only) shows typical performances of a digital MEMS microphone.

Table 67: Digital Microphone Input Performance

Parameter	Test conditions	Min	Typical	Max	Units
Sensitivity	@1kHz (0dB=1Pa)	-27	-26	-25	dBFS
Signal-to-noise ratio	94dBSPL@1kHz, A-weighted Bandwidth 20KHz	-	65	-	dB FS(A)
Maximum Acoustic Input	THD<10% @1kHz		122		dB SPL
Current consumption	VDD=1.8V, fclock =2.4MHz	390	560	730	μA
Power supply rejection ratio	200 mVPP sinewave @1kHz		64		dB FS/V

5.26.3 Earpiece

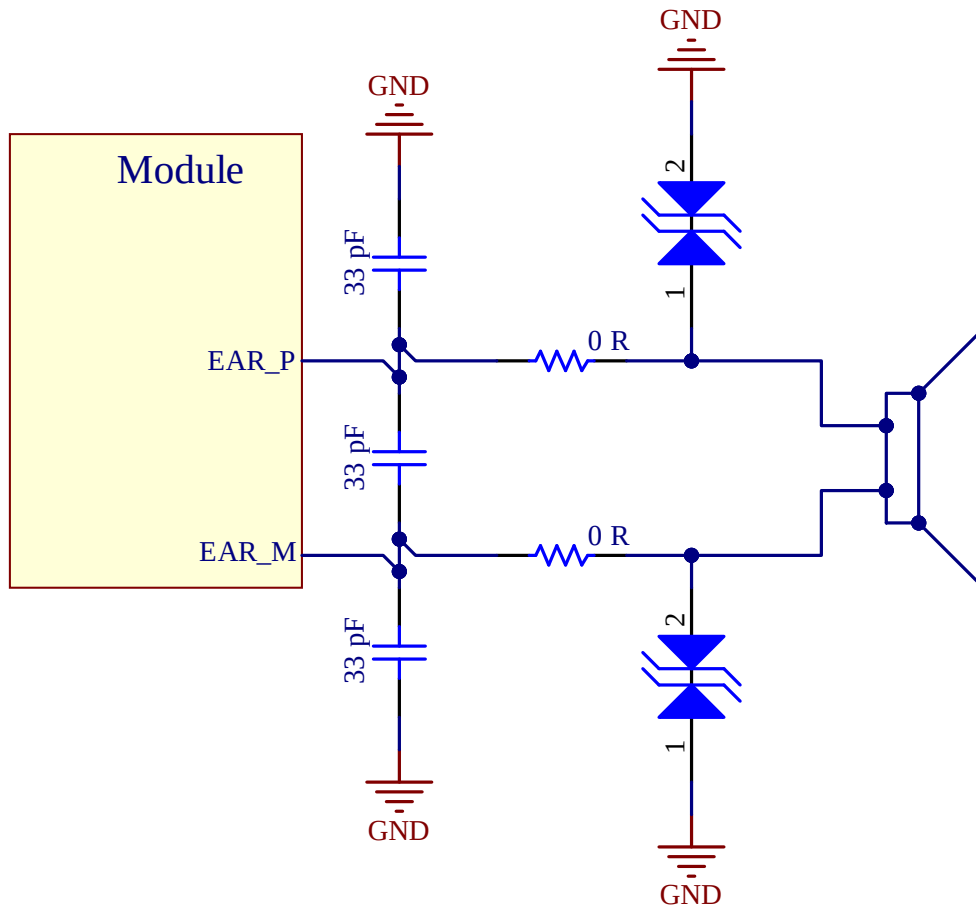


Figure 32: Reference Circuit Design for Earpiece Interface

Table 68: Earpiece output performances

Parameter	Test conditions	Min	Typical	Max	Units
Output power	32Ω load f = 1.02kHz, 6dB gain THD+N < 1%	120	124.5	-	mW
	16Ω load f = 1.02kHz, 6dB gain THD+N < 1%	235	243	-	mW
	10.67Ω load f = 1.02kHz, 6dB gain THD+N < 1%	310	320	-	mW
Full-scale output	6dB gain mode f = 1.02kHz	1.8	2	2.1	Vrms
Voltage	1.5dB gain mode f = 1.02kHz	1	1.2	1.3	Vrms
Output load		10.7	32	50000	Ω
Disabled output impedance	Measured externally, amplifier disabled	1	-	-	MΩ

5.26.4 Headset

Telit SE250B4 Smart module supports Stereo class-AB headphone with 16Ω, 32Ω impedance and up to 50kΩ loads.

Its typical output power at 1.02KHz and THD + N ≤ 1% is:

- 21.5 mW with 16Ω loads, 0 dBFS and -4.5dB gain

- 30.8 mW with 32Ω loads, 0 dBFS and 0dB gain

A 100kΩ pull-down resistor is integrated at CDC_HPH_L pin, which could be used for mechanical insertion or removal detection through CDC_HS_DET pin.

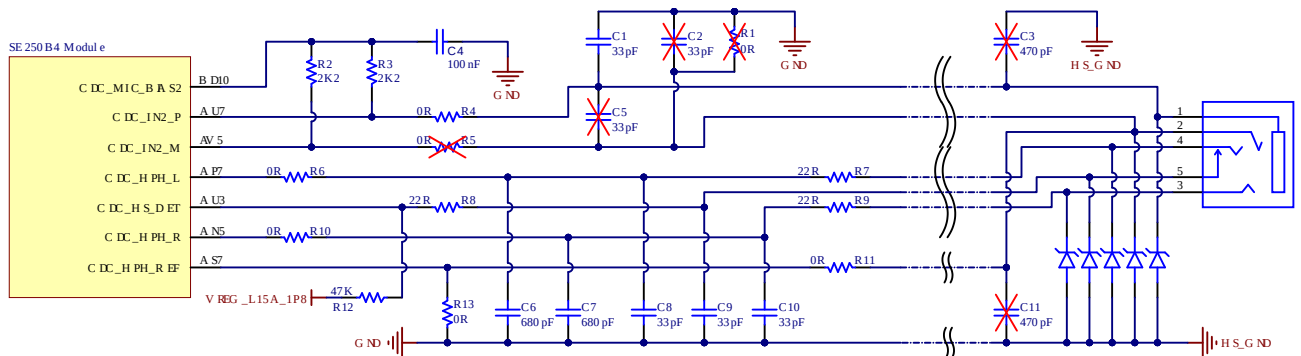


Figure 33: Reference Circuit Design for Headset Interface

Table 69:Headset performances

Parameter	Test conditions	Min	Typical	Max	Units
Output power	16Ω load f = 1.02kHz, 0 dBFS; 0 dB gain mode	-	60.5	-	mW
	32Ω load f = 1.02kHz, 0dBFS; 0dB gain mode	-	31.25	-	mW
Full-scale output Voltage	32Ω load f = 1.02kHz, 0dBFS;	0.95	1	-	Vrms
Ouput Load	Supported output load	16	32	-	Ω
Disabled Output	Measured externally, with amplifier disabled	-	20	-	Ω

5.26.5 Speaker

Telit SE250B4 Smart module can be used to pilot an external loudspeaker by the addition of an external amplifier. Following schematics is a design example using a Class K amplification stage. Please refer to amplifier and loudspeaker datasheet for output matching components.

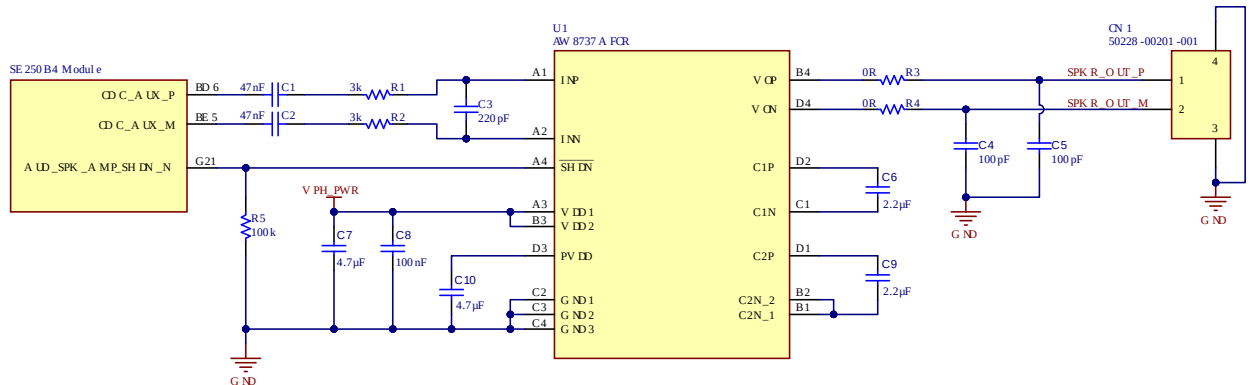


Figure 34: Reference Circuit Design for CDC_AUX Interface

5.26.6 Audio Signal Design Considerations

It is recommended to use EMC microphone with dual build in shunt capacitors (e.g. 10pF and 33pF) to filter RF interference, thus TDD noise.

Without this capacitor the TDD noise could be heard during voice calls, please note that the resonant frequency point of a capacitor largely depends on its material and manufacturing technique. Therefore, you should consult the capacitor vendors to choose the most suitable capacitor to filter out the high-frequency noises.

The severity of RF interference in the voice channel during GSM transmission largely depends on the application design. In some cases, EGSM900 TDD noise is more relevant, while in other cases the DCS1800 TDD is stronger. Therefore you need to select the more suitable capacitor according to the test results. Sometimes, depending also from the routing techniques adopted in your application, on material, or numbers of layers used in your stackup the capacitor might be not required.

In order to decrease radio or other signal interference, place RF antennas as far as possible from your audio interfaces and audio traces.

Power lines can act as a channel for RF interferences and then intermodulate noise on audio traces, the differential audio should be able to reject the common mode noise but it's a good practice to keep also power lines as far as possible from audio traces.

6 Antenna(s)

6.1 Antenna Requirements

Telit SE250B4 Smart module provides four antenna interfaces, two for cellular (main and diversity), one for WiFi/BT and another one for GPS antenna.

The characteristic impedance of all antenna ports inside the module and in our EVK is controlled to 50Ω impedance.

Table 70: Pin Definition for antenna connections

PIN	Signal	I/O	Function	Comment
A41	ANT_TRX	AIO	Main antenna interface	50Ω.
A33	ANT_DRX	AI	Diversity antenna interface	

6.1.1 Operating frequencies for Cellular Part

Table 71: SE250B4-EU operating frequencies

3GPP Bands	Downlink	Uplink	Unit
EGSM900 (B8)	925-960	880-915	MHz
DCS1800 (B3)	1805-1880	1710-1785	MHz
WCDMA B1	2110-2170	1920-1980	MHz
WCDMA B5	869-894	824-849	MHz
WCDMA B8	925-960	880-915	MHz
LTE-FDD B1	2110-2170	1920-1980	MHz
LTE-FDD B3	1805-1880	1710-1785	MHz
LTE-FDD B5	869-894	824-849	MHz
LTE-FDD B7	2620-2690	2500-2570	MHz
LTE-FDD B8	925-960	880-915	MHz
LTE-FDD B20	791-821	832-862	MHz
LTE-FDD B28	758-803	703-748	MHz
LTE-TDD B38	2570-2620	2570-2620	MHz
LTE-TDD B40	2300-2400	2300-2400	MHz
LTE-TDD B41	2496-2690	2496-2690	MHz

Table 72: SE250B4-NA Operating frequencies

3GPP Bands	Downlink	Uplink	Unit
LTE-FDD B2	1930-1990	1850-1910	MHz
LTE-FDD B4	2110-2155	1710-1755	MHz
LTE-FDD B5	869-894	824-849	MHz
LTE-FDD B7	2620-2690	2500-2570	MHz
LTE-FDD B12	729-746	699-716	MHz
LTE-FDD B13	746-756	777-787	MHz
LTE-FDD B14	758-768	788-798	MHz
LTE-FDD B17	734-746	704-716	MHz
LTE-FDD B25	1930-1995	1850-1915	MHz
LTE-FDD B26	859-894	814-849	MHz
LTE-FDD B66	2110-2200	1710-1780	MHz
LTE-FDD B71	617-652	663-698	MHz
LTE-TDD B41	2496-2690	2496-2690	MHz

6.1.2 Reference Design for Cellular Part

A reference circuit design for the main and Rx-diversity antenna interface is shown below. In your application board, please consider reserving a π -type matching circuit for each antenna to achieve better RF performances, and place matching components as close as possible to the antenna feeding point. Components must be chosen with RF measurements on the final device, so at the first stage of the design you can consider 0Ω series resistor and not mount shunt capacitors/inductors. After measurement adjust the component values for best antenna matching.

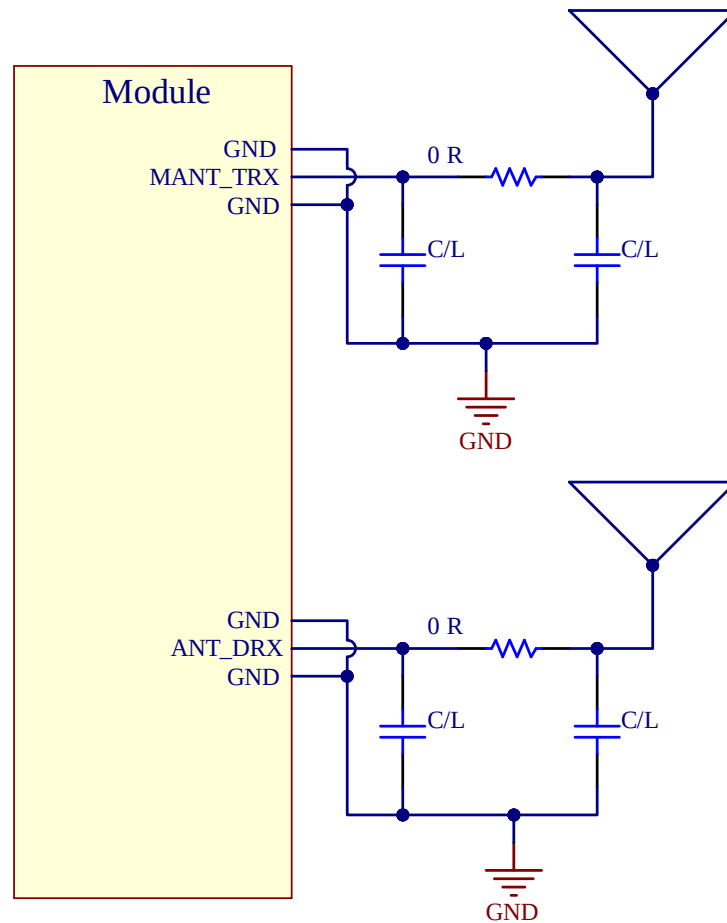


Figure 35: Recommended RF connection circuit for cellular part

6.2 Operating Frequencies for Wi-Fi/BT Part

The following tables show the pin definition and frequency specification for the Wi-Fi/BT antenna interface.

Table 73: Wi-Fi/Bluetooth Pin Definition

PIN	Signal	I/O	Function	Comment
AK1	ANT_WIFI	AIO	Wi-Fi/Bluetooth antenna interface	50 Ω .

Table 74: Wi-Fi/Bluetooth Frequencies

Type	Frequency	Unit
Wi-Fi (2.4GHz)	2402-2482	MHz
Wi-Fi (5GHz)	5180-5825	MHz
Bluetooth 5.0	2402-2480	MHz

6.2.1 Reference Design for Wi-Fi/BT part

A reference circuit design for Wi-Fi/Bluetooth antenna interface is shown below. Please consider adding the placeholder for these components in your PCB application board. Initially leave not mounted the reactive components C/L and use a default series resistor of 0Ω. After RF measurement, adjust the components values to obtain the best matching to 50Ω

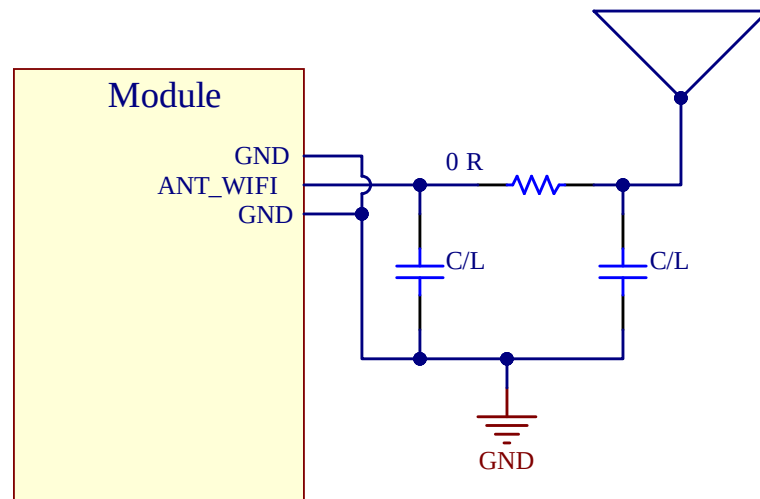


Figure 36: Recommended RF connection circuit for WiFi/BT Part

For antenna specification please refer to the table below.

Table 75: Antenna specification for WiFi/BT

Item	Value
Frequency range	2.402~2.484GHz / 4.9~5.925GHz
VSWR	< 2.2:1 recommended
Polarization	Vertical
Gain	ideally should be isotropic or omnidirectional practically <3.5dBi @ 2.4GHz, <2.5dBi @ 5GHz
Impedance	50Ω

Please consider that in general after the tuning the performance of WiFi should be as reported in the tables below.

Table 76: SE250B4-EU standard WiFi/ BT Performances @2.4GHz at 25°C. Maximum antenna gain reported in Table 99

Item	802.11b (1Mb/s)	802.11g (6Mb/s)	802.11n HT20 (MCS0)	802.11n HT40 (MCS0)	VHT20	VHT40	BT/BLE
Maximum output power dBm EIRP average (calculated)	19.2	19.6	19.7	19.5	19.4	19.3	15 / 9.7

Table 77: SE250B4-EU standard WiFi Performances @5GHz at 25°C. Maximum antenna gain reported in Table 99

Item	802.11a (6Mb/s)	802.11n HT20 (MCS0)	802.11n HT40 (MCS0)	802.11ac VHT20 (MCS0)	802.11ac VHT40 (MCS0)	802.11ac VHT80 (MCS0)
Maximum output power dBm EIRP average (calculated)	21.3	21.3	21.3	20.8	20.8	20.8

Table 78: SE250B4-NA standard WiFi/ BT Performances @2.4GHz at 25°C. Maximum antenna gain reported in Table 103

Item	802.11b (1Mb/s)	802.11g (6Mb/s)	802.11n HT20 (MCS0)	802.11n HT40 (MCS0)	VHT20	VHT40	BT/BLE
Maximum peak output power (dBm)	14.7	21.9	21.9	20.9	21.9	20.8	11.4 / 6.1

Table 79: SE250B4-NA standard WiFi Performances @5GHz at 25°C. Maximum antenna gain reported in Table 103

Item	802.11a (6Mb/s)	802.11n HT20 (MCS0)	802.11n HT40 (MCS0)	802.11ac VHT20 (MCS0)	802.11ac VHT40 (MCS0)	802.11ac VHT80 (MCS0)
Maximum peak output power (dBm)	19	19	19	18.4	18.4	18.5

6.3 GNSS Antenna Interface

The following tables show the pin definition and frequency specification for GNSS antenna interface.

Table 80: GNSS Pin definition

PIN	Signal	I/O	Function	Comment
A9	ANT_GPS	AI	GNSS antenna interface	50Ω.

Table 81: GNSS Antenna Interface

Type	Frequency	Unit
GPS L1	1575.42±1.023	MHz
GLONASS L1	1597.5-1605.8	MHz
BDS B1I	1561.098±2.046	MHz
Galileo E1	1575.42±2.046	MHz

6.3.1 Reference Design for GNSS part

GNSS antenna interface supports both active and ceramic/passive antennas.

Then we have two distinct reference design depending on which antenna type the customer wants in his application board.

The reference design for passive antenna is the same that we had for cellular and Wi-Fi part, we recommend adding the same π -type matching circuit network if you need to tune the characteristic impedance of the board.

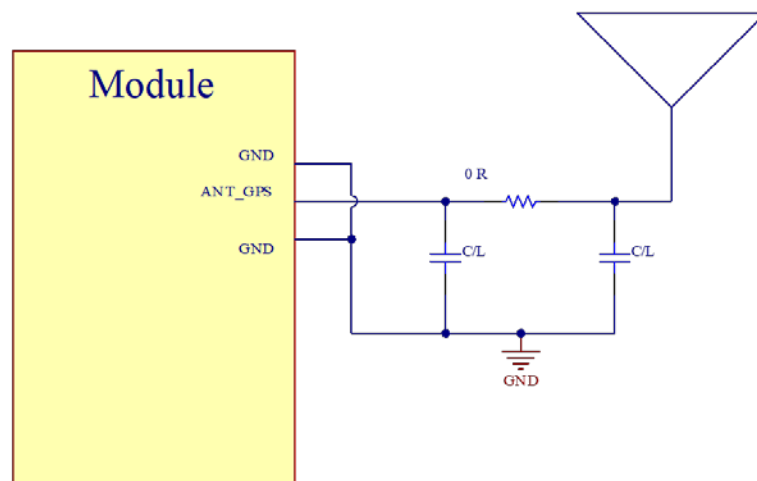


Figure 37 Recommended RF connection circuit for GNSS part

Active antennas require to provide the power for the LNA integrated inside the antenna. For these applications the designer need to add additional components, because a DC feed and DC block section is needed to avoid the dispersion of received signals through the supply path.

For Antenna requirement specification please refer to the table below

Table 82: GNSS Antenna specifications

Item	Value
Frequency range	1565-1607MHz
Polarization	RHCP or linear VSWR: < 2 (Typical)
Passive antenna gain	> 0dBi
Active antenna gain	> -2dBi
Active antenna noise figure	< 1.5dB
Active antenna embedded LNA gain	20dB (Typical)
Active antenna total gain	> 18dBi (Typical)

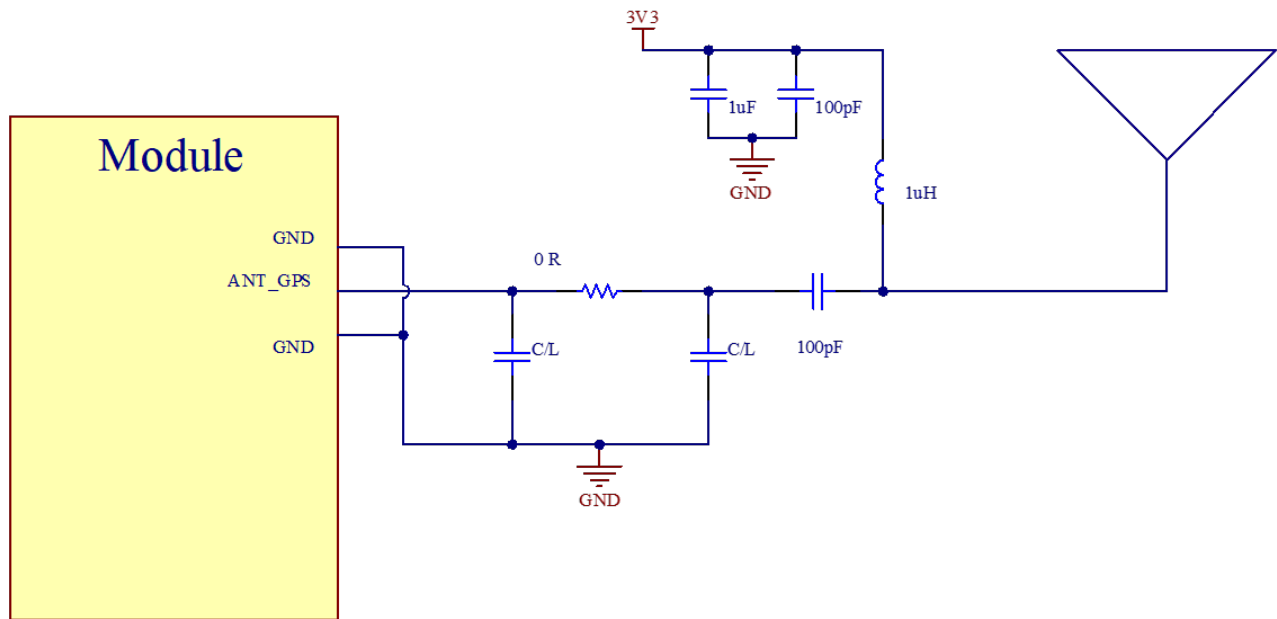


Figure 38 Recommended connection circuit for GNSS active antenna.

Note: We recommend using passive antennas. Active antennas can increase the signal, but an excessive gain can blind the receiver. Please make sure to follow the RF routing guideline for RF track routing.

Note: if GNSS works together with LTE in the field, please design the GNSS antenna by ensuring decoupling higher than 15dB with LTE main antenna.

6.4 PCB RF Design Guidelines

To ensure good RF performance, users must meet the following requirements:

- At least, 4-layer through-hole PCB must be chosen for good impedance control and signal shielding.
- Keep the RF traces at 50Ω.
- Maintain a complete and continuous reference ground plane from module antenna PAD to the RF connector.
- The RF traces should be away from any other noisy traces.
- Keep the RF traces as short as possible.
- Locate Telit SE250B4 Smart module in the center of PCB, rather than in the corner.
- Digital devices and traces should not be placed near sensitive signals like RF and clock.
- Keep SPKR and MIC away from sensitive RF lines.
- RF connectors should be placed close to the module's antenna PAD.
- Antenna matching circuit should be placed close to the antenna.
- If using a coaxial RF cable to connect the antenna, please avoid spanning over UIM cards, power circuits and high-speed digital circuits to minimize the impact on each other.

The characteristic impedance of RF signals should be controlled at 50Ω . In general, the impedance of RF signal is determined by the Permittivity (ER) of PCB material, line width (W), ground clearance (S), height of reference ground plane (H) and other factors.

Microstrip line and coplanar waveguide are usually used to control the characteristic impedance of RF wiring. The following illustrations show the structure design of microstrip line and coplanar waveguide.

The following is the structure diagram:

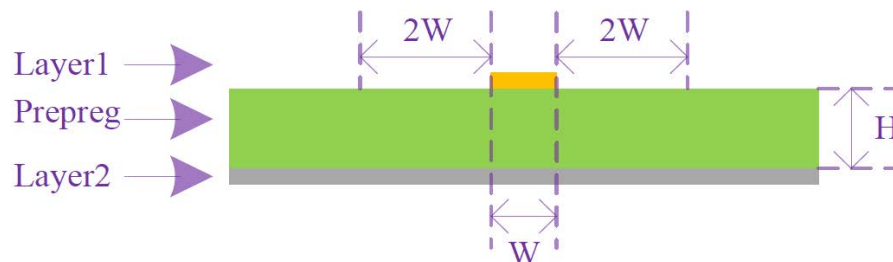


Figure 39 Two Layers PCB Microstrip Structure

Table 83: Impedance control of microstrip line structure

Item	Value	
PCB thickness	1mm	1.6mm
Permittivity (ER)	4.2	4.2
Line thickness	0.035mm	0.035mm
Layer	Layer1	Layer1
Reference plane	Layer2	Layer2
Target impedance	50Ω	50Ω
Expected linewidth W	1.7mm (67mil)	3mm (118mil)

The following is the structure diagram:

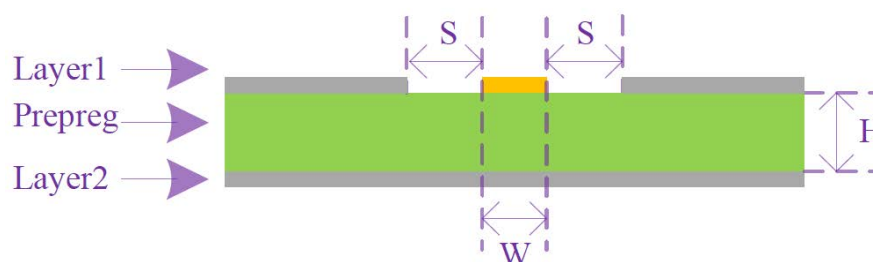


Figure 40 Two-layer PCB coplanar waveguide structure

Table 84: Impedance control of coplanar waveguide structure

Item	Value	
PCB thickness	1mm	1.6mm
Permittivity (ER)	4.2	4.2
Line thickness	0.035mm	0.035mm
Layer	Layer1	Layer1
Reference plane	Layer2	Layer2
Target impedance	50Ω	50Ω
Expected gap to ground S	0.65mm (25.6mil)	0.65mm (25.6mil)
Expected linewidth W	0.2mm (7.8mil)	0.15mm (5.9mil)

The following is the structure diagram:

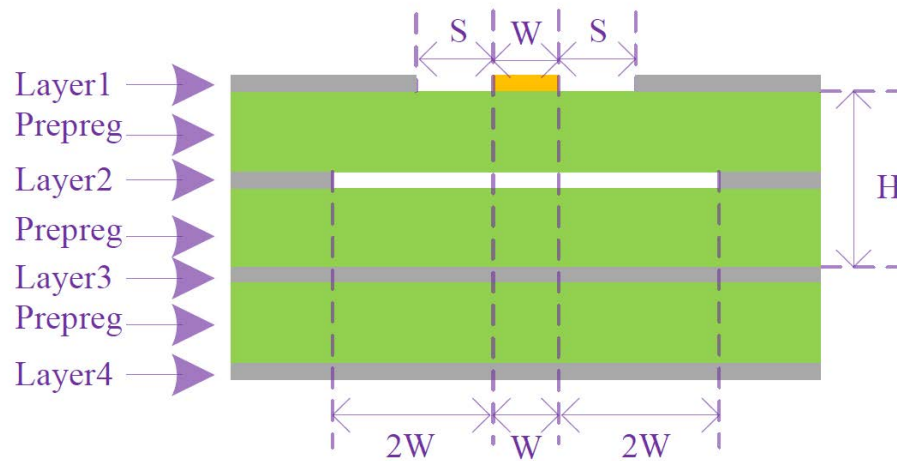


Figure 41 Four-layer PCB coplanar waveguide structure 1#

Four-layer PCB coplanar waveguide structure 2# is shown in following figure. The fourth layer is reference layer.

The following is the structure diagram:

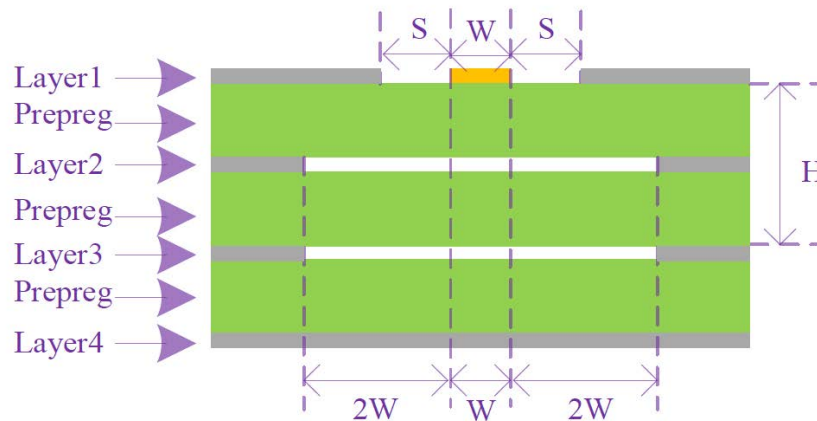


Figure 42 Four-layer PCB coplanar waveguide structure 2#

7 Recommended Peripherals List

Warning: The components listed in the recommended peripherals list are for information purpose only. For more information, see Terms of Use.

7.1 Camera

Table 85: Camera sensor Peripheral List

Items	Part Number	Vendor	Resolution
1	KPFC228 Sensor: BF2257CS Note: part used on Telit EVB	Kingcome	2M
2	CMF008-B5BA-E Sensor: HI556D Note: part used on Telit EVB	TRULYOPTO	5M
3	AAD12B Sensor: S5K3L6XX03-FGX9 Note: part used on Telit EVB	Shine.Tech	13M

7.2 LCD Driver

Table 86: LCD Driver Peripheral List

Item	Part Number	Vendor	Resolution
1	9A-2R057-1110B-K(T) LCD driver IC: FL7703N-G5-DP Touch IC: ST14348-B Note: part used on Telit EVB	Shenzhen DJN Optronics	720(R.G.B)*1520

7.3 Accelerometer and Gyroscope

Table 87: Accelerometer and Gyroscope Peripheral List

Item	Part Number	Vendor	Accelerometer	Gyroscope
1	ICM-42670-N Note: part used on Telit EVB	TDK	√	√
2	LSM6DSOQ Note: QCM2290 POR part	STM	√	√

7.4 E-Compass

Table 88: E-Compass Peripheral List

Item	Part Number	Vendor
1	AK09917D Note: QCM2290 POR part	AKM Semiconductor

7.5 Proximity and Ambient

Table 89: Proximity & Ambient Light Peripheral List

Number	Part Number	Vendor	Proximity	Ambient Light
1	MN78911DKDN Note: part used on Telit EVB	EMINENT	√	√

7.6 Pressure

Table 90: Pressure Peripheral List

Number	Part Number	Vendor
1	BMP285 Note: QCM2290 POR part	Bosch

7.7 Hall

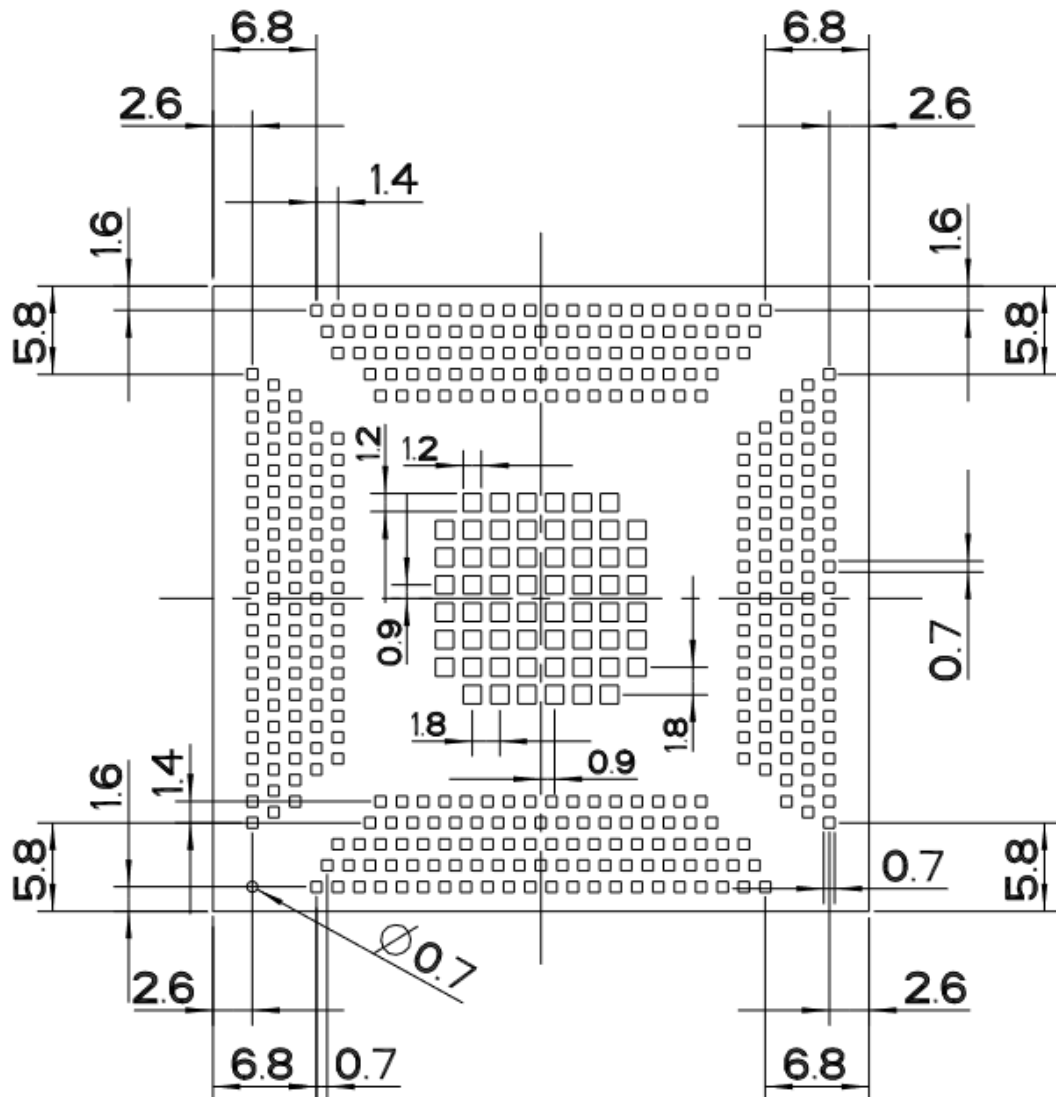
Table 91: Hall Peripheral List

Number	Part Number	Vendor
1	BU52053NVX Note: QCM2290 POR part	Rohm Semiconductor

8 Mechanical Design

The following is the bottom, top and side views of Telit SE250B4 Smart module.

8.1 Drawing



Footprint Pad Drawing
Soldermask Recommend

Figure 43 SE250B4 mechanical drawing (bottom view).

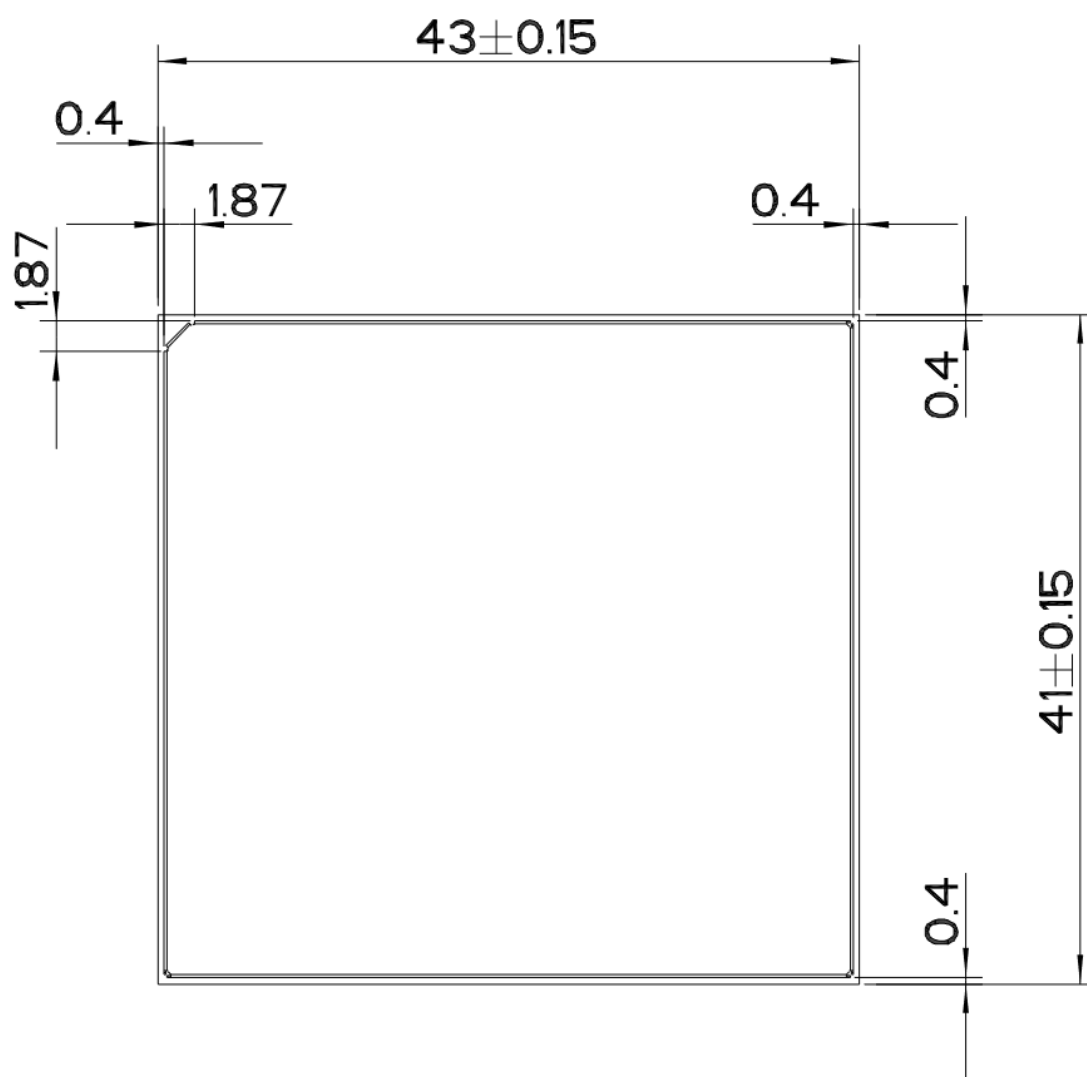


Figure 44 SE250B4 mechanical drawing (top view).

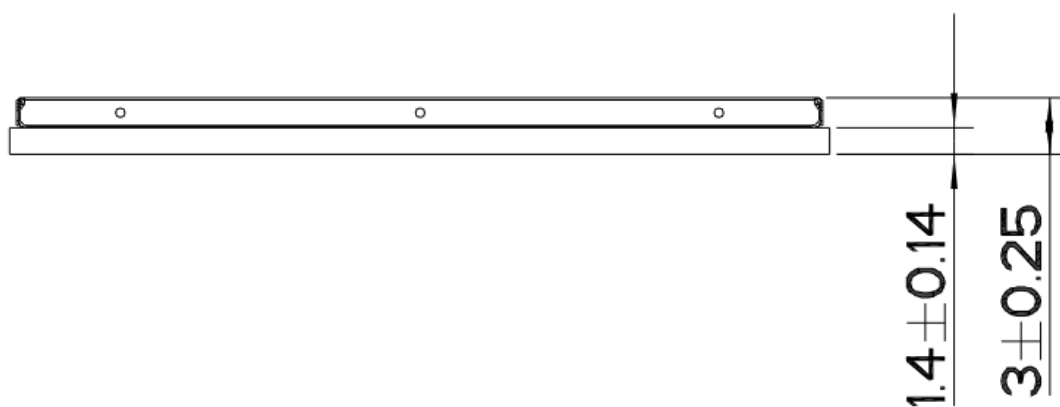
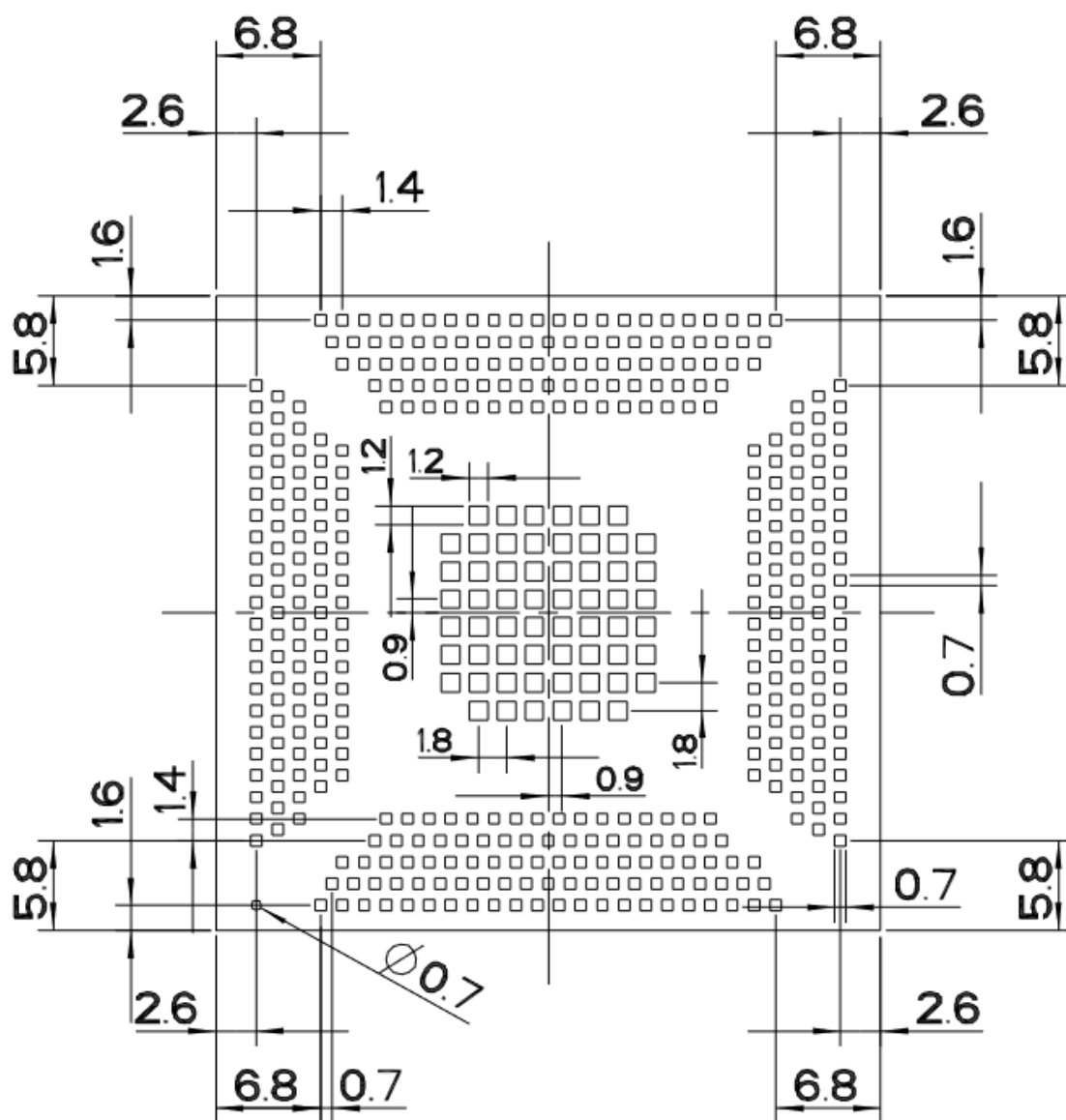


Figure 45 SE250B4 mechanical drawing (side view).

9 Application PCB Design

9.1 Footprint

The following is the footprint diagram:



Footprint Pad Drawing

The stencil and pad dimensions are identical

Figure 46 Footprint of the Telit SE250B4 Smart module

9.2 Solder reflows

The following is the SMT reflow of the Telit SE250B4 Smart module.

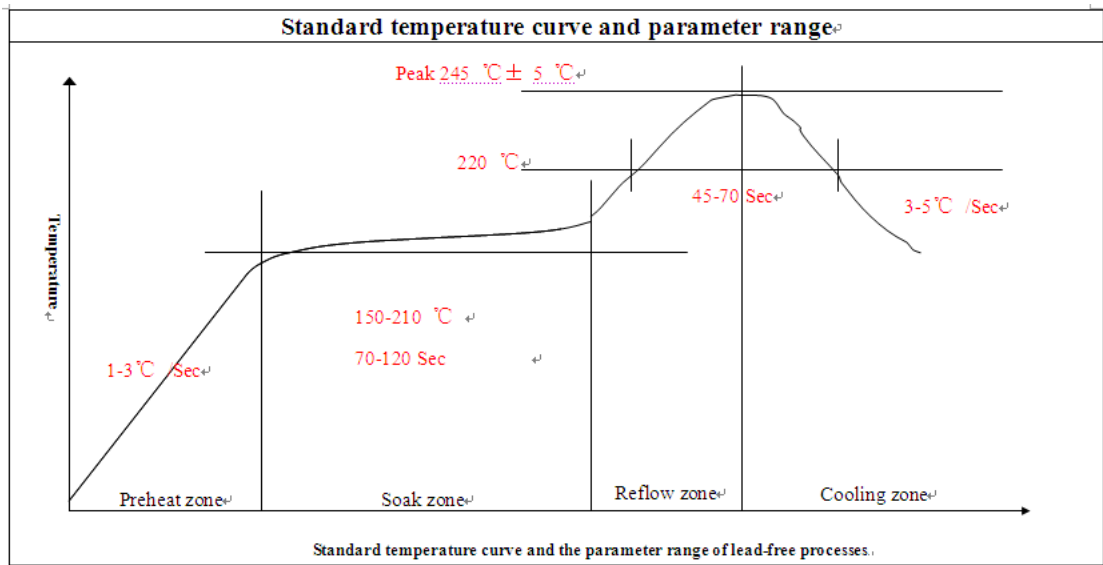


Figure 47 SMT reflow of the Telit SE250B4 Smart module

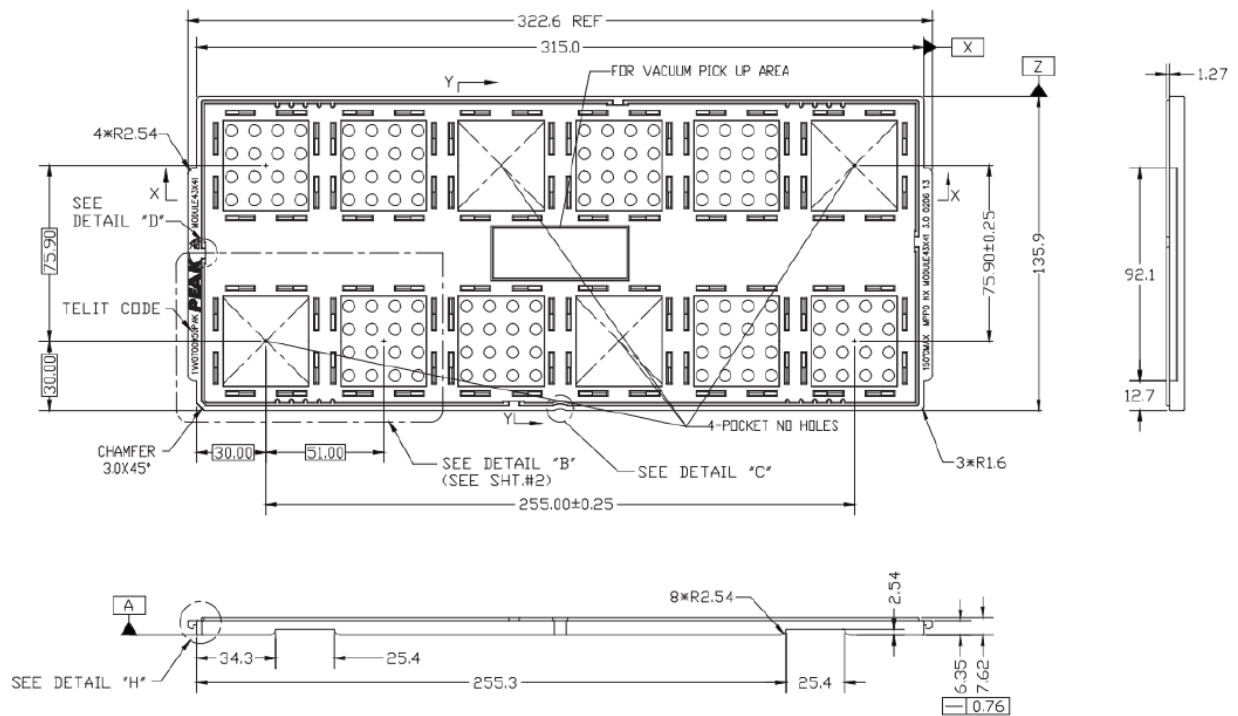
Warning: The SE250B4 module withstands one reflow process only

10 Packaging

10.1 Tray

Telit SE250B4 Smart module supports tray packaging.

The following is the packaging process:

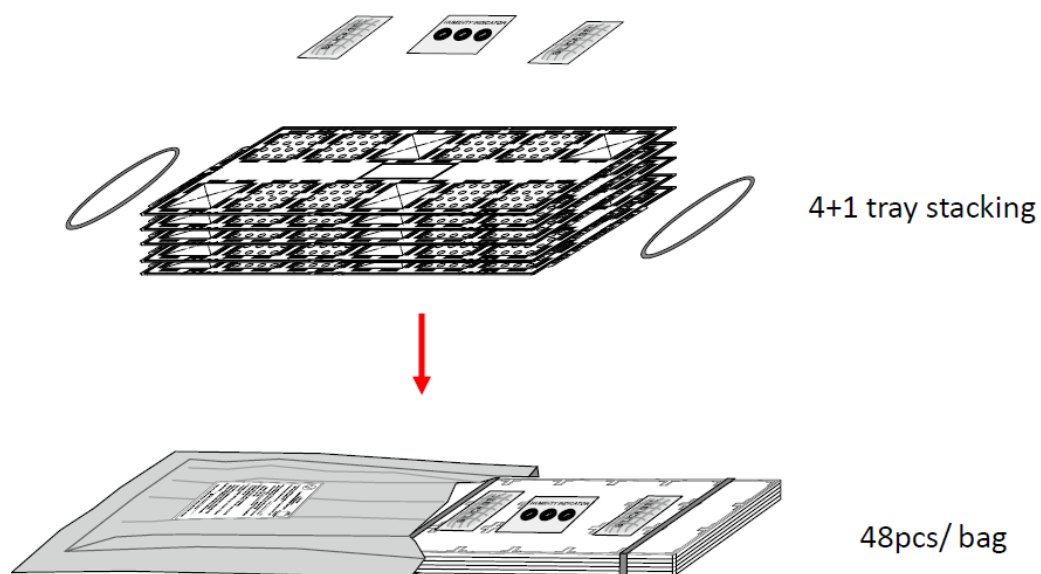


12pcs/ tray

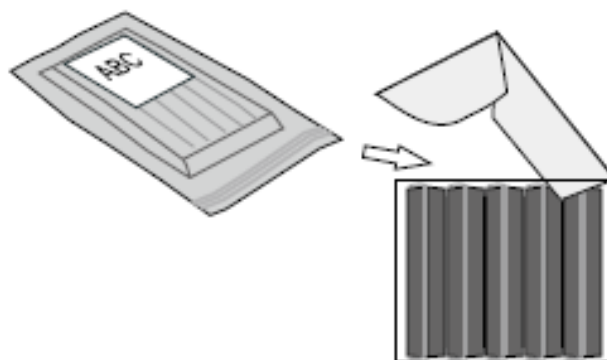
Figure 48 Tray

Table 92: Tray dimensions

Length (mm)	Width (mm)	Units per tray
322.6	135.9	12

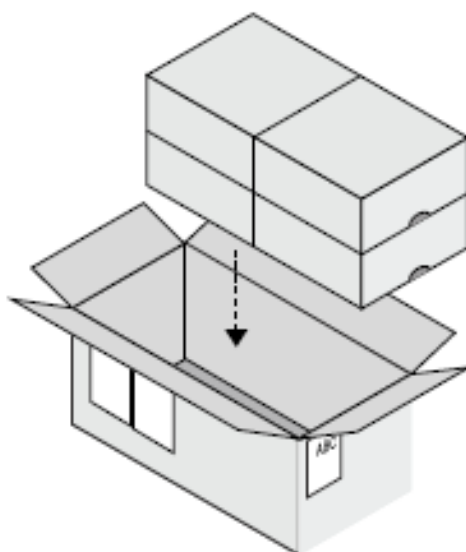


1pcs bag can put 5pcs tray



1 box 5pcs bag

Figure 49 Tray Packaging



1 carton 4pcs box

Figure 50 multi-packaging

10.2 Moisture Sensitivity

Telit SE250B4 Smart module is susceptible to damage induced by absorbed moisture and high temperature. A package's moisture-sensitivity level (MSL) indicates its ability to withstand exposure after it is removed from its shipment bag, while it is on the factory floor awaiting PCB installation. A low MSL rating is better than a high rating; a low MSL device can be exposed on the factory floor longer than a high MSL device.

10.2.1 MSL Rating Summary

Table 93: MSL rating summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	$\leq +30^{\circ}\text{C}/85\% \text{ RH}$
2	1 year	$\leq +30^{\circ}\text{C}/60\% \text{ RH}$
2a	4 weeks	$\leq +30^{\circ}\text{C}/60\% \text{ RH}$
3	168 hours	$\leq +30^{\circ}\text{C}/60\% \text{ RH}$
4	72 hours	$\leq +30^{\circ}\text{C}/60\% \text{ RH}$
5	48 hours	$\leq +30^{\circ}\text{C}/60\% \text{ RH}$
5a	24 hours	$\leq +30^{\circ}\text{C}/60\% \text{ RH}$
6	Mandatory bake before use. After bake, it must be reflowed within the time limit specified on the label.	$\leq +30^{\circ}\text{C}/60\% \text{ RH}$

SE250B4 device samples are currently classified as MSL3, in accordance with standard IPC/JEDEC J-STD-020.

10.3 Baking Requirements

It is necessary to bake modules if the prescribed time limit has been exceeded. The baking conditions are specified in the table mentioned below. Note that if baking is required, devices must be transferred into trays that can be baked to at least 125°C.

Table 94: Banking requirements

Baking conditions options	Duration
40°C±5°C, <5% RH	192 hours
120°C±5°C, <5% RH	4 hours

11 Conformity Assessment Issues

11.1 Approvals Summary

Table 95: EMEA Approvals summary

Region		EMEA
Country &	EU	UK
Type Approval	RED	UKCA
SE250B4-EU		
SE250B4-WE	-	-

Table 96: Americas Approvals summary

Region		Americas	
Country &	US	CA	BR
Type Approval	FCC	ISED	ANATEL
SE250B4-NA			-
SE250B4-WE	-	-	

	The equipment is compliant
	Type approval is in progress
-	The equipment is not compliant

11.2 EMEA Approvals

11.3 EU RED

11.3.1 EU Declaration of Conformity

In accordance with the above Approval Compliance Summary table, where applicable (green ball), hereby, Telit Cinterion declares that the equipment is in compliance with the Directive 2014/53/EU.

The full text of the EU declaration of conformity is available at the following internet address: <https://www.telit.com/red>

Text of 2014/53/EU Directive (RED) requirements can be found here:

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32014L0053>

11.3.2 RED Antennas

This radio transmitter has been approved under RED to operate with the antenna types listed below with the maximum permissible gain indicated. The usage of a different

antenna in the final hosting device may need a new assessment of host conformity to RED.

Table 97: RED Antenna used for testing

Model	Antenna Type
SE250B4-EU	2G/3G/4G: Model: ZTX S50, Type: Monopole with SMA connector, 50ohm impedance WiFi/Bluetooth® : Model: ZTX TG2, Type: Monopole with SMA connector, 50ohm impedance

Table 98: Licensed 2G/3G/4G Max antenna gain for RED

Licensed 2G/3G/4G Max Gain for RED (dBi)	
Band	SE250B4-EU
GSM 900	1.54
DCS 1800	2.06
WCDMA B1	2.2
WCDMA B8	2.31
LTE FDD 1	2.2
LTE FDD 3	2.06
LTE FDD 7	3.12
LTE FDD 8	1.54
LTE FDD 20	1.93
LTE FDD 28	1.81
LTE TDD 38	2.76
LTE TDD 40	2.76
LTE TDD 41	2.83

Table 99: Unlicensed 2.5G/5G Max Gain for RED (dBi)

Unlicensed 2.5G/5G Max Gain for RED (dBi)		
Band	Sub-band	SE250B4-EU
2.4G	2.400~2483.5MHz	3.757
5G	5150~5250MHz	2.475
	5250~5350MHz	1.879
	5470~5725MHz	2.122
	5725~5850MHz	1.672

11.4 Americas Approvals

11.4.1 USA FCC

11.4.1.1 FCC Certificates

The FCC Grants can be found here: <https://www.fcc.gov/oet/ea/fccid>

11.4.1.2 Applicable FCC Rules

Table 100: List of applicable FCC rules

Model <i>Modèle</i>	Applicable FCC Rules
SE250B4-NA	47 CFR Part 2, 22, 24, 27, 90 Part 15.247, 15.407

11.4.1.3 FCC Regulatory Notices

This module is intended for OEM integrators only. Per FCC KDB 996369 D03 OEM Manual, the following conditions must be strictly followed when using this certified module.

Modification statement

Telit has not approved any changes or modifications to this device by the user.

FCC Caution: Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate this equipment.

Interference statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) this device may not cause interference, and
- (2) this device must accept any interference, including interference that may cause undesired operation of the device.

RF exposure considerations

This equipment complies with FCC mobile radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20cm between the radiator and your body. If the module is installed in a portable host, a separate SAR evaluation is required to confirm compliance with relevant FCC portable RF exposure rules.

5G Operation restriction

Operations in the 5.15-5.25GHz band are restricted to indoor usage only. This device meets all the other requirements specified in Part 15E, Section 15.407 of the FCC Rules.

FCC Class B digital device notice

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio

communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by taking one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Information for the OEMs and Integrators

1. This device is intended for OEM integrators only.
2. Please see the full Grant of Equipment document for other restrictions.

OEM/Host manufacturers are ultimately responsible for the compliance of the Host and Module. The final product must be reassessed against all the essential requirements of the FCC rule such as FCC Part 15 Subpart B before it can be placed on the US market. This includes reassessing the transmitter module for compliance with the Radio and EMF essential requirements of the FCC rules. This module must not be incorporated into any other device or system without retesting for compliance as multi-radio and combined equipment.

The final host product must comply with the requirements specified in §15.203, §15.204(b) and §15.204(c).

Manual Information to the End User

The OEM integrator should be aware not to provide information to the end user on how to install or remove this RF module in the user's manual of the end product which integrates this module. The end user manual shall include all required regulatory information/warning as shown in this manual

Information on test modes and additional testing requirement

The module is tested for standalone mobile RF exposure use condition. Any other usage conditions such as co-location with other transmitter(s) or being used in a portable condition will need a separate reassessment through a class II permissive change application or new certification.

If this module is intended for use in a portable device, you are responsible for separate approval to satisfy the SAR requirements of FCC Part 2.1093.

Limited module procedures

Not applicable.

Trace antenna designs

Please, refer to Section 8 of this document for antenna connector trace layout design and to the Gerber files attached to this document. Note that only the unique antenna connectors as indicated in this manual can be used for all transmitter ports.

Any deviation(s) from the defined parameters of the trace layout to antenna connector(s), as described by the instructions, require that the host product manufacturer must notify the module grantee that they wish to change the antenna trace design. In this case, a Class II permissive change application is required to be filed by the grantee, or the host manufacturer can take responsibility through the change in FCC ID (new application) procedure followed by a Class II permissive change application.

Unique antenna connector

For situations where the host product manufacturer is responsible for an external connector, for example with an RF pin and antenna trace design, the integration instructions shall inform the installer that unique antenna connector must be used on the Part 15 authorized transmitters used in the host product. The module manufacturers shall provide a list of acceptable unique connectors.

Additional testing, Part 15 Subpart B disclaimer

The modular transmitter is only authorized by the FCC for the specific rule parts (for example, FCC transmitter rules) listed on the grant, and that the host product manufacturer is responsible for compliance to any other FCC rules that apply to the host not covered by the modular transmitter grant of certification.

This transmitter module is tested as a subsystem and its certification does not cover the FCC Part 15 Subpart B (unintentional radiator) rule requirement applicable to the final host. The final host will still need to be reassessed for compliance to this portion of rule requirements if applicable.

As long as all conditions above are met, further transmitter test will not be required. However, the OEM integrator is still responsible for testing their end-product for any additional compliance requirements required with this module installed.

IMPORTANT NOTE:

In the event that these conditions can not be met (for example certain laptop configurations or co-location with another transmitter), then the FCC authorization is no longer considered valid and the FCC ID can not be used on the final product. In these circumstances, the OEM integrator will be responsible for re-evaluating the end product (including the transmitter) and obtaining a separate FCC authorization.

EMI Considerations

Please follow the guidance provided for host manufacturers in KDB publications 996369 D02 and D04.

How to make changes

Only Grantees are permitted to make permissive changes. Please contact us should the host integrator expect the module to be used differently than as granted:

Grantee code: RI7

Grantee name: Telit Communications S.p.A.

Mailing address: Viale Stazione di Prosecco 5/b, 34010 Sgonico – Trieste, Italy

Website: <https://www.telit.com/>

Support contact: TS-EMEA@telit.com

11.4.1.4 FCC Antenna info

This radio transmitter has been approved by FCC to operate with the antenna types listed below with the maximum permissible gain indicated. Antennas of the same type with equal or lower gain may also be used with this module. Antennas having a gain greater than the maximum gain indicated for that type, are strictly prohibited for use with this device. The antenna must be installed such that 20 cm can be maintained between the antenna and users.

This radio module is sold without a specific antenna. Detailed electrical characteristics of the usable antenna are shown in Chapter 8.

For all transmitters (licensed and unlicensed), the final host product must comply with the requirements specified in §15.203, §15.204(b) and §15.204(c). As such, the host product must use a permanently attached antenna or an antenna that uses a unique coupling. The OEM integrator is responsible for using a unique external antenna connector in the host product, which must be selected from the types listed below.

List of unique antenna connectors

RP-SMA

Table 101: FCC Antenna used for testing

Model	Antenna Type
SE250B4-NA	4G: Model: ZTX S50, Type: Monopole with SMA connector, 50ohm impedance WiFi/Bluetooth® : Model: ZTX TG2, Type: Monopole with RP SMA connector, 50ohm impedance

Table 102: Licensed 4G Max Gain for FCC (dBi)

Licensed 4G Max Gain for FCC (dBi)	
Band	SE250B4-NA
LTE FDD 2	2.16
LTE FDD 4	2.06
LTE FDD 5	2.31
LTE FDD 7	3.12
LTE FDD 12	1.47
LTE FDD 13	1.29
LTE FDD 14	1.47
LTE FDD 17	1.47
LTE FDD 25	2.16
LTE FDD 26	2.31
LTE FDD 66	2.06
LTE FDD 71	1.58
LTE TDD 41	2.83

Table 103: Unlicensed 2.5G/5G Max Gain for FCC (dBi)

Unlicensed 2.5G/5G Max Gain for FCC (dBi)		
Band	Sub-band	SE250B4-NA
2.4G	2.400~2483.5MHz	3.757
5G	5150~5250MHz	2.475
	5250~5350MHz	1.879
	5470~5725MHz	2.122
	5725~5850MHz	1.672

The host product is required to include an antenna which complies with the requirements specified within these integration instructions.

FCC Labelling Requirements for the Host Device

The host device shall be properly labelled to identify the modules within the host device. The certification label of the module shall be clearly visible at all times when installed in the host device, otherwise the host device must be labelled to display the FCC ID of the module, preceded by the words "Contains transmitter module", or the word "Contains", or similar wording expressing the same meaning, as in the below table.

The grantee's FCC ID can be used only when all FCC compliance requirements are met.

Table 104: Host device FCC Label

Model	FCC ID
SE250B4-NA	Contains FCC ID: RI7SE250B4

11.4.2 Canada ISED

11.4.2.1 ISED Database

The product ISED certified can be found here:

Le certificat ISDE est disponible ici:

<https://sms-sgs.ic.gc.ca/equipmentSearch/searchRadioEquipments?execution=e1s1&lang=en>

11.4.2.2 Applicable ISED Rules / Liste des Règles ISDE Applicables

Table 105: Applicable ISED rules / Règles ISDE applicables

Model	Applicable ISED Rules
<i>Modèle</i>	<i>Règles ISDE applicables</i>
SE250B4-NA	RSS-130 Issue 2, RSS-132 Issue 4, RSS-133 Issue 7, RSS-139 Issue 4, RSS-140 Issue 1, RSS-199 Issue 4; RSS-247 Issue 3, RSS-Gen Issue 5, RSS-102 Issue 6

11.4.2.3 ISED Regulatory notices / Avis réglementaires de ISDE

Modification statement / Déclaration de modification

Telit has not approved any changes or modifications to this device by the user. Any changes or modifications could void the user's authority to operate the equipment.

Telit n'approuve aucune modification apportée à l'appareil par l'utilisateur, quelle qu'en soit la nature. Tout changement ou modification peuvent annuler le droit d'utilisation de l'appareil par l'utilisateur.

Interference statement / Déclaration d'interférence

This device complies with Industry Canada licence-exempt RSS standard(s). Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes : (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Wireless notice / Wireless avis

This device complies with ISED radiation exposure limits set forth for an uncontrolled environment and meets the RSS-102 of the ISED radio frequency (RF) Exposure rules. This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter. The antenna should be installed and operated with minimum distance of 20 cm between the radiator and your body.

Le présent appareil est conforme à l'exposition aux radiations ISDE définies pour un environnement non contrôlé et répond aux directives d'exposition de RSS-102 de la fréquence radio (RF) ISED règles d'exposition. L'émetteur ne doit pas être colocalisé ni fonctionner conjointement avec à autre antenne ou autre émetteur. L'antenne doit être installée de façon à garder une distance minimale de 20 centimètres entre la source de rayonnements et votre corps.

Caution / Avertissement:

- (i) the device for operation in the band 5150-5250 MHz is only for indoor use to reduce the potential for harmful interference to co-channel mobile satellite systems;
- (ii) for devices with detachable antenna(s), the maximum antenna gain permitted for devices in the bands 5250-5350 MHz and 5470-5725 MHz shall be such that the equipment still complies with the e.i.r.p. limit; (detachable antenna only)
- (iii) for devices with detachable antenna(s), the maximum antenna gain permitted for devices in the band 5725-5850 MHz shall be such that the equipment still complies with the e.i.r.p. limits as appropriate; (detachable antenna only)

(iv) where applicable, antenna type(s), antenna models(s), and worst-case tilt angle(s) necessary to remain compliant with the e.i.r.p. elevation mask requirement set forth in section 6.2.2.3 shall be clearly indicated.

Le guide d'utilisation des dispositifs pour réseaux locaux doit inclure des instructions précises sur les restrictions susmentionnées, notamment :

(i) les dispositifs fonctionnant dans la bande 5150-5250 MHz sont réservés uniquement pour une utilisation à l'intérieur afin de réduire les risques de brouillage préjudiciable aux systèmes de satellites mobiles utilisant les mêmes canaux;

(ii) pour les dispositifs munis d'antennes amovibles, le gain maximal d'antenne permis pour les dispositifs utilisant les bandes de 5 250 à 5 350 MHz et de 5 470 à 5 725 MHz doit être conforme à la limite de la p.i.r.e.; (detachable antenna only)

(iii) pour les dispositifs munis d'antennes amovibles, le gain maximal d'antenne permis (pour les dispositifs utilisant la bande de 5 725 à 5 850 MHz) doit être conforme à la limite de la p.i.r.e. spécifiée, selon le cas; (detachable antenna only)

(iv) lorsqu'il y a lieu, les types d'antennes (s'il y en a plusieurs), les numéros de modèle de l'antenne et les pires angles d'inclinaison nécessaires pour rester conforme à l'exigence de la p.i.r.e. applicable au masque d'élévation, énoncée à la section 6.2.2.3, doivent être clairement indiqués

Information for the OEMs and Integrators / Informations pour les OEM et les intégrateurs

This device is intended only for OEM integrators under the following conditions: (For module device use)

- 1) The antenna must be installed and operated with greater than 20cm between the antenna and users, and
- 2) The transmitter module may not be co-located with any other transmitter or antenna.

As long as 2 conditions above are met, further transmitter test will not be required. However, the OEM integrator is still responsible for testing their end-product for any additional compliance requirements required with this module installed.

Cet appareil est conçu uniquement pour les intégrateurs OEM dans les conditions suivantes: (Pour utilisation de dispositif module)

1) L'antenne doit être installée et exploitée avec plus de 20 cm entre l'antenne et les utilisateurs, et (if EUT is portable device, please delete this item)

2) Le module émetteur peut ne pas être coïmplanté avec un autre émetteur ou antenne.

Tant que les 2 conditions ci-dessus sont remplies, des essais supplémentaires sur l'émetteur ne seront pas nécessaires. Toutefois, l'intégrateur OEM est toujours responsable des essais sur son produit final pour toutes exigences de conformité supplémentaires requis pour ce module installé.

The OEM integrator has to be aware not to provide information to the end user regarding how to install or remove this RF module in the user's manual of the end product which integrates this module.

The end user manual shall include all required regulatory information/warning as show in this manual.

Manuel d'information à l'utilisateur final

L'intégrateur OEM doit être conscient de ne pas fournir des informations à l'utilisateur final quant à la façon d'installer ou de supprimer ce module RF dans le manuel de l'utilisateur du produit final qui intègre ce module.

Le manuel de l'utilisateur final doit inclure toutes les informations réglementaires requises et avertissements comme indiqué dans ce manuel.

IMPORTANT NOTE/ NOTE IMPORTANTE:

In the event that these conditions can not be met (for example certain laptop configurations or co-location with another transmitter), then the Canada authorization is no longer considered valid and the IC ID can not be used on the final product. In these circumstances, the OEM integrator will be responsible for re-evaluating the end product (including the transmitter) and obtaining a separate Canada authorization.

Dans le cas où ces conditions ne peuvent être satisfaites (par exemple pour certaines configurations d'ordinateur portable ou de certaines co-localisation avec un autre émetteur), l'autorisation du Canada n'est plus considéré comme valide et l'ID IC ne peut pas être utilisé sur le produit final. Dans ces circonstances, l'intégrateur OEM sera chargé de réévaluer le produit final (y compris l'émetteur) et l'obtention d'une autorisation distincte au Canada.

11.4.2.4 ISED Antenna Info / Informations sur l'antenne d'ISDE

This radio transmitter has been approved by ISED to operate with the antenna types listed below with the maximum permissible gain indicated. Antenna types not included in this list, having a gain greater than the maximum gain indicated for that type, are strictly prohibited for use with this device. This transmitter module is authorized only for use in device where the antenna may be installed and operated with greater than 20cm between the antenna and users.

Le présent émetteur radio a été approuvé par ISDE pour fonctionner avec les types d'antenne énumérés ci-dessous et ayant un gain admissible maximal. Les types d'antenne non inclus dans cette liste, et dont le gain est supérieur au gain maximal indiqué, sont strictement interdits pour l'exploitation de l'émetteur. Ce module émetteur est autorisé uniquement pour une utilisation dans un appareil où l'antenne peut être installée et utilisée à plus de 20 cm entre l'antenne et les utilisateurs.

This radio module is sold without a specific antenna. Detailed electrical characteristics of the usable antenna are shown in Chapter 8.

Ce module radio est vendu sans antenne spécifique. Les caractéristiques électriques détaillées de l'antenne utilisable sont présentées au chapitre 8.

Table 106: ISED Antenna used for testing

Model	Antenna Type
<i>Modèle</i>	<i>Type d'Antenne</i>
SE250B4-NA	4G: Model: ZTX S50, Type: Monopole with SMA connector, 50ohm impedance WiFi/Bluetooth® : Model: ZTX TG2, Type: Monopole with RP SMA connector, 50ohm impedance

Table 107: 2G/3G/4G Max Gain for ISED (dBi) / 2G/3G/4G Gain maximum pour ISDE (dBi)

Licensed 4G Max Gain for ISED (dBi)	
<i>4G Gain maximum pour ISDE (dBi)</i>	
Band	SE250B4-NA
LTE FDD 2	2.16
LTE FDD 4	2.06
LTE FDD 5	2.31
LTE FDD 7	3.12
LTE FDD 12	1.47
LTE FDD 13	1.29
LTE FDD 14	1.47
LTE FDD 17	1.47
LTE FDD 25	2.16
LTE FDD 66	2.06
LTE FDD 71	1.58
LTE TDD 41	2.83

Table 108 Unlicensed 2.5G/5G Max Gain for ISED (dBi) / Sans licence 2.5G/5G Gain maximum pour ISDE (dBi)

Unlicensed 2.5G/5G Max Gain for ISED (dBi)		
<i>Sans licence 2.5G/5G Gain maximum pour ISDE (dBi)</i>		
Band	Sub-band	SE250B4-NA
2.4G	2.400~2483.5MHz	3.757
5G	5150~5250MHz	2.475
	5250~5350MHz	1.879
	5470~5725MHz	2.122
	5725~5850MHz	1.672

11.4.2.5 ISED Label and compliance information / ISDE Étiquette et informations de conformité

The host product shall be properly labelled to identify the modules within the host product.

The ISED certification label of a module shall be clearly visible at all times when installed in the host product; otherwise, the host product must be labelled to display the ISED certification number for the module, preceded by the word "contains" or similar wording expressing the same meaning, as follows:

Contains IC: XXXXXX-YYYYYYYYYY

In this case, XXXXXX-YYYYYYYYYY is the module's certification number.

Le produit hôte devra être correctement étiqueté, de façon à permettre l'identification des modules qui s'y trouvent.

L'étiquette d'homologation d'un module d'ISDE devra être apposée sur le produit hôte à un endroit bien en vue, en tout temps. En l'absence d'étiquette, le produit hôte doit porter une étiquette sur laquelle figure le numéro d'homologation du module d'ISDE, précédé du mot « contient », ou d'une formulation similaire allant dans le même sens et qui va comme suit :

Contient IC : XXXXXX-YYYYYYYYYYY

Dans ce cas, XXXXXX-YYYYYYYYYYY est le numéro d'homologation du module.

Table 109: Host device ISED Label / Étiquette ISED du périphérique hôte

Model / Modèle	IC ID
SE250B4-NA	Contains IC: 5131A- SE250B4 Contient des IC: 5131A-SE250B4

11.5 Anatel Certification

Anatel labelling info

SE250B4-WE modules are certified in Brazil by Anatel.

Agência Nacional de Telecomunicações (ANATEL) do Brazil



"Este equipamento não tem direito à proteção contra interferência prejudicial e não pode causar interferência em sistemas devidamente autorizados"

"This equipment is not entitled to protection against harmful interference and must not cause interference in duly authorized systems."

SE250B4-WE Homologation # 08970-25-02618

11.6 RoHS and REACH Info

11.6.1 RoHS info

Any requests on information related to RoHS certifications can be addressed to Chemical.Certifications@telit.com.

11.6.2 REACH info

Any requests on information related to REACH certifications can be addressed to Chemical.Certifications@telit.com.

12 Acronyms and Abbreviations

Table 110: Acronyms and Abbreviations

Acronym	Definition
ADC	Analog – Digital Converter
CSI	Camera Serial Interface
CSD	Circuit Switched Data
CPE	Customer Premises Equipment
DSI	Display Serial Interface
EDGE	Enhanced Data rates for GSM Evolution
EVM	Error Vector Magnitude
FDD	Frequency Division Duplex
FR	Full Rate
GMSK	Gaussian Minimum Shift Keying
GNSS	Global Navigation Satellite System
GPIO	General-Purpose Input/Output
GPS	Global Positioning System
GPRS	General Packet Radio Service
GSM	Global System for Mobile communications
HDMI	High-Definition Multimedia Interface
HSPA+	High-Speed Packet Access Plus
I2C	Inter-Integrated Circuit
LTE	Long Term Evolution
M2M	Machine to Machine
MIMO	Multiple Input Multiple Output
NFC	Near Field Communication
QCM	Qualcomm Communication Module
QPSK	Quadrature Phase Shift Keying
RF	Radio Frequency
SIM	Subscriber Identification Module
SPI	Serial Peripheral Interface
SRDY	Slave Ready
UART	Universal Asynchronous Receiver Transmitter
UIM	User Identity Module
UMTS	Universal Mobile Telecommunication System
USB	Universal Serial Bus
WCDMA	Wideband Code Division Multiple Access
WLAN	Wireless Local Area Network

13 Related Documents

For current documentation and downloads, refer to <https://dz.telit.com/>.

Table 111: Related Documents

S.no	Doc Code	Document Title
1	1VW0301822	SE250B4 Smart EVB User Guide
2	1VW0301823	SE250B4 SDK User Guide

14 Document History

Table 112: Document History

Revision	Date	Changes
9	2026-07-08	Updated Battery charger pad section Updated 5.12 Power Off section Removed CSI pins from display interface Added notes 5.24 Camera interface section Added Camera MIPI pins in 4.2.7. Camera interface section
8	2026-02-17	Memory configuration updated Added SE250B4-WE variant Updated GNSS performance Added note about PM GPIOs and GPIOs update Updated Extended Temperature range in Table 5 Updated Conformity assessment section
7	2025-07-18	Minor format adjustments Updated GPIO information Added note on USB2.0 and GNSS
6	2025-01-16	Updated Peripheral List Migration to latest TC template Minor format adjustments
5	2024-07-29	Updated WiFi/BT performances
4	2024-06-07	Updated temperature range
3	2024-02-07	Added Current Consumption table (Table 33), added conformity assessment section
2	2023-08-07	Release to Downloadzone
1	2023-05-15	First issue
0	2023-05-23	Template Update

From Mod.0809 Rev.15

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